

# FLEXIBLE RETIREMENT AND OPTIMAL TAXATION\*

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## Abstract

Raising the retirement age is a common policy response when social security schemes face fiscal pressures. We develop and estimate a dynamic life cycle model to study optimal retirement and tax policy when individuals face health shocks and income risk and make endogenous retirement decisions. The model distinguishes three channels through which health affects retirement: nonconvexities in labor supply due to health-dependent fixed costs of working, earnings reductions, and mortality risk. Labor supply nonconvexities make labor supply elasticities endogenous and reverse the standard prediction that optimal labor wedges rise with age: Labor wedges eventually decline as hours fall with worker age and poor health. Mortality risk similarly overturns the standard positive savings distortion, making it optimal to subsidize savings. We estimate the model to match US microdata and compute the constrained-efficient allocation. In the optimum, labor wedges are hump-shaped, and lower for workers in poor health; retirement consumption increases with the claiming age more steeply than does the benefits top-up under the social security delayed retirement credit; and retirement savings is increasingly subsidized with worker age and poor health. These findings support flexible retirement policies rewarding delayed retirement while accommodating heterogeneity in health rather than uniform increases in the retirement age.

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# 1 Introduction

Social security reform remains one of the most significant policy debates across developed economies, driven primarily by concerns about the fiscal sustainability of retirement benefits. Most policy discussions focus on increasing the full retirement age or making social security benefits less generous. However, such policy proposals are socially unpopular and rarely materialize in bipartisan reform<sup>1</sup> because they cut costs at the price of hurting recipients' welfare and retirement choices. In this paper, we posit that flexible retirement policies may better align with individuals' life cycle decisions than a uniform increase in the retirement age, and we characterize what those optimal policies look like.

Central to this thesis is the recognition that retirement is an endogenous decision in which the deterioration of health as a person ages plays a determinant role. Poor health affects retirement decisions through several channels. It introduces labor supply nonconvexities by making work substantially less pleasant or feasible, leads to wage reductions due to productivity losses, and increases mortality risks (Blundell, French and Tetlow 2016). Each of these channels distinctly affects an individual's labor supply and retirement choices, such that any comprehensive retirement policy analysis must account for health, ability, and mortality shocks.

The paper aims to answer some key questions: How should retirement benefits, labor income taxes, and savings incentives be designed jointly when retirement is endogenous and health deteriorates? How do health-induced nonconvexities, productivity losses, and mortality risks shape the dynamics of optimal labor and savings distortions? What do constrained-efficient optimal retirement policies look like, and how do they compare with the incentives embedded in status quo US policies?

To answer these questions, we develop a dynamic life cycle model featuring heterogeneous agents who face uncertain health transitions, wage risk, mortality risk, and medical expenditures. Agents choose their labor force participation, hours worked, consumption, and savings subject to realistic nonconvexities in their labor supply as in state-of-the-art retirement studies (French 2005; French and Jones 2011; Rogerson and Wallenius 2013). Our model explicitly integrates endogenous retirement choices with exogenous health shocks for the optimal tax analysis (Farhi and Werning 2013; Golosov, Troschkin and Tsyvinski 2016; Stantcheva 2017), and we derive how labor supply nonconvexities, productivity losses, and mortality risks from poor health shape the optimal policies. In short, we deliver the first dynamic taxation analysis with an estimated life cycle model and compute optimal policies

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<sup>1</sup>For instance, as of April 2025, the retirement reform package France adopted in April 2023, whose showcase reform is an increase in the retirement age from 62 to 64, remained politically controversial, with multiple efforts to repeal it (Le Monde 2023).

when retirement is endogenous.

In our framework, wages from work vary because of health shocks and stochastic and persistent ability. We study optimal nonparametric policies targeting income histories and health but not ability. The government or planner maximizes social welfare subject to incentive compatibility constraints that reflect individuals’ unobservable ability and participation constraints as in the standard [Mirrlees \(1971\)](#) income taxation model. Nonparticipation is reversible at an estimated reentry cost, so retirement is an outcome of the model rather than an assumption.

We characterize optimal policies through the use of labor and savings wedges, which are implicit distortions or markdowns in wages and returns on savings that arise in equilibrium. We derive three key theoretical results.

First, for various widely used classes of preferences with nonconvexities in labor supply, the optimal labor wedge decreases for workers in poor health. The intuition behind this result lies in the endogenous adjustment of labor supply elasticities to health and age. In contrast to the isoelastic preferences used in [Farhi and Werning \(2013\)](#) and [Stantcheva \(2017\)](#), preferences generating nonconvexities lead to labor supply elasticities that increase in poor health. Consequently, the planner finds it optimal to reduce labor distortions for unhealthy individuals. This result qualifies and generalizes [Saez’s \(2002\)](#) finding—in a static setting—that if behavioral responses are concentrated on the extensive margin, the optimal tax is an earned income tax credit (EITC) with negative marginal tax rates. In our dynamic setting, the labor wedge is lower for workers in poor health but remains positive in our calibration, in contrast to what [Saez \(2002\)](#) finds. The intuition, in the spirit of [Prescott, Rogerson and Wallenius’s \(2009\)](#) idea of *time averaging*, is that agents in a dynamic model can flexibly choose the *fraction of their lifetime* during which they work. The planner can therefore provide extensive-margin work incentives through promises of higher retirement consumption rather than through negative marginal distortions.

Second, the effect of poor health on wages, the productivity loss channel, alters the law of motion of the labor wedge primarily through the elasticity of wages to ability rather than through the level of wages itself. Specifically, a higher wage elasticity with respect to ability increases wage inequality among individuals with similar health status and thus enhances the insurance value of labor distortions. Under the wage specification estimated in the retirement literature ([French and Jones 2011](#)) and in this paper, log wages are additive in a health–age profile and an ability component. The wage elasticity with respect to ability is therefore one in every health state, implying that health-induced wage levels have no direct effect on the dynamics of the labor wedge. Wage levels nevertheless matter indirectly because they reduce hours, which raises the endogenous elasticity ratio.

Third, by incorporating mortality risk from poor health in our model, we overturn standard results on the optimal savings wedge. While dynamic taxation models that abstract from mortality risk (Farhi and Werning 2013; Golosov et al. 2016) typically imply that positive distortions on savings enhance work incentives, we show that in the presence of mortality risk, a subsidy for savings can be optimal. As survival probabilities decrease because of poor health, the optimal savings wedge becomes negative, indicating a need for a subsidy. This occurs because mortality risk, when uninsured under incomplete annuity markets, induces households to shift consumption disproportionately toward the present. Consequently, the government optimally intervenes by subsidizing savings, facilitating consumption smoothing, and insuring against mortality risk. This result provides a rationale for real-world policies such as preferential tax treatment for retirement accounts (e.g., traditional and Roth individual retirement accounts (IRAs) and 401(k)s), emphasizing that heterogeneous uninsurable mortality risk can justify such policies under time-consistent preferences.

However, these channels do not operate in isolation and can indirectly shape labor wedges through interactions with each other. When they interact, the implications for the optimal labor distortions become richer and more nuanced. An increase in the labor supply elasticity from nonconvexities reduces the optimal labor wedge because labor distortions become costlier. At the same time, productivity declines associated with poor health reduce equilibrium work hours, amplifying labor supply elasticities among unhealthy workers (a level effect) and strengthening incentives for insurance-based redistribution (an elasticity effect). Moreover, consumption smoothing over time increases the relative importance of the autoregressive term in the wedge formula, which reinforces the long-term effects of the productivity and nonconvexity channels.

To quantitatively assess the complex interactions among the three channels, we estimate our model to match key empirical features of retirement and Social Security claiming incentives in the US economy. We build on the established frameworks of French (2005) and French and Jones (2011) with a rich description of status quo US taxes, Social Security, disability insurance, medical expenditures, pensions, and incomplete markets. We estimate health transitions, mortality, medical expenditures, wages, a persistent AR(1) ability process, and French–Jones preferences with health-dependent participation costs and a reentry cost. The model matches observed patterns in labor force participation, retirement, and asset accumulation, aligning closely with prior quantitative studies in the literature (e.g., De Nardi, French and Jones 2010; French and Jones 2011; De Nardi, Pashchenko and Porapakarm 2024).

Our quantitative decompositions show that labor supply nonconvexities, driven by participation and reentry costs, explain most retirement behavior. Removing both costs signif-

icantly raises labor force participation, leads low-productivity workers to work a few hours instead of retiring, and makes labor force exit cease to be absorbing. Productivity losses due to poor health also play a notable role, whereas mortality differences have a more modest impact on participation, though they are important for welfare considerations.

The presence of nonconvexities in labor supply has long been recognized as critical to aggregate labor market outcomes (Holter, Ljungqvist, Sargent and Stepanchuk 2025). Lucas (1987), in his praise of Kydland and Prescott (1982), highlighted the value of incorporating microeconomic realism into macroeconomic models. However, earlier models relying on employment lotteries and complete markets, such as the model of Rogerson (1988), faced criticism for being inconsistent with individual employment histories (Browning, Hansen and Heckman 1999). Prescott (2006a) later advocated for life cycle models with incomplete markets, emphasizing the importance of labor indivisibility as a determinant of lifetime labor supply and career length and bringing macroeconomic analysis into closer alignment with empirical reality and microeconomic evidence (Prescott 2006a,b). Recent studies, such as Graves, Gregory, Ljungqvist and Sargent (2023), further show that these nonconvexities significantly affect career length and labor supply responses to taxation.

We conduct our complete normative analysis by computing the full dynamic Mirrlees optimum at the estimated parameters. We first characterize the retirement allocation and the labor and savings wedges with all three health channels operating. We then examine the roles of nonconvexities and ability persistence, allow the cohort to finance legacy intergenerational transfers, and compare the optimal distortions with those generated by status quo US policies.

First, the optimal labor wedges are hump-shaped in age, and ordered by health, with healthy workers facing the highest wedges and workers with a disability the lowest. The mean wedge for healthy workers peaks near age 40 at approximately 0.20 and falls below 0.08 at ages 56–58 as the exit margin becomes more important. The mild increase among the oldest workers reflects positive selection among those who remain employed rather than a return of the standard age drift. A state-matched analysis shows that composition explains the mild increase in labor wedge among the oldest workers. Unhealthy workers who remain employed are concentrated in high-wedge states, and when the planner’s state variables are held fixed, unhealthy workers face still a lower labor wedge.

Second, the optimal savings wedges become more favorable as workers age and for those who are unhealthy. The mean wedge is negative over much of the life cycle, while the median agent faces a savings subsidy at every age; a right tail of histories with large information rents continues to face positive savings taxation. This history dependence implies that an unconditional subsidy on all assets is likely suboptimal. A statutory tax schedule based on

age and health status alone captures little of the variation in the optimal savings wedge, whereas an earnings-conditioned savings instrument captures nearly two-thirds of it. These results point toward preferential treatment of retirement savings, such as 401(k) and IRA accounts, that becomes more generous with worker age and poor health while retaining earnings- or means-tested clawbacks.

Third, both the persistence of ability and the cohort’s budget requirement matter, but in different ways. Replacing the estimated mean-reverting process with the random-walk benchmark from the dynamic Mirrlees literature substantially increases the optimal late-life participation rates. By contrast, requiring the cohort to fund legacy transfers benchmarked to Social Security’s actuarial deficit and net private bequest flows barely changes the optimal retirement profile. Most of the adjustment to fund intergenerational transfers therefore occurs through consumption and hours, and the utility cost of an additional present-value dollar transferred to other generations increases with the size of the transfers.

Finally, the comparison with the status quo US system identifies three directions for reform. The social security benefit top-up that rewards later retirement under the US’s statutory delayed retirement credit rises less steeply than the retirement-consumption schedule in the optimum. The status quo labor wedges fall with age over most of the life cycle but increase with age near the retirement margin, and disability insurance introduces an abrupt work disincentive at age 50 where the optimal gradient of incentives in relation to health is smooth. The status quo savings distortions are also largest for disabled and retired individuals—precisely those who would optimally receive increasingly favorable saving incentives. Retirement benefits should therefore depend more strongly on the claiming age, labor distortions should decline more smoothly as health deteriorates, and retirement saving incentives should become more favorable with age and poorer health.

Overall, our results suggest that flexible retirement policies, rather than rigid increases in the retirement age alone, would better approximate the constrained-efficient allocation. They would accommodate early retirement for individuals for whom continued work is costly while preserving incentives for healthier and more productive individuals to remain employed.

We extend our analysis along several key dimensions in our online appendix. We develop a tractable continuous-time model to derive analytical insights into the optimal retirement choices under labor supply nonconvexities. Unhealthy workers optimally retire earlier than their healthy counterparts. An option value of working under uncertainty, similar to what arises in real-options models (Dixit 1989; Leland 1994), naturally emerges and decreases with age, increasing endogenous labor supply elasticities. We also propose a simplified implementation of optimal Pareto-improving policies for consumption-risk-neutral agents as studied by Golosov et al. (2016). This implementation features retirement benefits similar

to Social Security benefits that depend only on lifetime income and the retirement age, complemented by history-independent taxes and lump-sum transfers. Finally, we examine robustness to the absence of fixed and reentry costs, to a random-walk ability process, and to different budget requirements.

## 1.1 Literature Review

This paper builds on studies of the empirics of health shocks and retirement incentives; it contributes to the literature on dynamic taxation with and without endogenous retirement and to the literature on quantitative life cycle models of retirement.

**Empirics of Health Shocks and Retirement Incentives:** A large empirical literature documents that health and program incentives shape retirement (McGarry 2004; Blau and Goodstein 2010; Maestas and Zissimopoulos 2010; Coile 2019), that social security provisions generate strong early-retirement incentives across countries (Gruber and Wise 1998; Gruber and Wise, eds 2004), and that claiming and labor force exit are tightly linked, with about two-thirds of workers claiming benefits upon stopping work (Shoven et al. 2017). Liebman and Luttmer (2015) find that information interventions alone do not shift claiming, and Seibold (2021) shows that statutory retirement ages, over and above the financial incentives they impose, act as reference points for retirement behavior. Our normative results speak to this evidence: The optimum contains no kink or notch and none of the bunching that Seibold (2021) documents at statutory claiming ages. Blundell et al. (2016) survey the evidence on health and retirement and identify the three channels our model formalizes. A growing macro-health literature models health dimensions such as medical treatments (Papageorge 2016), frailty (Hosseini et al. 2022), and mental health (Abramson et al. 2024) without systematically studying their implications for optimal policy; our results suggest that the labor supply and mortality channels of health effects may importantly shape these policy implications. Complementarily to structural approaches, Haller (2022) evaluates pension reforms with sufficient statistics in the Bailey–Chetty tradition.

**Dynamic Income Taxation Without Retirement:** The modern dynamic Mirrlees literature characterizes the optimal distortions under persistent private ability shocks (Goloso et al. 2003, 2006; Farhi and Werning 2013; Goloso et al. 2016), with extensions to human capital (Stantcheva 2017) and learning-by-doing (Makris and Pavan 2021); Goloso and Tsyvinski (2015), Stantcheva (2020), and Kaplow (2024) survey this work. These models assume exogenous retirement and abstract from health, and they typically find that the optimal labor wedges rise with age. We show that both the modeling and the finding change with endogenous retirement: The elasticity ratio that these papers hold constant becomes



an endogenous object that changes with health status, the rise in the labor wedge is capped and reversed near the labor force exit margin, and the key policy instrument becomes the claiming-age dependence of retirement consumption—an income-history-dependent instrument outside the scope of the age-dependent linear taxes that this literature finds nearly optimal.

**Dynamic Income Taxation with Endogenous Retirement:** The first analysis of retirement in an optimal social insurance framework is [Diamond and Mirrlees \(1978\)](#), and [Golosov and Tsyvinski \(2006\)](#) characterize disability insurance with asset testing when disability is an absorbing state. Static extensive-margin analyses include [Saez \(2002\)](#), [Jacquet et al. \(2013\)](#), [Gomes et al. \(2017\)](#), and [Rothschild and Scheuer \(2013\)](#). Closer to us, [Cremmer et al. \(2004\)](#), [Choné and Laroque \(2014\)](#), and [Michau \(2014\)](#) study retirement timing in settings without persistent lifetime shocks, and [Shourideh and Troshkin \(2015\)](#) study pension design when a permanent type pins down the productivity history. [Hosseini and Shourideh \(2019\)](#) and [Nishiyama and Smetters \(2007\)](#) study the privatization and funding of social security in overlapping-generation economies. Relative to this work, our environment carries persistent stochastic ability, observable stochastic health with all three retirement channels, and costly reversible exit, and our quantification solves the resulting mechanism design problem at estimated parameters rather than in an illustrative calibration. On the savings margin, [Moser and Olea de Souza e Silva \(2019\)](#) study optimal paternalistic savings policies under present bias, and [Bastani and Ring \(2024\)](#) study optimal delayed taxation of savings under financial frictions; our mortality result shows that savings subsidies arise under fully time-consistent preferences and incomplete markets once mortality is heterogeneous.

**Quantitative Life Cycle Models of Retirement:** Our estimated model builds directly on [French \(2005\)](#), [De Nardi et al. \(2010\)](#), and [French and Jones \(2011\)](#), and on the subsequent literature that has enriched health and social insurance in life cycle models ([Karabarbounis 2016](#); [Jones and Li 2023](#); [Bairoliya and McKiernan 2023](#); [Borella et al. 2023](#); [Yu 2024](#); [Fan et al. 2024](#)). Two ingredients of our estimation matter most for the normative analysis and follow methods and standards in this literature: health-dependent fixed participation costs and a reentry cost estimated jointly with the French–Jones preference parameters, and a jointly estimated persistent AR(1) ability process. The first disciplines the nonconvexity in labor supply that makes retirement optimal; the second is quantitatively consequential for optimal policy, as the random-walk comparison above shows. Unlike this literature, which evaluates parametric reforms within the status quo system, we compute the constrained-efficient benchmark that such reforms approximate; [Huggett and Parra \(2010\)](#) is closest in spirit, characterizing efficient allocations with a fixed retirement age, whereas retirement



choice and optimal distortion are our objects of study.

**Outline:** The paper is structured as follows. Section 2 sets up the life cycle model of endogenous retirement with health and wage shocks. Section 3 develops the planning problem and its recursive formulation. Section 4 presents our theoretical results on the optimal labor and savings wedges and on the welfare cost of legacy budget requirements. Section 5 uses the estimated economy to decompose how each health channel shapes allocations. Section 6 computes the full dynamic Mirrlees optimum in the estimated economy, quantifies each theoretical result, and contrasts the optimum with the distortions of the status quo US system. Section 7 concludes. All proofs are presented in Theory Appendix A, which also contains a continuous-time complement to our discrete-time model. Data Appendix B and Quantitative Appendix C contain the data, the model’s estimation and fit, the computational method, the incentive-compatibility verification, and additional quantitative exhibits.

## 2 A Model of Endogenous Retirement Choice

### 2.1 Environment

Time is discrete, a period is a year, and workers live up to  $T$  years.<sup>2</sup>

**Technology:** Agents work  $n_t \geq 0$  hours at time  $t$  at a wage rate  $w_t$ . Technology is linear, as in Mirrlees (1971):

$$y_t = w_t n_t \tag{1}$$

Wages are a function of two shocks that affect workers: ability  $\theta_t$  and health  $h_t$ :

$$w_t = w_t(\theta_t, h_t) \tag{2}$$

where  $w_t(\cdot, h_t)$  is strictly increasing and concave in ability (for each health status  $h_t$ ,  $\partial w_t(\theta, h_t)/\partial \theta > 0$ ,  $\partial^2 w_t(\theta, h_t)/\partial \theta^2 \leq 0$ ). This wage determination equation, as in the similar frameworks of Erosa et al. (2016), French (2005), and French and Jones (2011), allows health, age, and idiosyncratic ability shocks to affect wages. The notation  $\varepsilon_{w\theta} \equiv d \ln(w_t(\theta, h_t)) / d(\ln \theta_t)$  represents the elasticity of wages to ability.

**Preferences:** Workers have preferences in consumption and labor that are impacted by their health status  $u(c_t, n_t; h_t)$ . This specification captures many preferences, such as those described in French and Jones (2011) and Rogerson and Wallenius (2013), that can lead, with fixed costs of working, to nonconvexities in labor supply. For instance, in French and

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<sup>2</sup>In our estimation, survival to the terminal age is negligible for every health group, so the bound  $T$  operates as a mortality probability that rises sharply late in life. None of our qualitative results depend on the value of  $T$ .

Jones (2011), the utility from consumption and leisure is

$$u(c_t, n_t; h_t) = \frac{1}{1-\nu} \left( c_t^\gamma l_t (n_t; h_t)^{1-\gamma} \right)^{1-\nu} \quad (3)$$

where  $\gamma \in [0, 1]$  is the weight on consumption and  $\nu$  the coefficient of relative risk aversion (CRRA). Leisure in period  $t$  is given by:

$$l_t = L - n_t - \phi_n I\{n_t > 0\} - \phi_{RE} RE_t - \phi_h(h_t) \quad (4)$$

where  $L$  is the total time endowment,  $\phi_n$  the fixed time cost of work incurred when hours worked are positive,  $\phi_{RE}$  a reentry cost incurred by individuals who reenter the labor market  $RE_t = 1$ , and  $\phi_h(h_t)$  the time lost due to unhealthy status.

**Nonconvexities and Endogenous Labor Supply Elasticities:** Under the preferences in (3)–(4), the compensated  $\varepsilon_t^c$  and uncompensated  $\varepsilon_t^u$  labor supply elasticities to the net wage, at constant savings, combine into the ratio

$$\frac{\varepsilon_t^c}{1 + \varepsilon_t^u} = \frac{l_t}{l_t + n_t}$$

in which the preference parameters  $\gamma$  and  $\nu$  cancel while the fixed costs remain through the leisure term in (4). This elasticity ratio varies with hours, health, and age, and will be an important statistic in our analysis.

**Health, Disability, and Mortality:** In each period, individuals face uncertainty in health,  $h_t \in \mathcal{H} \equiv \{0, 1, 2\}$ , which takes three values: 0 for good health, 1 for poor health, and 2 for a disability. The initial health status in period 1 has probability  $\pi_{j,t=1} = Pr(h_1 = j)$ . Health in the next period,  $h_{t+1}$ , depends on the individual's current health and age and evolves according to a Markov chain across the three states, with an age-dependent Markov transition matrix  $\pi$ . A typical element of the health transition matrix at age  $t$  is given by:

$$\pi_{j,i,t+1} = Pr(h_{t+1} = j | h_t = i, t + 1), \quad i, j \in \mathcal{H}^2 \quad (5)$$

Blundell et al. (2016) conclude in their study of retirement incentives that “falling health is an important determinant of retirement and it is a feature worth capturing in a retirement model.” Our model captures the three properties of health that they find important for retirement decisions.

**1. Labor supply nonconvexity channel:** *“Declining health makes work less pleasant”.*

We consider several forms of nonconvexities in labor supply: a time cost of work as in French and Jones (2011), setup time costs as in Prescott et al. (2009), and fixed utility costs of work. These nonconvexities can depend on health, with fixed costs being larger for unhealthy workers.

2. **Productivity channel:** Declining health “*can reduce an individual’s productivity and thus the individual’s wage.*” For each ability  $\theta$ ,  $w(\theta, 0) \geq w(\theta, 1) \geq w(\theta, 2)$ .
3. **Mortality channel:** “*Health shocks might reduce life expectancy.*” The parameter  $s_{t+1}$  denotes the probability that an individual is alive at age  $t + 1$  conditional on her being alive at age  $t$ . The survival probability depends on age and previous health status, as  $s_{t+1}(h_t) = S(h_t, t + 1)$ . Because individuals live up to a maximum age  $T$ ,  $s_{T+1} = 0$  for any health status  $h$ .

**Ability and Its Evolution:** Following [Farhi and Werning \(2013\)](#), agents are born in period 1 with heterogeneous earning ability  $\theta_1$  with distribution  $f_1(\theta_1)$  and at time  $t$  with distribution  $f^t(\theta_t|\theta_{t-1})$  over a fixed support  $\Theta = [\underline{\theta}, \bar{\theta}]$ . Let the probability of a history  $\{\theta^t\} \in \Theta^t$  be  $P(\theta^t) = f^t(\theta_t|\theta_{t-1}) \cdots f^2(\theta_2|\theta_1)f_1(\theta_1)$ .

We introduce notation for the health and mortality shocks. Denote by  $\{h^t\} \equiv (h_1, \dots, h_t)$  the history of health shocks up to period  $t$ ,<sup>3</sup> and by  $\Pi(h^t)$  the probability of a history  $\{h^t\}$ ,  $\Pi(h^t) \equiv \pi_{h_t, h_{t-1}, t} \cdots \pi_{h_2, h_1, 2} \pi_{h_1, t=1}$ . Similarly, let  $S(h^0) = 1$  and  $S(h^{t-1}) \equiv s_t(h_{t-1}) \cdots s_2(h_1)$  be the probability of survival until  $t$ . The expected utility, with discount factor  $\beta$ , from an allocation  $\mathbf{x}$  that specifies consumption and hours as a function of the history of ability  $\{\theta^t\}$  and health  $\{h^t\}$  is

$$U(\mathbf{x}) = \sum_{t=1}^T \beta^{t-1} \sum_{h^t \in \mathcal{H}^t} S(h^{t-1}) \Pi(h^t) \int_{\Theta^t} u(c(\theta^t, h^t), n(\theta^t, h^t); h_t) P(\theta^t) d\theta^t \quad (6)$$

The planner’s problem over these allocations is the subject of the next section.

### 3 Planning Problem

In every period, the planner observes consumption  $c_t$ , work status, health status, and income from work  $y_t$ . However, the planner does not observe  $\theta_t$  and therefore does not observe labor  $n_t = y_t/w_t(\theta_t, h_t)$ , either. This private information friction reflects the fact that the planner cannot set policies that depend on unobservable ability but can set income-dependent policies such as tax policy and health status-dependent policies such as disability insurance and Medicare. Health status can be certified: Medical diagnoses and formal disability determinations are how these programs condition transfers on health in practice. Ability is not verifiable, so the planner, as in the [Mirrlees \(1971\)](#) tradition, conditions on its observable consequence, income.<sup>4</sup>

<sup>3</sup>We use the underscore notation  $h_t$  for realizations at time  $t$  and the superscript  $h^t$  for the history of realizations  $h_1, h_2, \dots$  until  $h_t$ .

<sup>4</sup>If health were instead private information, the planner would face a two-dimensional screening problem in which agents report both ability and health, and health-misreporting constraints would attenuate, though

This section first sets up the planning problem and explains the structure of the incentive constraints. Second, we relax the incentive problem using the first-order approach (FOA) procedure developed in [Farhi and Werning \(2013\)](#), and we incorporate health status and the retirement decision. Third, through a redefinition of the state space, we write a recursive program for the planner's relaxed problem.

### 3.1 Incentive Compatibility

In constrained efficient allocations, we focus on direct revelation mechanisms and suppose, on the basis of the revelation principle, that agents report their ability  $\theta_t \in \Theta$  in each period. A reporting strategy, denoted by  $r = \{r_t : \Theta^t \times \mathcal{H}^t \rightarrow \Theta\}_{t=1}^T$ , is a sequence of maps that specify a reported type  $\hat{\theta}_t = r_t(\theta^t, h^t)$  after each history. The period- $t$  report may depend on the full history of true ability shocks up to and including  $t$  and on the observed health history. A strategy  $r$  induces a history of reports  $\hat{\theta}^t = r^t(\theta^t, h^t) \equiv (r_1(\theta^1, h^1), \dots, r_t(\theta^t, h^t))$ . Allocations  $c : \Theta^t \times \mathcal{H}^t \rightarrow \mathbb{R}_+$ ,  $y : \Theta^t \times \mathcal{H}^t \rightarrow \mathbb{R}_+$  determine consumption  $c$  and output  $y$  as functions of the history of reports  $\hat{\theta}^t$  and observed health  $h^t$ . The continuation utility after history  $(\theta^t, h^t)$  under a reporting strategy  $r$ , denoted by  $\omega^r(\theta^t, h^t)$ , depends on the true history both through the wage at which assigned output must be produced  $w_t(\theta_t, h_t)$  and through the reports the strategy has generated. It solves

$$\begin{aligned} \omega^r(\theta^t, h^t) = & u \left( c(r^t(\theta^t, h^t), h^t), \frac{y(r^t(\theta^t, h^t), h^t)}{w_t(\theta_t, h_t)}; h_t \right) \\ & + \beta s_{t+1}(h_t) \int \sum_{h_{t+1} \in \mathcal{H}} \omega^r(\theta^{t+1}, h^{t+1}) \pi_{h_{t+1}, h_t, t+1} f^{t+1}(\theta_{t+1} | \theta_t) d\theta_{t+1} \end{aligned}$$

with  $\omega^r(\theta^{T+1}, h^{T+1}) = 0$ . The continuation utility after truthful revelation,  $\omega(\theta^t, h^t)$ , corresponds to the truthful strategy  $r_t(\theta^t, h^t) = \theta_t$  for all  $t$  and solves

$$\begin{aligned} \omega(\theta^t, h^t) = & u \left( c(\theta^t, h^t), \frac{y(\theta^t, h^t)}{w_t(\theta_t, h_t)}; h_t \right) \\ & + \beta s_{t+1}(h_t) \int \sum_{h_{t+1} \in \mathcal{H}} \omega(\theta^{t+1}, h^{t+1}) \pi_{h_{t+1}, h_t, t+1} f^{t+1}(\theta_{t+1} | \theta_t) d\theta_{t+1} \end{aligned}$$

An allocation  $c(\theta^t, h^t), y(\theta^t, h^t)$  is incentive compatible if and only if truth-telling yields a weakly higher continuation utility than any reporting strategy  $r$ :

$$[\text{IC}]: \omega(\theta_1, h_1) \geq \omega^r(\theta_1, h_1) \quad \forall \theta_1, h_1, r \quad (7)$$

Denote by  $X^{\text{IC}}$  the set of incentive-compatible allocations. To solve this dynamic problem, we now relax the planner's incentive constraints using the FOA, which requires the following

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not necessarily eliminate, the health-contingent insurance that Propositions 4 and 6 characterize. See [Golosov and Krasikov \(2023\)](#) and [Boerma et al. \(2022\)](#) on multidimensional screening.

assumptions:

**Assumption 1.** *1.A The utility function  $u(c, y; h, \theta)$  is bounded and twice continuously differentiable. The partial derivative  $u_\theta(c, y; h, \theta)$  is bounded.*

*1.B The density  $f^t(\theta_t|\theta_{t-1})$  has full support and has a continuously differentiable derivative with respect to its second argument  $\partial f^t(\theta_t|\theta_{t-1})/\partial \theta_{t-1}$  that is also bounded.*

Suppose the agent has reported ability truthfully until time  $t-1$  and deviates to  $\hat{\theta}$  at time  $t$ ; that is, the reporting strategy satisfies  $r_s(\theta^s, h^s) = \theta_s$  for all  $s \leq t-1$  and  $r_t(\theta^t, h^t) = \hat{\theta} \neq \theta_t$ , with truthful reporting resumed thereafter. The continuation utility under this reporting strategy solves

$$\begin{aligned} \omega^r(\theta^t, h^t) = & u\left(c(\theta^{t-1}, \hat{\theta}, h^t), \frac{y(\theta^{t-1}, \hat{\theta}, h^t)}{w_t(\theta_t, h_t)}; h_t\right) \\ & + \beta s_{t+1}(h_t) \int \sum_{h_{t+1} \in \mathcal{H}} \omega^r((\theta^{t-1}, \hat{\theta}, \theta_{t+1}), h^{t+1}) \pi_{h_{t+1}, h_t, t+1} f^{t+1}(\theta_{t+1}|\theta_t) d\theta_{t+1} \end{aligned}$$

where the first argument of the continuation value records that the report history is  $(\theta^{t-1}, \hat{\theta})$ .

Incentive compatibility in (7) implies that for almost all  $\theta^t, h^t$ , temporal incentive compatibility holds:

$$\omega(\theta^t, h^t) = \max_r \omega^r(\theta^t, h^t) \quad (8)$$

Inversely, if equality (8) holds after all  $\theta^{t-1}, h^t$  and almost all  $\theta_t$ , then (7) also holds (Kapička 2013). Applying the envelope theorem of Milgrom and Segal (2002) under Assumption 1 to the agent's reporting problem, we obtain the marginal variation in continuation utility with respect to reports:

$$\begin{aligned} \dot{\omega}(\theta^t, h^t) \equiv \frac{\partial \omega(\theta^t, h^t)}{\partial \theta_t} = & -\frac{w_{\theta, t}(\theta_t, h_t)}{w(\theta_t, h_t)} n(\theta^t, h^t) u_n(c(\theta^t, h^t), n(\theta^t, h^t); h_t) \\ & + \beta s_{t+1}(h_t) \int \sum_{h_{t+1} \in \mathcal{H}} \omega(\theta^{t+1}, h^{t+1}) \pi_{h_{t+1}, h_t, t+1} \frac{f^{t+1}(\theta_{t+1}|\theta_t)}{\partial \theta_t} d\theta_{t+1} \end{aligned} \quad (9)$$

Let  $X^{\text{FOA}}$  be the set of allocations that satisfy the necessary condition for incentive compatibility (9). This expression characterizes the slope of promised utility with respect to ability types at incentive-compatible allocations. The first term is the static rent as in Mirrlees (1971), and the second term a dynamic rent (Pavan et al. 2014) that summarizes the agent's advance information about his future ability (this second term vanishes when ability is not persistent).

### 3.2 Government's Objective

The analysis is in partial equilibrium, and intergenerational transfers are exogenous. The planner can save aggregate resources in a small open economy at a gross return  $R \equiv 1 + r$ . Her objective is to minimize the net present value of the cost of providing an allocation, subject to incentive compatibility (7) and the expected lifetime utility of each initial type  $(\theta, h)$  being above a utility floor  $\underline{U}(\theta, h)$ . Let  $U(\{c, y\}; \theta, h)$  be lifetime utility as defined in (6) for an agent of initial type  $(\theta, h)$ . In our quantitative analysis, the utility floor is chosen so that the planner's expected discounted net resource cost is zero: The optimum is self-financing and involves no outside resources or intergenerational transfers. The relaxed planning problem, denoted by  $P^{\text{FOA}}$ , replaces the incentive constraint with the envelope condition (9) and is given by

$$\begin{aligned} K(\underline{U}(\theta, h)_\Theta) &\equiv \min_{\{c, y\}} \sum_{t=1}^T \left(\frac{1}{R}\right)^{t-1} \sum_{h^t \in \mathcal{H}^t} S(h^{t-1}) \Pi(h^t) \int_{\Theta^t} [c(\theta^t, h^t) - y(\theta^t, h^t)] P(\theta^t) d\theta^t \\ &\text{subject to } U(\{c, y\}; \theta, h) \geq \underline{U}(\theta, h), \\ &y(\theta^t, h^t) \geq 0, c(\theta^t, h^t) \geq 0, \\ &\{c, y\} \in X^{\text{FOA}} \end{aligned}$$

### 3.3 Recursive Problem

To write the problem recursively, we introduce the following notation for the expected future continuation utility and the future marginal rent:

$$\begin{aligned} v(\theta^t, h^t) &\equiv s_{t+1}(h_t) \int \sum_{h_{t+1} \in \mathcal{H}} \omega(\theta^{t+1}, h^{t+1}) \pi_{h_{t+1}, h_t, t+1} f^{t+1}(\theta_{t+1} | \theta_t) d\theta_{t+1} \\ \Delta(\theta^t, h^t) &\equiv s_{t+1}(h_t) \int \sum_{h_{t+1} \in \mathcal{H}} \omega(\theta^{t+1}, h^{t+1}) \pi_{h_{t+1}, h_t, t+1} \frac{f^{t+1}(\theta_{t+1} | \theta_t)}{\partial \theta_t} d\theta_{t+1} \end{aligned}$$

The continuation utility  $\omega(\theta^t, h^t)$  and envelope condition can be rewritten as:

$$\begin{aligned} \omega(\theta^t, h^t) &= u(c(\theta^t, h^t), n(\theta^t, h^t); h_t) + \beta v(\theta^t, h^t) \\ \dot{\omega}(\theta^t, h^t) &= -\frac{w_{\theta, t}(\theta_t, h_t)}{w(\theta_t, h_t)} n(\theta^t, h^t) u_n(c(\theta^t, h^t), n(\theta^t, h^t); h_t) + \beta \Delta(\theta^t, h^t) \end{aligned}$$

We can then write a recursive formulation of the relaxed planning problem for the states of promised utility (to keep track of the participation constraint), marginal future rents (to



account for dynamic incentives), current ability, health, and time  $(v, \Delta, \theta_-, h_-, t)$ :

$$K(v, \Delta, \theta_-, h_-, t) = \min_{X^{\text{FOA}}} s_t(h_-) \sum_{h \in \mathcal{H}} \pi_{h, h_-, t} \int_{\Theta} \left[ c(\theta, h) - w_t(\theta, h) n(\theta, h) + \frac{1}{R} K(v(\theta, h), \Delta(\theta, h), \theta, h, t+1) \right] f^t(\theta | \theta_-) d\theta$$

subject to

$$\begin{aligned} \omega(\theta, h) &= u(c(\theta, h), n(\theta, h); h) + \beta v(\theta, h) \\ \dot{\omega}(\theta, h) &= -\frac{w_{\theta, t}(\theta, h)}{w(\theta, h)} n(\theta, h) u_n(c(\theta, h), n(\theta, h); h) + \beta \Delta(\theta, h), \\ v &= s_t(h_-) \int \sum_{h \in \mathcal{H}} \omega(\theta, h) \pi_{h, h_-, t} f^t(\theta | \theta_-) d\theta, \\ \Delta &= s_t(h_-) \int \sum_{h \in \mathcal{H}} \omega(\theta, h) \pi_{h, h_-, t} \frac{f^t(\theta | \theta_-)}{\partial \theta_-} d\theta, \\ \omega(\theta, h) &\geq \underline{U}(\theta, h) \text{ if } t = 1 \end{aligned}$$

where the minimization is over the functions  $c(\theta, h), n(\theta, h), \omega(\theta, h), v(\theta, h), \Delta(\theta, h)$ . We obtain the solution of the relaxed problem by minimizing the problem at time 1, treating  $\Delta$  as a free variable:

$$K(\underline{U}(\theta, h)_{\Theta}) = \min_{\Delta} K(\underline{U}(\theta, h)_{\Theta}, \Delta, 1, 0, 1)$$

This recursive formulation allows the dimensionality of the problem to be reduced.

Because the envelope condition is only a necessary condition, the solution to the relaxed problem need not solve the full program; we therefore check numerically the incentive compatibility of the computed allocation of Section 6, following [Farhi and Werning \(2013\)](#). First, at each date, the pair of promises  $(v, \Delta)$  ranges over an endogenous feasible set that is not known in closed form. We discretize a rectangle for the corresponding multipliers and verify that the computed solution remains in the interior. Appendix C.5 reports these checks. Second, we simulate large misreporting strategies on different time frames (one-period and permanent) and bound their welfare gains against the numerical tolerance of our solution algorithm. Appendix C.6 reports these checks and shows that the state where the FoA faces the most potential deviation is that of the poorest disabled workers. The computed allocation passes the checks within numerical accuracy.

## 4 Optimal Policies: Theory

In this section, we characterize the optimal allocations, obtained as solutions to the relaxed program above, using wedges, or implicit taxes, and subsidies, distortions, or markups. To

simplify the exposition, from here on, the dependence on the full history is often left implicit; for example,  $c_t = c(\theta^t, h^t)$  and  $\tau_{L,t} = \tau_{L,t}(\theta^t, h^t)$ . Similarly, function arguments are sometimes left out; for example,  $w_{\theta,t} = (\partial/\partial\theta)w_t(\theta_t, h_t)$ . The term  $\mathbb{E}_t$  denotes the expectation as of time  $t$ , conditional on  $(\theta_t, h_t)$ .

**Definition 1.** (*Wedges*) The labor wedge is the gap between the marginal rate of substitution of consumption for leisure and the marginal product of labor (or the wage).

$$\tau_L(\theta^t, h^t) = 1 + \frac{1}{w_t} \frac{u_n(c_t, n_t; h_t)}{u_c(c_t, n_t; h_t)} \quad (10)$$

The savings wedge is the gap between the marginal rate of substitution in consumption across  $t+1$  and  $t$  and the real return on savings (or the interest rate).

$$\tau_K(\theta^t, h^t) = 1 - \frac{1}{R\beta} \frac{u_c(c_t, n_t; h_t)}{\mathbb{E}_t u_c(c_{t+1}, n_{t+1}; h_{t+1})} \quad (11)$$

The expectation is taken over next-period states including death, so  $\mathbb{E}_t u_c(c_{t+1}, n_{t+1}; h_{t+1}) = s_{t+1}(h_t) \mathbb{E}_t [u_c \mid \text{alive at } t+1]$ .

Wedges are akin to locally linear subsidies and taxes and would all be zero in the first-best economy. In this section, we characterize these wedges for allocations that solve the relaxed planning problem.

We first study in isolation each of the three channels through which health affects retirement—nonconvexities, productivity losses, and mortality—and then characterize their interactions. We emphasize the role of the endogenous labor supply elasticities and the wage elasticity with respect to ability introduced in Section 2.

## 4.1 Labor Supply Nonconvexity Channel

In this section, we study the effect of labor supply nonconvexities on the optimal policies. We isolate this channel with the nonconvexity of utility from leisure but assume no wage difference due to differences in health or survival probability differences due to health,  $w_t(\theta_t, h_t) = w_t(\theta_t, 0) = \theta_t$ ,  $s(h_t) = s(0) = 1$ ,  $\forall h_t \in \mathcal{H}$ . We call this setting Economy 1.

**Assumption 2.** 2.A Utility is separable in consumption and labor:

$$u(c_t, n_t; h_t) = \tilde{u}(c_t) - \psi(n_t; h_t)$$

where  $\tilde{u}$  is concave and  $\psi(\cdot, h_t)$  is bounded and twice continuously differentiable in  $n_t$ .

2.B Ability is a log autoregressive process with persistence  $\rho \leq 1$ :

$$\ln(\theta_t) = \rho \ln(\theta_{t-1}) + \epsilon_t$$

where  $\epsilon_t$  has density  $f^t(\epsilon|\theta_{t-1})$  with  $\mathbb{E}[\epsilon|\theta_{t-1}] = 0$ .

The disutility from labor  $\psi$  in Assumption 2.A need not be convex in labor or separable in health. Let  $\varepsilon_t^u(h)$  and  $\varepsilon_t^c(h)$  be the uncompensated and compensated labor supply elasticities to the net wage, with savings held fixed.<sup>5</sup>

In [Farhi and Werning \(2013\)](#), utility is separable, and the disutility of labor is isoelastic, so the elasticity ratio  $\frac{\varepsilon^c}{1+\varepsilon^u}$  is a constant. Their labor wedge formula has two parts. A drift term, the positive covariance between consumption growth and productivity, increases the labor wedge over time. That is because providing incentives requires promising higher consumption growth to abler agents, which raises the value of insurance. An autoregressive term creates persistence in the labor wedge with a coefficient equal to the persistence of ability  $\rho$  times the constant elasticity ratio. With persistent shocks, the optimal labor wedge therefore rises with age. With nonconvex labor supply, the elasticity ratio is no longer a constant: It is an endogenous, health-dependent state variable. Lemma 2 gives the resulting law of motion.

**Lemma 2.** *Suppose that Assumptions 1 and 2 hold in Economy 1. The optimal labor wedge evolves according to*

$$\begin{aligned} E_{t-1} \left[ \frac{\tau_{L,t}}{1 - \tau_{L,t}} \frac{\varepsilon_t^c(h_t)}{1 + \varepsilon_t^u(h_t)} \frac{1}{\beta R} \frac{u'(c_{t-1})}{u'(c_t)} | h_t \right] = \\ Cov \left( \frac{1}{R\beta} \frac{u'(c_{t-1})}{u'(c_t)}, \ln(\theta_t) \right) + \rho \frac{\tau_{L,t-1}}{1 - \tau_{L,t-1}} \frac{\varepsilon_{t-1}^c(h_{t-1})}{1 + \varepsilon_{t-1}^u(h_{t-1})} \end{aligned} \quad (12)$$

for all  $h_t \in \mathcal{H}$ , where  $\varepsilon^u$  and  $\varepsilon^c$  are the endogenous uncompensated and compensated labor supply elasticities to the net wage with savings held fixed.

The proof and expressions of the elasticities are in Appendix A.1. The expectation is taken for the history up until  $t - 1$  and for each health status  $h_t \in \mathcal{H}$ , i.e., for given  $\{\theta^{t-1}, h^t\}$  and conditional on survival. Equation (12) has the same structure as its [Farhi and Werning \(2013\)](#) counterpart. The covariance term captures the insurance value created by long-run incentives, and the autoregressive term creates persistence with coefficient  $\rho$  times the elasticity ratio.<sup>6</sup> The difference is that the elasticity ratio now multiplies the current wedge on the left-hand side and the inherited wedge on the right-hand side as a history-dependent and health-dependent object. With the elasticity ratio held fixed, the formula reproduces

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<sup>5</sup>We define these two elasticities, respectively, as in [Saez \(2001\)](#) at constant savings:

$$\varepsilon^u = \frac{\frac{\psi_n(n;h)}{n} + \frac{\psi_n(n;h)^2}{u'(c)^2} u''(c)}{\psi_{nn}(n;h) - \frac{\psi_n(n;h)^2}{u'(c)^2} u''(c)}, \quad \varepsilon^c = \frac{\frac{\psi_n(n;h)}{n}}{\psi_{nn}(n;h) - \frac{\psi_n(n;h)^2}{u'(c)^2} u''(c)}$$

For preferences that are nonseparable within a period, such as (3), the same objects are defined from the agent's static labor supply problem at constant savings; Appendix A.2 contains the derivations.

<sup>6</sup>The endogenous elasticities depend on the full history  $\{\theta^t, h^t\}$ . For ease of notation, we emphasize the dependence on health, as this will be the focus of our analysis.

the familiar upward trend of the labor wedge with age. Once the ratio is endogenous, a new effect appears. An increasing elasticity ratio  $\frac{\varepsilon_t^c(h_t)}{1+\varepsilon_t^u(h_t)}$  pushes in the direction of a reduction in the optimal labor wedge. Lemma 3 gives a closed-form expression for the elasticity ratio and comparative statics when hours decline, for explicit preferences that satisfy the assumptions of Lemma 2 and for the French and Jones (2011) preferences that we use in our quantitative analysis.

**Lemma 3** (The elasticity ratio in closed form). *Consider an interior allocation with  $n_t > 0$ , and let  $l_t$  denote leisure net of hours and fixed costs as in (4).*

- (i) *For the separable nonconvex preferences  $u(c_t, n_t; h_t) = \tilde{u}(c_t) + \frac{1}{1-\gamma} l_t^{1-\gamma}$  with leisure  $l_t$  defined in (4) as in (French and Jones 2011), and exponent  $\gamma > 0$ ,*

$$\frac{\varepsilon_t^c}{1 + \varepsilon_t^u} = \frac{l_t}{l_t + \gamma n_t}$$

- (ii) *For the French and Jones (2011) preferences (3)–(4), at constant savings,*

$$\frac{\varepsilon_t^c}{1 + \varepsilon_t^u} = \frac{l_t}{l_t + n_t}$$

*independently of the consumption weight  $\gamma$  and the curvature  $\nu$ .*

*In both cases, the ratio is strictly decreasing in hours  $n_t$ , and it converges to one as hours vanish. In contrast, for isoelastic convex disutility of labor  $\psi(n) = \frac{n^{1+1/\varepsilon}}{1+1/\varepsilon}$ , the ratio is the constant  $\frac{\varepsilon}{1+\varepsilon}$ .*

The proof is in Appendix A.2. For isoelastic disutility of labor, the elasticity ratio is constant, equal to  $\frac{\varepsilon}{1+\varepsilon}$ . This is the case in Farhi and Werning (2013) and the numerical exercise of Stantcheva (2017), where the labor wedge increases with age. However, for preferences that lead to nonconvexities in labor supply, as hours fall because of age or poor health, the elasticity ratio rises toward one. This increasing elasticity ratio, all else equal, ultimately leads to a declining labor wedge. However, leisure and consumption are endogenous and vary over the life cycle and with health. Proposition 4 formalizes the reduction in the optimal labor wedge for health deteriorations for the standard families of nonconvex preferences in the retirement literature.

Indeed, our autoregressive expression (12) applies to preferences that are separable in consumption and labor, which covers the limiting cases of the prevalent forms of labor supply nonconvexity. First, it allows us to analyze nonconvexities in the form of lost time out of leisure that can worsen with poor health in equation (4) of French and Jones (2011) at the separable limit where  $\nu \rightarrow 1$  or generally under preferences of the form:

$$u(c_t, n_t; h_t) = \tilde{u}(c_t) + \frac{1}{1-\gamma} l_t^{1-\gamma} \quad (13)$$

for  $\gamma > 0$ , CRRA utility of consumption  $\tilde{u}$ , and leisure  $l_t$  defined in (4). Second, it can capture nonconvexities in the form of a smooth version of a production setup time cost  $g(n) = \max\{0, n - n_h\}$  as in [Prescott et al. \(2009\)](#) and [Rogerson and Wallenius \(2013\)](#).<sup>7</sup>

**Proposition 4.** *Under the assumptions of Lemma 2, then, for all these preferences,*

- (i) *the separable limit of [French and Jones \(2011\)](#) fixed time cost preferences (3) and (4) at  $\nu \rightarrow 1$ ,*
- (ii) *general separable and nonconvex labor supply preferences with time fixed costs of participation defined by (13),*
- (iii) *preferences with fixed setup costs of production as in [Prescott et al. \(2009\)](#),*

*the labor wedge ultimately decreases in poor health, that is,  $\exists t' \geq 1$  such that*

$$E_{t-1} \left[ \frac{\tau_{L,t}}{1 - \tau_{L,t}} \frac{1}{\beta R} \frac{u'(c_{t-1})}{u'(c_t)} | h_t \right] \leq \frac{\tau_{L,t-1}}{1 - \tau_{L,t-1}}$$

*for all  $t \geq t'$  and (almost) all  $\{\theta^{t-1}, h^{t-1}, h_t\}$  with  $h_t > h_{t-1}$ .*

The proof is in Appendix A.3. With nonconvexities, poor health raises the elasticity ratio so the planner delivers promised utility to unhealthy workers by lowering their labor distortions.

This result qualifies [Saez's \(2002\)](#) finding in a static setting that when behavioral responses are concentrated on the extensive margin, the optimal schedule switches from a negative income tax with high positive marginal tax rates to an earned income tax credit with negative or small marginal tax rates. The parallel is that nonconvexities lead to a smaller labor wedge for unhealthy workers, whose behavioral responses are more concentrated on the extensive margin. The difference is that in our life cycle economy, agents can choose the fraction of their lifetime spent working, as in ([Rogerson and Wallenius 2013](#)). Therefore, the planner's per-period incentive provision tradeoff is much milder than the work-forever-vs.-never-work choice in [Saez \(2002\)](#). Because individuals eventually supply labor, the planner need not make a large unconditional transfer to lifelong nonworkers, and extensive-margin incentives are provided through the level of retirement consumption rather than through negative marginal distortions. In our calibration in Section 6, we find, consistent with this logic,

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<sup>7</sup>Our strict analog in the spirit of [Prescott et al. \(2009\)](#) and [Rogerson and Wallenius \(2013\)](#) considers health-dependent setup costs  $n_2 \geq n_1 \geq n_0$  so that production from labor is  $g(n_t) = \max\{0, n_t - n_h\}$ . This function and preferences with a fixed utility cost at  $\phi_h I\{n > n_h\}$  are not differentiable at  $n_h$ . However, a smooth family of functions that approximates  $g$  is the tanh function  $g_{\tanh}(n_t) = \frac{1}{k} \tanh(k(n - n_h))$ , where  $k > 0$  is the smoothing parameter and  $g_{\tanh}(n_t) \rightarrow_{k \rightarrow 0} g(n_t)$ . This smoothing satisfies the properties of  $g$  in [Prescott et al. \(2009\)](#) (see Fig. 4 in [Prescott et al. 2009](#), p. 27, for a graphical discussion of  $g$ ). Our result applies to general smoothed nonconvexities that satisfy two simple conditions presented in Appendix A.3.

that the labor wedge is hump-shaped but remains positive and that retirement consumption increases in the age of the last labor market exit.

**A Two-Period Illustration:** Let  $T = 2$ . All agents work in period 1 with common ability  $\theta_1$  and good health. At the start of period 2, health  $h_2 \in \{G, B\}$  is observed, with fixed time cost  $\phi(1) \geq \phi(0)$ , and ability  $\theta_2 = \theta_1^\rho e^\epsilon$  is private, after which the agent either works hours  $n_2$  or retires. Preferences are the separable nonconvex ones in (13). Applying Lemma 2 between the two periods, for each realized  $h_2$ ,

$$E_1 \left[ \frac{\tau_{L,2}}{1 - \tau_{L,2}} \frac{l_2}{l_2 + \gamma n_2} \frac{1}{\beta R} \frac{u'(c_1)}{u'(c_2)} \middle| h_2 \right] = \text{Cov} \left( \frac{1}{R\beta} \frac{u'(c_1)}{u'(c_2)}, \epsilon \right) + \rho \frac{\tau_{L,1}}{1 - \tau_{L,1}} \frac{l_1}{l_1 + \gamma n_1}$$

To gauge the magnitudes of the variation of the labor wedge in old age, specialize this equation to log utility  $u(c) = \ln c$ , set  $\gamma = 1$ ,  $\rho = 1$  and  $\beta R = 1$ , and write  $X_t \equiv \tau_{L,t}/(1 - \tau_{L,t})$ ,  $Z_t \equiv l_t/(l_t + n_t)$ , and  $\Delta_t^c \equiv \ln c_t - \ln c_{t-1}$ , so that  $u'(c_1)/u'(c_2) = e^{\Delta_2^c}$ . Evaluating  $X_2$ ,  $Z_2$ , and  $e^{\Delta_2^c}$  at the conditional mean allocation given bad old-age health  $h_2 = B$ , approximating  $\mathbb{E}_1[e^{\Delta_2^c}] \approx 1 + \frac{1}{2}\sigma_{\Delta^c}^2$  and  $e^{\Delta_2^c} \approx 1 + \Delta_2^c$  inside the covariance gives, to first order,

$$\frac{X_2}{X_1} \approx 1 - \underbrace{\frac{\Delta(n/l)}{1 + n/l}}_{\text{drop in hours}} - \underbrace{\frac{\frac{1}{2}\sigma_{\Delta^c}^2}{}}_{\text{consumption growth var.}} + \underbrace{\frac{\varphi \sigma_\epsilon^2}{X_1 Z_1}}_{\text{pass-through}} \quad (14)$$

where  $\Delta(n/l) \equiv (n/l)_1 - (n/l)_2 > 0$  denotes the fall in hours per unit of leisure upon a health deterioration,  $\sigma_{\Delta^c}^2$  is the variance of consumption growth, and  $\varphi \equiv \text{Cov}(\Delta^c, \epsilon)/\sigma_\epsilon^2$  is the pass-through of permanent income shocks to consumption.

This approximation has a transparent economic interpretation. The first highlighted term arises from the nonconvexity channel. When hours relative to leisure time fall because of bad health, distorting labor there is expensive, so the planner shrinks the labor wedge. The second term captures the fact that the more consumption dispersion the optimum already features, the less additional labor distortion is required to provide incentives. The third term is the drift of [Farhi and Werning \(2013\)](#). Incentive provision requires consumption growth to covary positively with innovations in ability, and those information rents raise the labor wedge with age. The magnitude of the drift is governed by how much of a permanent shock is passed through to consumption under the optimal allocation.

The labor wedge ratio in (14) falls below one when the drop in hours to leisure time and the dispersion in consumption outweigh this drift. It is an approximation for *optimal allocations*, but we can calibrate it with empirical estimates to illustrate plausible magnitudes of optimal labor wedge changes. [Podor and Halliday \(2012\)](#) find, using the American Time Use Survey, that bad health status results in about a 304-hour decrease in market work, which—at the annual time endowment net of participation costs of around 3700 hours in our

calibration—corresponds to  $\Delta(n/l) \approx 0.082$  around a baseline  $n/l \approx 0.55$ . The nonconvexity term is then  $-5.3\%$ . Consumption inequality data can be noisy, and insurance should lead to less consumption inequality in the optimum than in the data. In [Deaton and Paxson \(1994\)](#), the variance of log consumption within each cohort rises by 0.07 per decade in the US, that is,  $\sigma_{\Delta c}^2 \approx 0.007$ . Therefore, we set a conservative value of  $\sigma_{\Delta c}^2 \approx 0$  since a positive dispersion term would lead to an even larger decline in the labor wedge. Setting  $\varphi = 0.64$ ,<sup>8</sup> the permanent-shock pass-through estimated by [Blundell et al. \(2008\)](#), together with  $\sigma_\epsilon^2 \approx 0.016$  from our calibration using the Panel Study of Income Dynamics (PSID), which is consistent with [Storesletten et al. \(2004\)](#) and a first-period wedge  $\tau_{L,1}$  of 30%, so that  $X_1 Z_1 \approx 0.27$ , the drift term contributes  $+3.7\%$ . The three forces net to  $X_2/X_1 \approx 0.98$ , and the labor wedge shrinks with health deteriorations.

## 4.2 Productivity Loss Channel

In this section, we study the effect of the fact that declining health lowers wages. We isolate this channel with a wage that decreases in poor health but assume no labor supply nonconvexity or differences in survival probability due to poor health,  $w_t(\theta_t, h_t) \leq w_t(\theta_t, h')$ ,  $s(h_t) = s_t(0) \forall h_t \in \mathcal{H}, h' \leq h_t$  and  $u(c_t, n_t) = \tilde{u}(c_t) - \psi(n_t)$ , where  $\psi$  is isoelastic with Frisch elasticity  $\varepsilon$ . We call this setting Economy 2.

**Lemma 5.** *Suppose that Assumptions 1 and 2 hold in Economy 2; then, the labor wedge evolves according to*

$$E_{t-1} \left[ \frac{\tau_{L,t}}{1 - \tau_{L,t}} \frac{1}{\varepsilon_{w\theta,t}(h)} \frac{1}{\beta R} \frac{u'(c_{t-1})}{u'(c_t)} | h \right] = \rho \frac{\tau_{L,t-1}}{1 - \tau_{L,t-1}} \frac{1}{\varepsilon_{w,\theta-,t-1}(h_-)} + \left( 1 + \frac{1}{\varepsilon} \right) \text{Cov} \left( \frac{1}{R\beta} \frac{u'(c_{t-1})}{u'(c_t)}, \ln(\theta_t) \right)$$

The proof is in Appendix A.1. This expression is similar to the one derived in [Stantcheva \(2017\)](#) for the effect of human capital on the optimal labor wedge. The level of the wage loss from poor health does not appear in the law of motion of the labor wedge. Health-driven wage differences enter the labor wedge formula only through the wage elasticity with respect to ability,  $\varepsilon_{w\theta,t}(h)$ . A higher elasticity  $\varepsilon_{w\theta,t}(h)$  increases the magnitude of the labor wedge because it increases the value of insurance and redistribution. The intuition for this is that if a certain health status is associated with a higher elasticity of wage to ability, then agents with that health status have higher wage inequality among themselves after ability realizations.

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<sup>8</sup>Note that  $\varphi = 0.64$  is a conservative input here because it measures pass-through in the data rather than at the constrained optimum, where insurance would compress it further.



Estimates of the variation in the wage elasticity with respect to ability are uncommon since ability is often empirically treated as a residual. [French and Jones \(2011\)](#) estimate a specification where the logarithm of hourly wages at age  $t$ ,  $\ln w_t$ , is a function of health status and age,  $W(h_t, t)$ , and an autoregressive component of wages,  $\theta_t$ , as follows:  $\ln(w_t(\theta_t, h_t)) = W(h_t, t) + \ln(\theta_t)$ . For such specifications, which we adopt in our quantitative analysis, the elasticity of wage to ability is constant and independent of health status,  $\varepsilon_{w\theta,t}(h) = 1$ . The direct adverse productivity effect of poor health in [Lemma 5](#) is therefore inactive under our estimated wage process. Wage levels still affect the optimal policy through hours. Hours fall when wages fall during worker spells of poor health because of the substitution effect, and the endogenous elasticity ratio of [Lemma 3](#) increases at the same time. [Section 4.4](#) characterizes this interaction, which shapes the health gap in labor wedges in the quantification of [Section 6](#).

### 4.3 Mortality Channel

In this section, we study the effect of the fact that health shocks might increase mortality. Our results here on the savings wedge apply to our full baseline economy, without assumptions that rule out a specific channel.

**Proposition 6.** *Suppose that Assumptions 1 and 2 hold. Then, the savings wedge satisfies*

$$1 - \frac{1}{s_{t+1}(h_t)} \left( 1 - \frac{\sigma_{c,t+1}^2 u_c(c_t)}{\beta R u_c(\underline{c}_{t+1})^3} \right) \leq \tau_K \leq 1 - \frac{1}{s_{t+1}(h_t)} \left( 1 - \frac{\sigma_{c,t+1}^2 u_c(c_t)}{\beta R u_c(\overline{c}_{t+1})^3} \right)$$

where  $\sigma_{c,t+1}^2$  is the variance of the marginal utility of consumption in the next period  $\sigma_{c,t+1}^2 \equiv \text{Var}_t[\tilde{u}_c(c_{t+1}) | \theta_t, h_t]$  and, respectively,  $\underline{c}_{t+1}$  and  $\overline{c}_{t+1}$  are the minimum and maximum values of future consumption  $\underline{c}_{t+1} \equiv \min_{\{\theta \in \Theta, h \in \mathcal{H}\}} c_{t+1}(\theta^t, \theta, h^t, h)$  and  $\overline{c}_{t+1} \equiv \max_{\{\theta \in \Theta, h \in \mathcal{H}\}} c_{t+1}(\theta^t, \theta, h^t, h)$ .

The proof is in [Appendix A.4](#). The result generalizes the standard characterizations of the savings wedge in the dynamic taxation literature.

The left-hand inequality reproduces the standard positive wedge result. If survival is certain,  $s_{t+1}(h_t) = 1$ , it yields  $0 \leq \frac{\sigma_{c,t+1}^2 u_c(c_t)}{\beta R u_c(\underline{c}_{t+1})^3} \leq \tau_K$ . That is, without mortality risk, the savings wedge is positive ([Farhi and Werning 2013](#); [Golosov et al. 2016](#)). This result follows from applying Jensen's inequality to [Rogerson's \(1985\)](#) inverse Euler equation (IEE). Savings are taxed because the income effect of private saving undermines the provision of work incentives.

The right-hand inequality is an upper bound, obtained from [Liao and Berg's \(2019\)](#) sharpening of Jensen's inequality. With mortality risk  $1 - s_{t+1}(h_t) > 0$ , the upper bound can turn negative, and the optimum subsidizes savings.

To see this transparently, consider the log utility setting of the back-of-the-envelope calculation (14),  $\beta R = 1$  and lognormal consumption growth. With  $\text{Var}_t(u_c(c_{t+1})) \approx \sigma_{\Delta c}^2/c_t^2$  and substituting  $u_c(c_t) = 1/c_t$  and  $u_c(\bar{c}_{t+1})^{-3} = \bar{c}_{t+1}^3$ , the upper bound becomes

$$\tau_K \leq 1 - \frac{1}{s_{t+1}(h_t)} \left( 1 - \sigma_{\Delta c}^2 \bar{g}^3 \right) \quad (15)$$

where  $\bar{g} \equiv \bar{c}_{t+1}/c_t$ . The upper bound is negative if and only if the mortality rate exceeds the consumption dispersion term,

$$\underbrace{1 - s_{t+1}(h_t)}_{\text{mortality rate}} > \underbrace{\sigma_{\Delta c}^2 \bar{g}^3}_{\text{dispersion in consumption}} \quad (16)$$

The two sides are the two competing forces that determine the sign of the savings wedge. On the one hand, mortality risk lowers the effective return on nonannuitized savings and pushes the planner toward subsidizing them. On the other hand, variance in consumption growth is the standard IEE leading to a positive savings wedge.

Retaining the conservative<sup>9</sup> value  $\sigma_{\Delta c}^2 = 0.007$  from [Deaton and Paxson \(1994\)](#) and truncating the growth distribution at its 99th percentile gives  $\bar{g} = 1.21$  at mean consumption  $c_t$  and  $\sigma_{\Delta c}^2 \bar{g}^3 = 0.012$ , so that the savings wedge is negative once annual mortality exceeds 1.2%. In our calibrated survival process, this threshold is crossed during retirement for workers in poor health and more than a decade earlier for workers with disabilities. Evaluated at mortality rates of 2.4 and 6%, (15) yields  $\tau_K \leq -1.2\%$  and  $\tau_K \leq -5.1\%$ , respectively. The optimal savings subsidy steepens sharply with age and with health deterioration, and the subsidy region covers earlier ages when workers are in worse health since survival probabilities fall faster. The quantification in Section 6 shows this occurs at realistic mortality rates.

A broad interpretation of our result is that subsidies for savings, such as preferential treatment of 401(k) retirement plans, can be motivated by mortality risk despite the standard savings distortions early in life that facilitate work incentives. Given that the savings distortion from the IEE is small ([Farhi and Werning 2013](#)), we should expect savings subsidies to be the norm at older ages, and heterogeneity in mortality should be part of canonical quantitative models of retirement (cf., [Jones and Li 2023](#)).

## 4.4 Interactions

The previous sections considered separately three distinct channels through which health affects the optimal policies: nonconvexities in labor supply, productivity losses, and mortality risk. Here, we characterize their combined effects.

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<sup>9</sup>A smaller variance of log consumption makes the condition easier to satisfy at low mortality rates.

**Lemma 7.** *Suppose that Assumptions 1 and 2 hold.*

$$E_{t-1} \left[ \frac{\tau_{L,t}}{1 - \tau_{L,t}} \frac{1}{\varepsilon_{w\theta,t}} \frac{\varepsilon_t^c(h)}{1 + \varepsilon_t^u(h)} \frac{1}{\beta R} \frac{u'(c_{t-1})}{u'(c_t)} | h \right] = \\ \rho \frac{\tau_{L,t-1}}{1 - \tau_{L,t-1}} \frac{1}{\varepsilon_{w,\theta_-}} \frac{\varepsilon_{t-1}^c(h_-)}{1 + \varepsilon_{t-1}^u(h_-)} + Cov \left( \frac{1}{R\beta} \frac{u'(c_{t-1})}{u'(c_t)}, \ln(\theta_t) \right)$$

The proof is in Appendix A.1. This lemma nests Lemmas 2 and 5 and shows how the nonconvexity and productivity channels combine. The law of motion of the labor wedge is governed by the product of the inverse wage–ability elasticity and the endogenous elasticity ratio. An increase in the labor supply elasticity—particularly through nonconvex labor supply—shrinks the labor wedge by making distortions of labor choices costlier. Productivity losses in poor health further reduce equilibrium hours worked by unhealthy workers as a share of disposable leisure hours; by Lemma 3, this raises their elasticity ratio endogenously, even though the wage–ability elasticity itself is constant under our estimated wage process. A higher wage elasticity with respect to ability would amplify the insurance motive of the government and thus increase the magnitude of labor wedges. Finally, the optimal allocations tend to smooth consumption over time, thereby dampening the intertemporal variation in the marginal utility of consumption. As a result, the autoregressive term in the labor wedge formula becomes relatively more important as workers age and reinforces the impact of the productivity and nonconvexity channels in subsequent periods.

## 4.5 Intergenerational Transfers

The planner minimizes the net resource cost  $K$  of an incentive-compatible allocation; our baseline quantification chooses the utility floor at which this cost is zero and the optimum is self-financing. Current cohorts, however, transfer resources across generations publicly (through the legacy financing of pay-as-you-go programs) and privately (through bequests). Legacy debt burdens the current cohort with benefits already paid to earlier ones, while bequests pass resources forward, but both are net outflows from the cohort’s budget. We therefore define the legacy budget requirement  $G \equiv -K \geq 0$  as the net present value of resources the cohort hands to other generations and trace out the family of optima indexed by  $G$ .

**Proposition 8.** *Suppose that Assumptions 1 and 2 hold, and let  $W(G)$  denote maximized ex ante welfare along a family of optima indexed by the budget requirement  $G$  and  $\lambda_1(G)$  be the multiplier on the initial promised utility. Then,*

$$\frac{dW}{dG} = -\frac{1}{\lambda_1(G)}.$$

*The marginal utility cost of intergenerational transfers increases with the budget requirement.*

*Proof.* The value  $W(G)$  is defined by  $K(W(G)) = -G$ . Differentiating and applying the envelope theorem of [Milgrom and Segal \(2002\)](#) to the planner’s Lagrangian, which gives  $K'(U) = \lambda_1$ , yields  $dW/dG = -1/K'(W(G)) = -1/\lambda_1(G)$ . Under Assumptions 1 and 2, convexity of the cost function makes  $K'$  increasing, and since  $W$  decreases in  $G$ ,  $\lambda_1(G) = K'(W(G))$  decreases in  $G$ .  $\square$

Every present-value dollar the cohort hands to other generations costs it  $1/\lambda_1$  units of ex ante utility, and the cost rises as the budget requirement grows. A larger  $G$  leaves the cohort with less output to distribute, consumption is lower at every history, and the marginal utility of consumption is correspondingly higher, so each additional dollar transferred leads to a larger welfare loss. Section 6.6 quantifies this frontier for intergenerational transfers benchmarked against Social Security’s legacy debt and private bequest flows.

## 4.6 Discussion

**Exit, Reentry, and Retirement:** Some form of late-life labor decline arises in the workhorse dynamic income taxation model without nonconvexities. In that setting, work hours decline smoothly with age and can become arbitrarily small, while everyone participates in the labor market. Our model therefore features retirement as observed in the data, with workers moving from full-time work to nonparticipation. In the quantification of Section 6, removing the estimated fixed costs together with the reentry cost leads to workers working few hours and shrinks the nonparticipation period by over fifteen years.

The model of [Grochulski and Zhang \(2017\)](#) characterizes optimal contracts with termination and temporary worker exit for workers with exogenous outside options. In our setting, workers exit, reenter, and have terminal exit from the labor market at retirement age because of the fixed costs of work and reentry costs they face. Importantly, the workers’ outside option at retirement is endogenous and part of the mechanism.

**Nonseparability:** Our estimated [French and Jones \(2011\)](#) preferences (3) are nonseparable between consumption and leisure within each period. Retirees and workers differ in their marginal utility of consumption at the same consumption level. Our theoretical results in this section are derived under within-period separability for tractability (including the separable limits of [French and Jones \(2011\)](#) nonseparable preferences), and our quantification solves the full nonseparable model. Across periods, our preferences are time separable, and this is a substantive restriction. [Grochulski and Kocherlakota \(2010\)](#) show that, with preferences that are nonseparable over time, the optimal savings wedge can be negative and history-dependent transfers at retirement, such as social security benefits, are optimal. In our recursive formulation, cross-period nonseparability would add an additional state variable (for instance, a habit stock) to the planner’s problem. Extending our analysis in such a

direction is beyond the scope of this paper. However, the results of Proposition 6 on retirement savings subsidies under mortality risk and the history-dependence of post-retirement consumption are consistent with the results of Grochulski and Kocherlakota (2010).

## 5 Nonconvexities as Main Driver of Retirement

The theoretical results of Section 4 make the optimal policy depend on how strongly each health-related channel operates in isolation and how they interact. How large are the labor supply nonconvexities? How much does wage variation in health affect participation? How about mortality risk?

To answer these questions, we estimate the primitives of the economy of Section 2 (with the nonseparable French and Jones (2011) preferences from (3) and health-dependent time costs for labor force participation and reentry,<sup>10</sup> so that

$$l_t = L - n_t - \phi_n(h_t)I\{n_t > 0\} - \phi_{RE}RE_t, \quad (17)$$

the ability process, and the health, wage, and mortality shocks) by embedding them in the status quo US policy economy with Social Security, medical expenditures, disability insurance, pensions, taxes, and incomplete markets.

Our estimation proceeds in two steps. First, health transitions, mortality rates, medical expenses, and wage profiles are estimated from the PSID and the Medical Expenditure Panel Survey. Second, preference parameters are estimated by the method of simulated moments, where we match 280 moments from the PSID over worker ages 30–69: labor force participation and hours by health status, mean and median assets, and labor market reentry rates. Appendix C.1 presents the full setup, institutional details, estimation procedure, and fit of the model to the data. Table 2 in Section 6 collects the estimated parameters where our quantitative analysis uses them.

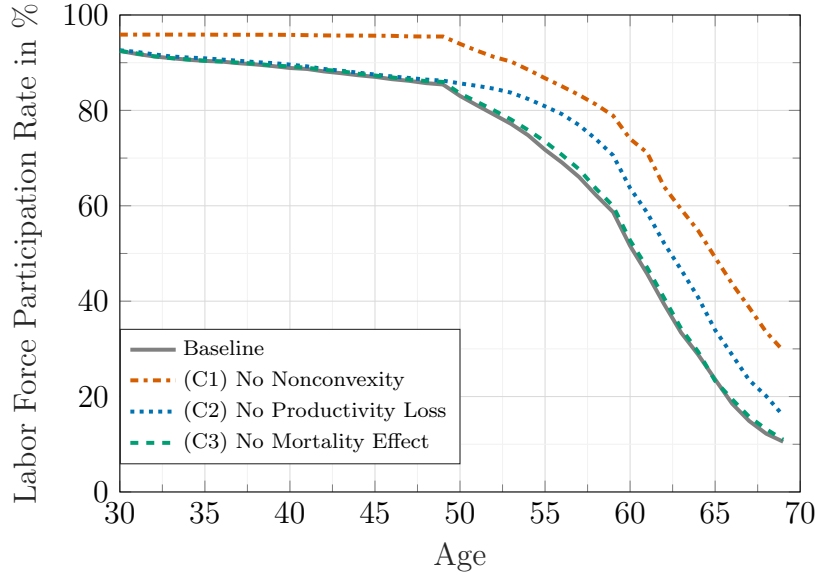
With the estimated economy in hand, to quantify the extent to which the nonconvexity, productivity loss, and mortality channels affect labor force participation and retirement, we run simulations that eliminate one channel at a time. The *No Nonconvexity* benchmark (C1) sets the fixed costs of participation and reentry costs to zero, the *No Productivity Loss* benchmark (C2) equalizes wages across health categories to the good-health wage, and the *No Mortality* benchmark (C3) sets survival probabilities to those of individuals in good health. The excess participation rates relative to rates in the baseline estimated model capture the extent to which each channel explains retirement.

Figure 1 illustrates the result of these decompositions. Nonconvexities in labor supply are

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<sup>10</sup>French and Jones (2011) feature a time cost of bad health that does not depend on participation and a time cost of participation that does not depend on health. We consider a richer time cost of participation that depends on health to allow for nonconvexities in labor supply that more directly depend on health.

Figure 1: Effects of Preferences and Technological Factors on Labor Force Participation



*Notes:* Labor force participation rates as a function of worker age for the estimated quantitative model and counterfactual simulations in which the different channels are individually eliminated: *No Nonconvexity* (C1) sets the fixed costs of participation to zero; *No Productivity Loss* (C2) sets wages equal to those of healthy individuals; *No Mortality Effect* (C3) sets survival probabilities equal to those of healthy individuals.

the primary preference and technological factor explaining labor force nonparticipation in old age, followed by health-related productivity differences. Absent the nonconvexities, labor force participation would be larger at each working age. Productivity differences related to health start affecting labor force participation among workers aged 50 and older. Mortality differences related to health, however, hardly affect labor force participation.

When we keep the nonconvexities but eliminate the health differences in the time costs,  $\phi_n(h) = \phi_n(0)$  for all  $h$ , participation changes only slightly. This qualifies the [Blundell et al. \(2016\)](#) statements in that the *presence* of the nonconvexities in labor supply matters more for retirement behavior than the *variation* in these nonconvexities across health states.

Table 1: Effects on Labor Force Participation Through the Three Channels, by Worker Age Group and Health Status

| Age Group |                      | Good Health |       |       | Poor Health |       |       | With a Disability |       |       |
|-----------|----------------------|-------------|-------|-------|-------------|-------|-------|-------------------|-------|-------|
|           |                      | 30-49       | 62-64 | 65-69 | 30-49       | 62-64 | 65-69 | 30-49             | 62-64 | 65-69 |
| (C1)      | No Nonconvexity      | 6.64        | 28.69 | 26.65 | 7.99        | 21.82 | 19.75 | 11.07             | 12.50 | 12.53 |
| (C2)      | No Productivity Loss | 0.42        | 11.06 | 7.56  | 1.63        | 15.52 | 9.92  | 3.15              | 17.84 | 10.93 |
| (C3)      | No Mortality Effect  | 0.20        | 0.66  | 0.29  | 0.28        | 1.92  | 0.97  | 0.47              | 1.00  | 1.37  |

*Notes:* Percentage-point increases relative to model-estimated labor force participation under the three benchmarks, broken down by health status and for age categories 30–49, 62–64, and 65–69. *No Nonconvexity* (C1) sets the fixed costs of participation to zero; *No Productivity Loss* (C2) sets wages equal to those of healthy individuals; *No Mortality Effect* (C3) sets survival probabilities equal to those of healthy individuals.

Table 1 breaks down the excess percentage-point increases in participation for each benchmark by worker health status and age (young/prime age: 30–49; early retirement: 62–64; late retirement: 65–69). Nonconvexities consistently explain labor participation across age and health categories, while health-dependent mortality rates have much lower effects. Health-related wage differences—though they affect overall participation less than nonconvexities do—dominate in the participation decisions of individuals aged 62–64 with a disability, when disability insurance benefits are available. The mortality channel matters least for participation but is central to the savings distortions. Section 6 computes the optimum with all three channels operating.

## 6 Optimal Policies: Quantitative Analysis

We now solve for the full dynamic Mirrlees optimum in the economy that we estimate in Appendix C.1. Individuals enter the model at age 30 and live up to age 85 with the estimated nonseparable French and Jones (2011) preferences, estimated AR(1) ability process, three health states with estimated health transitions, health-dependent mortality, health- and age-dependent wage profiles, endogenous exit and reentry at the estimated reentry cost, and health-dependent fixed costs of labor market participation.<sup>11</sup>

### 6.1 Computation and Calibration

**Method:** We compute the optimum with a recursive dual formulation in the spirit of Farhi and Werning (2013), extended in four directions: (i) the productivity level  $\theta$  enters the state space unnormalized, which is necessary to accommodate the general AR(1) ability process ( $\rho < 1$ ) and nonseparable CRRA preferences; (ii) health is an observable state (good health, poor health, or disability) affecting wages, fixed time costs, and survival; (iii) participation is an extensive-margin choice with fixed time costs and a reentry cost; and (iv) nonparticipation is always reversible at the reentry cost, so retirement in our results is an outcome rather than an assumption. We work in the dual space  $(\lambda, \Gamma, \theta_-, h_-, P_-, t)$ , where  $\lambda$  and  $\Gamma$  are the multipliers on promised utility and marginal promised utility. The initial promised utility is set so that the optimum is self-financing with a zero expected cost of allocations. Section 6.6 performs sensitivity analysis on the budgetary cost. Appendix C.5 details the algorithm and its implementation.

**Calibration:** Table 2 collects the parameters of the Mirrlees economy. The preference parameters, time endowment, fixed costs of participation by health, and reentry cost are

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<sup>11</sup>Our baseline quantitative specification with French and Jones (2011) preferences does not satisfy the differentiability and separability conditions of Assumptions 1 and 2. However, our theoretical results obtained under tractable assumptions, including at the separable limit of French and Jones (2011) preferences, are useful for interpreting our quantitative results.



the second-step estimates. The ability process, health–age wage profiles, health transition matrices, mortality rates by health, and initial distribution are the first-step estimates. The interest rate follows French (2005).

Table 2: Calibration of the Mirrlees economy

| Parameter                | Definition                    | Value     | Source                |
|--------------------------|-------------------------------|-----------|-----------------------|
| $\gamma$                 | Consumption weight            | 0.5354    | Table 13              |
| $\nu$                    | CRRA for flow utility         | 1.7029    | Table 13              |
| $\beta$                  | Discount factor               | 0.9797    | Table 13              |
| $L$                      | Time endowment                | 4490.5    | Table 13              |
| $r$                      | Interest rate                 | 4%        | French (2005)         |
| $\rho$                   | Ability persistence           | 0.981     | PSID (Appendix C.1.2) |
| $\sigma_\epsilon^2$      | Innovation variance           | 0.016     | PSID (Appendix C.1.2) |
| $W(h, t)$                | Wage profiles                 | Figure 12 | PSID                  |
| $\pi_{t+1}(h_{t+1} h_t)$ | Health transitions            | Figure 12 | PSID                  |
| $s_{t+1}(h_t)$           | Survival probabilities        | Figure 12 | PSID                  |
| $\phi_n(0)$              | Participation cost, healthy   | 791.07    | Table 13              |
| $\phi_n(1, 2)$           | Participation cost, unhealthy | 867.54    | Table 13              |
| $\phi_{RE}$              | Reentry cost                  | 3311.5    | Table 13              |
| $(h_1, P_0)$             | Initial conditions            | Table 11  | PSID                  |

Notes: Calibration values. Data profiles are estimated from the Panel Study of Income Dynamics (PSID); the panel of unhealthy individuals pools workers in poor health and workers with a disability using the age-30 population shares (4.17% and 1.09%).

## 6.2 Labor Force Participation and Retirement

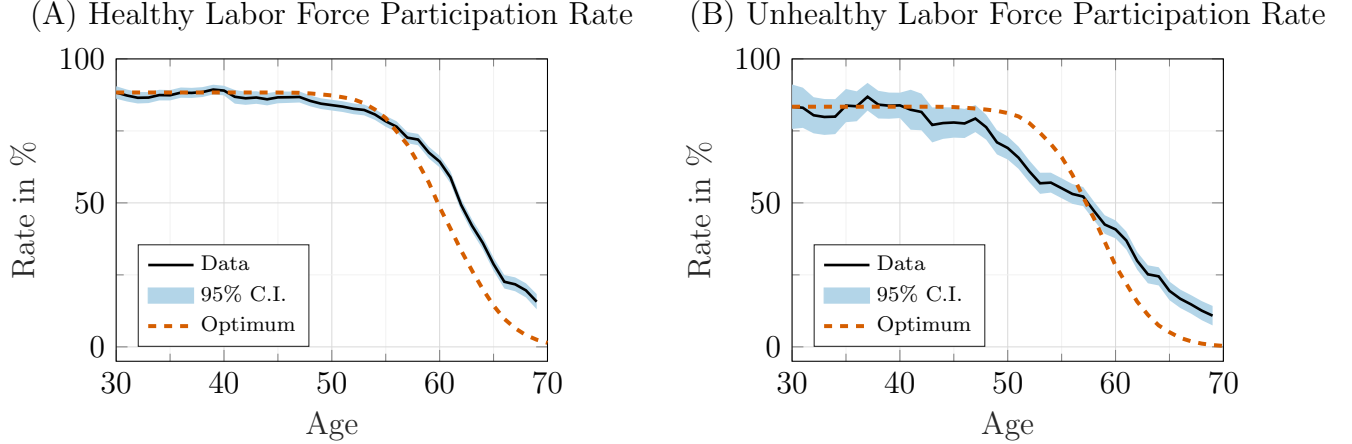
Figure 2 plots labor force participation by age in the optimal allocation against the profiles estimated from the data, for healthy individuals (panel A) and unhealthy individuals (panel B), where the latter group pools workers in poor health and those with a disability.<sup>12</sup> The model profile in each panel is normalized to the initial level in the data for that health group at age 30.<sup>13</sup> Healthy individuals optimally exit the labor force later than unhealthy ones, and the gradient of labor force exit by health status in the data mirrors that implied by our model. The participation profile of unhealthy workers begins its decline at ages several years younger than does the participation profile of healthy workers in both the optimum and the data. Where model and data differ is in how the exits are spread. In the optimum,

<sup>12</sup>Throughout the rest of the paper, we use the labels *unhealthy/in bad health* to refer to the pooled group of both workers in poor health and workers with a disability and *healthy/in good health* to refer to workers without health problems.

<sup>13</sup>The planner’s problem has essentially full participation at young ages because it abstracts from the unemployment, schooling, and caregiving margins that hold measured participation below one in the data. The normalization absorbs this level difference and makes the timing of exit comparable between model and data.

participation profiles remain at or above their counterparts in the data through working ages into the early 50s and then fall more steeply. The profiles in the data decline more gradually and show a thin tail of participation at working ages into the 70s.

Figure 2: Labor Force Participation in the Optimum and the Data



*Notes:* Labor force participation rates by age for healthy (panel A) and unhealthy (panel B) individuals: the optimal allocation profiles against their counterparts in the data, with 95% confidence bands (shaded). Data profiles are estimated from the Panel Study of Income Dynamics (PSID); the panel of unhealthy individuals pools workers in poor health and workers with a disability using the age-30 population shares (4.17% and 1.09%). The model profile in each panel is normalized to the age-30 participation level for that health group in the data, so the panels compare the timing of exit rather than levels.

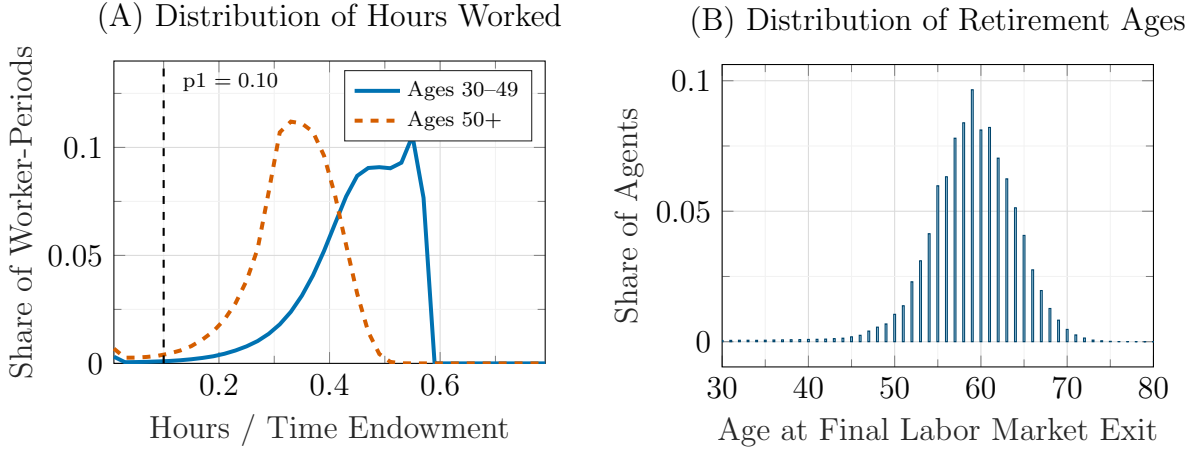
Virtually no one works fewer than 10% of their time endowment. Hours conditional on working average 41% of the annual time endowment for healthy workers at ages 35–55 (30% for unhealthy workers) and are bounded away from zero because short hours cannot make the fixed participation cost worth paying (Panel A of Figure 3). In addition, the distribution of hours worked shifts downward with worker age.

The work option remains open until age 84. Panel B of Figure 3 shows the distribution of ages at final labor force exit. The exits are spread over more than a decade, without a spike at any age. At the estimated reentry cost, reentry after exit occurs in well under 0.05% of out-of-work agent-periods, so exits are effectively absorbing, consistent with the small reentry rates in the data.<sup>14</sup>

Agents choose the fraction of life spent working, and the planner incentivizes the extensive margin of work with promises of greater retirement consumption. Figure 4 plots the mean consumption of new retirees by claiming age, normalized to 100% at the NRA of 65, against the statutory early retirement factor and delayed retirement credit in the US Social

<sup>14</sup>A counterfactual simulation of an economy without fixed costs and reentry cost ( $\phi_n(h) = \phi_{RE} = 0$ ) and with everything else held fixed confirms the mechanism: The mean retirement age rises from 59 to 76. With the fixed costs removed, workers hold small jobs. With the reentry cost also removed, exits stop being absorbing, and nearly half of age-70 nonworkers return to work (15% of all agents in the mid-70s). Appendix C.9 reports the simulation results for this economy.

Figure 3: Distribution of Optimal Hours Worked and Retirement Ages



Notes: Panel A: Hours among workers in the optimal allocation. Only 0.58% of worker-periods at ages 30–49 and 1.89% at ages 50 and above spend less than 10% of the time endowment on work. Panel B: Distribution of retirement (final exit) ages in the optimal allocation.

Security system.<sup>15</sup> The optimal new-retiree consumption is steeply increasing in the retirement age in our calibration and all the counterfactual economies we analyze. This result is particularly robust to several alternative choices in our modeling of preferences (Ndiaye 2018). The claiming-age dependence of retirement consumption is the model counterpart of actuarial adjustments. Policies that implement the constrained optimum would promise those who retire later higher retirement consumption, whether through higher retirement benefits, subsidized retirement savings, or government-provided health insurance.

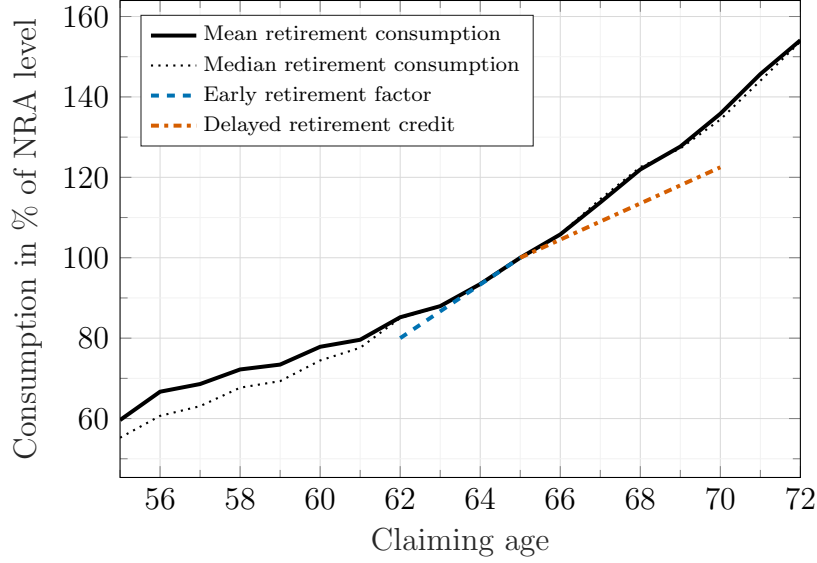
We now turn to the labor and savings distortions that arise from the lack of complete markets and full insurance in our quantitative model. These distortions can be summarized formally by the labor and savings wedges in equations (10) and (11) computed for each state and whose averages over our estimated model are reported next.

### 6.3 Labor Wedge

Figure 5 plots the optimal labor wedge by age and health among workers with positive hours. First, the wedge is positive in our baseline calibration. Because extensive-margin incentives to work are delivered through the retirement-consumption promises captured in Figure 4, participation responses need not push the optimal marginal distortions to turn negative as in Saez (2002), which illustrates the discussion following Proposition 4. Second, the wedge is ordered by health at every age, with the wedge for workers in good health above that for workers in poor health, which is in turn above the wedge for workers with

<sup>15</sup>These adjustments to social security benefits could be interpreted as the retirement consumption of a hand-to-mouth worker. In incomplete markets, limited consumption smoothing would imply that the gradient of retirement consumption with respect to claiming age would be close to the gradient of social security benefits with respect to claiming age.

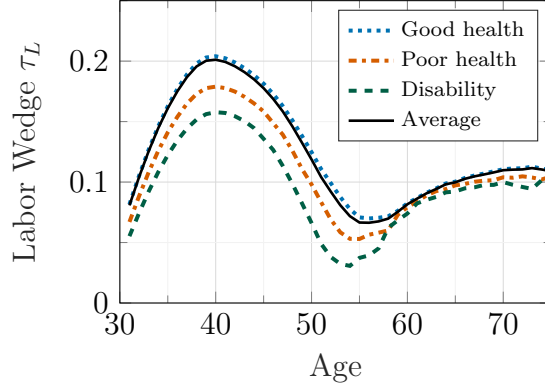
Figure 4: Retirement Consumption as a Function of Claiming Age



*Notes:* Mean consumption of new retirees by claiming (final exit) age, normalized to 100% at the normal retirement age (NRA) of 65, together with the Social Security benefit schedules as discounted by the statutory early retirement factor and as augmented by the delayed retirement credit.

disabilities. Worse health raises the endogenous elasticity and lowers the optimal distortion at the same state, the sign that Proposition 4 predicts. Third, the profile peaks near age 40 (at 0.20 for good health), declines to a trough below 0.08 at ages 56–58 as the exit margin becomes more important over time with disability risk, and rises mildly at the oldest working ages, where only a small, positively selected group remains.

Figure 5: Optimal Labor Wedge by Worker Age and Health Status



*Notes:* Optimal labor wedge among workers with positive hours, by age and health status. The *Average* line pools all workers. The optimal labor wedges are ordered by health status and are lower for unhealthy workers than for healthy workers.

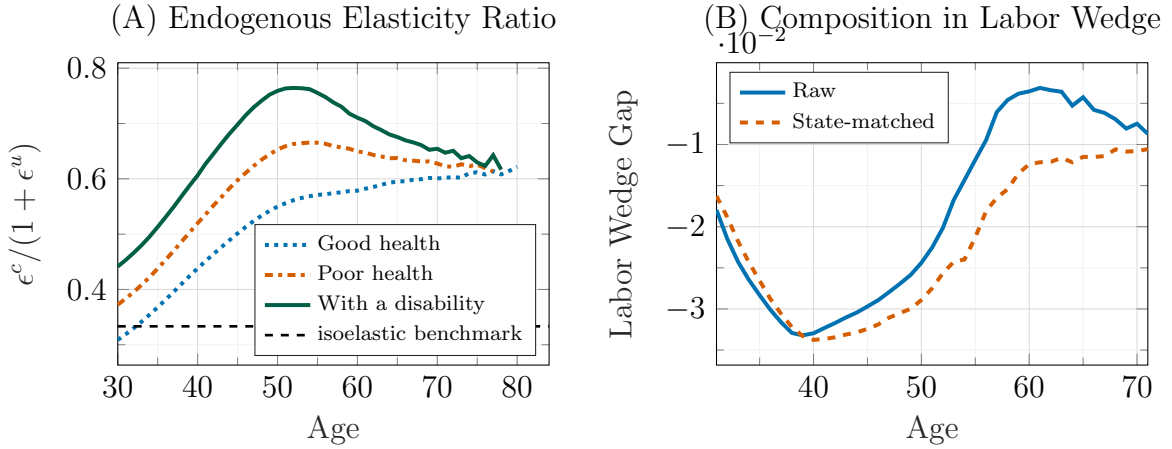
### 6.3.1 Endogenous Elasticity Ratio

The law of motion of the wedge (Lemma 7) contains two terms through which health can act: the elasticity ratio  $\varepsilon^c/(1 + \varepsilon^u)$ , introduced in Lemma 2, and the elasticity of wage to

ability,  $\varepsilon_{w\theta}(h)$ , isolated in Lemma 5. Lemma 5 states that wage differences across health enter the dynamics only through  $\varepsilon_{w\theta}$ . Under the wage process we estimate,  $\ln w = W(h, t) + \ln \theta$ , that elasticity equals one for every health state, so the direct wage channel is idle.

Although the direct channel is inactive, wages continue to affect the labor wedge through hours. Hours enter both the within-period optimality condition, which determines the level of the wedge, and the elasticity ratio  $\varepsilon^c/(1 + \varepsilon^u) = l/(l + n)$  of Lemma 3, which governs the wedge's persistence in the law of motion. The level effects are visible in the state-matched gaps of Table 3; here, we quantify the persistence channel by computing the elasticity ratio along the simulated optimum.

Figure 6: Endogenous Elasticity Ratio and Labor Wedge Gap



Notes: Panel A: The endogenous elasticity ratio  $\varepsilon^c/(1 + \varepsilon^u) = l/(l + n)$  along the simulated optimum (left) and the labor wedge dynamics it governs (right). The dashed horizontal line in the left panel is the constant ratio  $\varepsilon/(1 + \varepsilon) = 1/3$  that isoelastic-convex disutility with a Frisch elasticity of 0.5 would imply. Panel B: The raw versus state-matched labor wedge gaps by health status (where the latter reweights the planner's state values  $(\lambda, \Gamma, \theta)$  of the bad-health workers to match the good-health distribution). The matched gap is the same-state health effect. The difference between the raw and matched gaps reflects composition.

Under our preferences, the constant-savings elasticities have closed forms at the allocation (Lemma 3), with the leisure share of the time budget net of fixed costs given by  $l = L - n - \phi_n(h) - \phi_{RE} \mathbf{1}\{P_- = 0\}$ .<sup>16</sup> Panel A of Figure 6 plots the persistence multiplier  $\rho \times l/(l + n)$  alongside the wedge dynamics it governs. The ratio is determined endogenously: It rises with age as hours fall and is highest for workers with disabilities at every age, whereas under isoelastic-convex disutility, it would be the constant benchmark drawn in the figure. The rising persistence multiplier toward  $\rho$  is what caps and then reverses the wedge's upward drift in Figure 5. The mild rise of the cross-sectional mean at the oldest working ages in Figure 5 reflects the positive selection in the group that remains employed, as we see next.

<sup>16</sup>Since equilibrium reentry is negligible, the ratios reflect  $\phi_n(h)$ , while  $\phi_{RE}$  operates on the extensive margin by making exits effectively irreversible.

### 6.3.2 Composition and the Direct Effect of Health

The difference in labor wedges among workers of different health statuses is quantitatively meaningful, but the raw gap conflates two forces: the direct effect of health at a given state and a composition effect arising from the fact that the participating unhealthy workers at a given age have different histories of ability and utility promises than the participating healthy workers. To separate the two, we conduct a state-matched comparison. We reweight the planner’s state variables  $(\lambda, \Gamma, \theta)$  for unhealthy workers to match the distribution for healthy workers at each age.<sup>17</sup> Under this reweighting, any remaining difference between the labor wedge of the unhealthy and that of the healthy workers is the effect of health, with the underlying state held constant. Panel B of [Figure 6](#) plots the raw and state-matched wedge differences across ages, and [Table 3](#) reports both at four ages.

When we account for composition, the difference in labor wedge between healthy and unhealthy workers grows. At age 65, for example, the raw gap between the wedges of workers in bad and those in good health is  $-0.004$ , whereas the state-matched gap is  $-0.012$ . Unhealthy workers still employed at 65 are of higher income and concentrated in high-wedge states, which makes the raw gap smaller than the state-matched gap.

Table 3: Raw vs. State-Matched Labor Wedge Gaps (Wedge for Workers in Bad Health Minus Wedge for Workers in Good Health)

|                               | 45       | 55       | 60       | 65       |
|-------------------------------|----------|----------|----------|----------|
| Raw gap                       | $-0.030$ | $-0.012$ | $-0.004$ | $-0.004$ |
| State-matched gap             | $-0.032$ | $-0.021$ | $-0.012$ | $-0.012$ |
| Difference due to composition | $-0.003$ | $-0.009$ | $-0.009$ | $-0.007$ |

*Notes:* Raw = mean wedge of workers in bad health minus mean wedge of workers in good health. State-matched = the same difference after the bad-health group is given the good-health group’s mix over  $(\lambda, \Gamma, \theta)$  cells at that age; because the two groups then differ only in health, this row is the health effect at equal states.

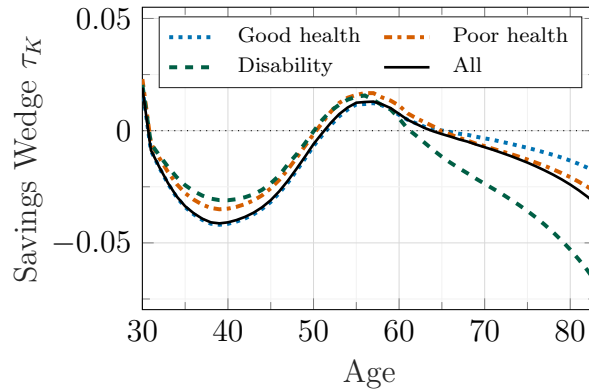
## 6.4 Savings Wedges

[Figure 7](#) reports the mean optimal savings wedge by worker age and health. The savings wedge is negative over much of the life cycle, although its mean is positive at the initial age and again over a window near the retirement margin. Health and mortality change the ordering of the wedge profiles with age. At older ages, the effect of mortality on the wedges becomes more important. The profile for workers with disabilities falls below the profile for healthy workers at about age 59, and the profile for workers in poor health does so around age 65. Groups with higher mortality therefore receive larger savings subsidies sooner.

<sup>17</sup>Lagged participation is not among the matching variables because reentry is negligible, so essentially all workers have  $P_- = 1$ .

The means obscure heterogeneity across histories. A right tail of states with large information rents generates positive savings taxation, while the median agent faces a savings subsidy at every age (cf. Figure 20 of Appendix C.8). Statutory schedules based only on age and health capture little of this state-by-state variation. The relevant policy implication is therefore not that all assets be subsidized unconditionally. The optimum prescribes preferential treatment of retirement savings, such as 401(k) and IRA accounts, that is increasing with age and poorer health, with retention of clawbacks (potentially means-tested) for some histories. A savings tax conditioned on current earnings captures nearly two-thirds of the variation in the simulated optimum (cf. Appendix C.8).<sup>18</sup> Earnings therefore provide a practical statistic for targeting savings incentives even when the full history-dependent mechanism cannot be implemented.

Figure 7: Optimal Mean Savings Wedge by Age and Health Status



Notes: Mean of optimal savings wedge among all surviving agents, by age and health status. The optimal savings wedges are negative in most of workers' lifetime.

## 6.5 Role of Nonconvexities and Ability Persistence

We highlighted the role of nonconvexities in driving observed retirement behavior. Without the nonconvexities, we expect the productivity loss and mortality channels to still be operating. We remove fixed time costs of participation and reentry costs, holding every other parameter at the headline calibration, and summarize the results in the second column of Table 4, with additional profiles discussed in Appendix C.9.

Setting the fixed costs and reentry cost to zero raises participation far above the headline. Labor force participation at age 60 rises from 0.549 to 0.989 for the healthy and from 0.338

<sup>18</sup>We compare optimal wedges with the distortions that the best tax function in a few age and income variables produces, with no functional form imposed on taxes. This exercise compares allocations rather than their welfare effects. Comparing the welfare effects of simpler policies to those of optimal history-dependent allocations would be beyond the scope of this paper and would require an alternative analysis focusing more on the status quo economy than the dynamic Mirrlees economy.



to 0.961 for workers in poor health or with a disability. With no fixed cost to cover and no reentry costs, low-productivity workers work a few hours instead of exiting, and retirement loses its absorbing character. The labor wedges, which peak at a value similar to the headline (0.201 against 0.201), keep their hump shape and ordering by health status but decline less steeply than in the baseline. The reason is that the productivity channel in levels still operates. Wages still fall with old age and bad health. By the substitution effect, hours fall as a share of disposable leisure time, the elasticity ratio  $l/(l+n)$  rises, and the labor wedge falls. Saving subsidies are still optimal because of the mortality risk.

The baseline estimates ability persistence at  $\rho = 0.981$ . Much of the dynamic Mirrlees literature, like [Farhi and Werning \(2013\)](#) and [Shourideh and Troshkin \(2015\)](#), instead assumes permanent shocks ( $\rho = 1$ ). Although the numerical difference appears small, the shocks compound and may affect policy optimality.<sup>19</sup> We therefore re-solve the economy with a random-walk ability process while holding the remaining parameters fixed. The third column of [Table 4](#) reports the main comparison, with additional exhibits in [Appendix C.10](#). The random-walk economy produces substantially more labor force participation. Participation among healthy workers at age 60 rises from 0.549 to 0.904. For workers in poor health or with a disability, it rises from 0.338 to 0.767.

The mechanism operates through the upper tail of late-life productivity. Remaining employed at older ages is attractive primarily for workers whose productivity remains high. With permanent shocks, favorable productivity realizations persist, keeping this upper tail thick. Mean reversion at the estimated  $\rho$  erodes the value of those realizations and makes earlier exit optimal.

Mean reversion also raises the labor wedge: Its peak is 0.201 in the headline against 0.171 under the random walk. The savings wedge is, however, larger in the random-walk economy (its peak is +0.036 against the headline’s +0.013), while the mean and median savings subsidies at older ages are of similar magnitude in the two economies.

## 6.6 Legacy Debt, Intergenerational Transfers, and Bequests

In the baseline, we used promised utility so that the cohort’s allocation would be self-financing. Actual cohorts face legacy obligations. Social Security’s legacy financing obliges current workers to fund benefits already paid to earlier generations, with a 75-year actuarial deficit of 3.50% of taxable payroll ([Board of Trustees, OASDI 2024](#)), and private bequests move resources across cohorts on a comparable scale, with net flows on the order of 1.5–3% of lifetime earnings ([Feiveson and Sabelhaus 2018](#); [Alvaredo et al. 2017](#)). We therefore ask

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<sup>19</sup>The two economies center the yearly ability shock differently: With  $\rho = 1$ , the mean shock is one, as in the random-walk setup of [Farhi and Werning \(2013\)](#), while our baseline with  $\rho < 1$  holds the median shock at one, as in the estimation. The persistence change is therefore combined with these changes in moments.

Table 4: Role of Fixed Costs and Ability Persistence

|                                       | Headline | No fixed costs | Random walk |
|---------------------------------------|----------|----------------|-------------|
| LFPR at 60 (healthy)                  | 0.549    | 0.989          | 0.904       |
| LFPR at 60 (unhealthy)                | 0.338    | 0.961          | 0.767       |
| Peak mean labor wedge                 | 0.201    | 0.201          | 0.171       |
| Peak mean savings wedge, ages 50–65   | +0.013   | +0.018         | +0.036      |
| Mean savings wedge at 65 (disabled)   | −0.012   | −0.006         | −0.011      |
| Median savings wedge at 65 (disabled) | −0.019   | −0.021         | −0.018      |

*Notes:* Headline is the estimated Mirrlees economy. *No fixed costs* sets  $\phi_n(h) = \phi_{RE} = 0$ , holding every other parameter at the headline values. *Random walk* sets  $\rho = 1$ , holding every other parameter fixed. LFPR at 60 is the share working at age 60 among survivors in the stated health group. Peak mean labor wedge is the maximum over ages of the mean among workers with positive hours. Peak mean savings wedge is the maximum no-annuity wedge over ages 50–65 only. The panel of unhealthy individuals pools those in poor health and disability using their age-30 population shares (4.17% and 1.09%).

how legacy obligations and intergenerational transfers affect the results.

To answer this question, we recompute the optimum imposing that the planner’s net resource cost  $K$  must equal a fixed share of the present value of the cohort’s simulated output,  $K/PV(y) \in \{-1, -2, -3.5, -5, -10, -20\}\%$ , with output discounted at the planner’s interest rate ( $r = 4\%$ ). The  $-3.5\%$  economy is the Social Security Trustees’ solvency benchmark. [Table 5](#) reports the results, and [Figure 8](#) plots the change in participation by age and the budget-side objects across the grid.

Table 5: Optimum Under Legacy Budgetary Requirements

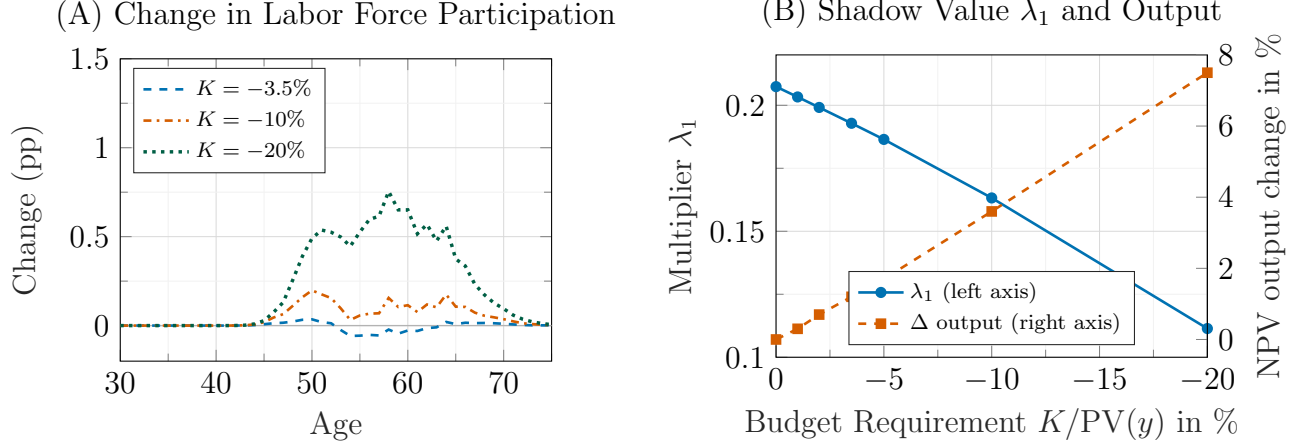
| $K/PV(y)$    | $\lambda_1$ | $\Delta$ output (%) | $\Delta$ LFPR at 60 (pp) | $\Delta$ hours at 40 (%) | $\Delta$ share exit $\geq 62$ (pp) | $\Delta$ share exit $\geq 65$ (pp) |
|--------------|-------------|---------------------|--------------------------|--------------------------|------------------------------------|------------------------------------|
| 0 (baseline) | 0.2050      | —                   | [0.487]                  | [0.4498]                 | [0.303]                            | [0.119]                            |
| −1%          | 0.2008      | +0.3                | −0.02                    | +0.4                     | −0.01                              | −0.00                              |
| −2%          | 0.1966      | +0.7                | −0.03                    | +0.9                     | −0.02                              | +0.00                              |
| −3.5%        | 0.1902      | +1.2                | −0.03                    | +1.6                     | −0.01                              | +0.01                              |
| −5%          | 0.1835      | +1.7                | −0.02                    | +2.3                     | −0.00                              | +0.02                              |
| −10%         | 0.1599      | +3.6                | +0.13                    | +4.7                     | +0.13                              | +0.11                              |
| −20%         | 0.1051      | +7.5                | +0.74                    | +9.2                     | +0.61                              | +0.40                              |

*Notes:* Each row is the optimum with the initial promise set so that the planner’s expected discounted net resource cost divided by the net present value of the cohort’s simulated output is equal to the negative values that the cohort finances.  $\lambda_1$  is the initial multiplier on promised utility, in levels. Bracketed entries in the baseline row are levels; the remaining rows report changes relative to the baseline. The  $-3.5\%$  economy is the Social Security Trustees’ solvency benchmark.

The retirement margin barely responds in our counterfactuals. At the 3.5% benchmark, output rises by 1.2%, labor-force participation at age 60 changes by only 0.03 percentage points, and the shares retiring at age 62 or later and age 65 or later each change by only 0.01 percentage points. Even at a 20% requirement, labor-force participation at age 60 rises by only 0.74 percentage points. The share retiring at age 62 or later increases from 30.3 to

about 30.9%, and the share retiring at age 65 or later rises by roughly 0.4 percentage points.

Figure 8: Participation Responses and the Planner's Budget Under Budgetary Requirements



Notes: Panel A plots the change in labor force participation at each worker age, in percentage points, relative to participation in the self-financing baseline ( $K = 0$ ), for  $K/PV(y) \in \{-3.5, -10, -20\}\%$ . Labor force participation barely increases as a response to a higher budget requirement. Panel B plots the initial multiplier  $\lambda_1$  and the output change against the budget requirement. Output increases and welfare decreases with a larger budget requirement.

Most of the adjustment instead occurs through consumption and intensive-margin labor supply. Consumption absorbs roughly two-thirds of the resource requirement. Relative to the near-self-financing 1% economy, median consumption in the first retirement year for healthy retirees falls by about 0.5% under a 5% requirement and by about 3% under the 20% stress test. Mean hours at age 40 rise by 1.6% at the 3.5% benchmark and by 9.2% at the 20% requirement.

This difference between the intensive- and extensive-margin responses follows from the driver of retirement in the model. The participation threshold is determined largely by the fixed cost of work. A pure resource requirement generates an income/wealth effect. Workers respond by consuming less and, conditional on employment, working more. The same wealth effect moves the extensive margin of participation threshold much less. Panel A of Figure 8 shows that even under the 20% requirement, the increase in labor force participation in old age remains below two percentage points.

Panel B reports how welfare and output change for different budget requirements. The multiplier on initial promised utility,  $\lambda_1$ , falls monotonically from 0.2050 in the self-financing allocation to 0.1051 under the 20% requirement. By Proposition 8, the marginal utility cost of transferring one additional present-value dollar is  $1/\lambda_1$ . The declining multiplier therefore implies that the utility cost of an additional legacy dollar rises as the resource requirement becomes larger.

Output rises at the same time, by 1.2% at the 3.5% benchmark and 7.5% in the 20% stress test. Additional work finances part of the transfer, but it does so mainly through hours rather than through a large postponement of retirement.

## 6.7 Status Quo Wedges and Directions for Reform

We finally use the model estimated under current US policies to compute the labor and savings wedges generated by the status quo and propose directions for reform of retirement policies. The status quo economy includes taxes, Social Security benefits, DI, medical expenditures, and incomplete insurance markets.

Figure 9 reports the average wedges by worker age and health status.



*Notes:* Panel A shows the labor wedge in the status quo US economy by worker age and health status. Panel B plots the savings wedge by worker age and health status. The black solid lines represent people on average, and the blue/red/green nonsolid lines represent people in good health/in poor health/with a disability.

**Labor Wedge:** The labor wedge (panel A) is largest for individuals in good health, lower for those in poor health, and lowest for those with a disability for most of the life cycle. This ranking by health status aligns with the properties of the optimal distortions. The introduction of DI at age 50, however, generates a much larger labor wedge for individuals with a disability. This distortion diminishes gradually and nearly disappears by age 65, with the relative ranking of the wedges across health states unchanged.

For ages from 30 to 60, the status quo labor wedge shows a decreasing time trend, while the optimal wedge of Figure 5 is hump-shaped with a much smoother profile: The optimum concentrates distortions in prime age, when elasticities are low, and relieves them near the exit margin, when elasticities are high. The status quo wedge shows the opposite behavior around the disability and claiming ages.

**Savings Wedge:** The savings wedges (panel B) for healthy and unhealthy individuals do not differ much. Individuals with a disability, however, and the newly retired face larger savings distortions. This is explained by the nonseparability of consumption and leisure: As individuals with a disability and retirees have more leisure, they gain more value from present consumption, which increases the savings distortions.

The savings distortions in the status quo economy (in rates akin to wealth taxes) exceed those at the optimum and, importantly, differ in that they do not subsidize savings on net as in [Figure 7](#). The larger status quo wedge reflects individuals' limited opportunities to smooth consumption because of borrowing constraints. Where the optimum calls for savings subsidies that are mostly increasing in age and poorer health, the status quo delivers savings distortions that are largest for people who are disabled and retirees.

**Directions for Reform:** The comparisons of optimal retirement consumption and statutory delayed retirement credits ([Figure 4](#)) and of the optimal and status quo wedges ([Figure 9](#)) point to three margins on which policy can move toward the constrained-efficient allocation.

First, retirement benefits should depend more strongly on the claiming age. In the optimum, consumption upon retirement rises more steeply with delayed exit than do the benefit schedules topped up with the delayed retirement credit and continues to rise beyond age 70. A steeper delayed-claiming schedule would strengthen the reward for remaining employed without requiring a uniformly higher retirement age.

Second, labor distortions should decline more smoothly with deteriorating health. The optimum does not produce a discrete disincentive to work at a particular disability age. The optimal labor wedge falls as the cost of continued work rises. Disability policy that phases eligibility or generosity more gradually with age and work capacity would better approximate this margin than an abrupt eligibility threshold.

Third, retirement-saving incentives should become more favorable as workers age and their health worsens. The optimum subsidizes saving once mortality is sufficiently high, and this point arrives earliest for disabled individuals. Preferential treatment through 401(k)- or IRA-type instruments can approximate this feature, particularly when eligibility or clawbacks depend on earnings. Such conditioning is important because the optimal savings wedge remains strongly history dependent.

Together, the quantitative results do not support a uniform increase in the NRA to resolve the solvency issues of US Social Security. Optimal exit ages are heterogeneous and depend strongly on health and productivity histories. The planner would instead combine a stronger reward for delayed retirement with lower labor distortions near the labor force exit margin and increasingly favorable savings incentives as mortality rises. This allocation

would accommodate early retirement among workers for whom continued work is costly while preserving incentives among those for whom working longer remains valuable.

## 7 Conclusion

In this paper, we have integrated endogenous retirement choices and health shocks into a dynamic life cycle income taxation model to study optimal retirement and tax policy. Our theoretical analysis highlights that policy design must account for labor supply non-convexities, productivity loss, and mortality risk. Nonconvexities make the relevant labor supply elasticities endogenous, causing the standard rise in the optimal labor wedge with age to be capped and reversed and the wedge to fall among unhealthy workers. Health-driven wage losses affect the wedge dynamics directly only through the elasticity of wage to ability, whereas mortality risk makes the upper bound on the savings wedge negative and requires subsidies earliest for the workers in the worst health.

Quantitatively, our estimation shows that labor supply nonconvexities are the dominant driver of retirement behavior in the US. In the constrained optimum, retirement occurs earlier when workers are in worse health, and retirement consumption rises with the claiming age more steeply than the statutory delayed retirement credit. Labor wedges are hump-shaped and lower for unhealthy workers, while savings incentives become more favorable as workers age and for those in poorer health. The retirement profile barely changes when the cohort must fund legacy transfers, with adjustment occurring through consumption and hours. Relative to this benchmark, the status quo rewards delayed claiming too little, creates an abrupt work disincentive around the disability eligibility threshold, and imposes the largest savings distortions on the groups whom the optimum would subsidize.

Overall, our results suggest that flexibility—rather than rigid increases in the retirement age alone—characterizes optimal retirement policy. Future research should study general equilibrium feedback effects, private health information, and empirical responses to reforms of the claiming-age dependence of retirement benefits.

## References

- Abramson, Boaz, Job Boerma, and Aleh Tsyvinski**, “Macroeconomics of mental health,” Technical Report, National Bureau of Economic Research 2024. [6](#)
- Alvaredo, Facundo, Bertrand Garbinti, and Thomas Piketty**, “On the Share of Inheritance in Aggregate Wealth: Europe and the USA, 1900–2010,” *Economica*, 2017, *84* (334), 239–260. [35](#)
- Bairoliya, Neha and Kathleen McKiernan**, “Revisiting Retirement and Social Security Claiming Decisions,” *Unpublished working paper, Princeton University*, 2023. [7](#)
- Bastani, Spencer and Marius A. K. Ring**, “Optimal Delayed Taxation in the Presence of Financial Frictions,” 2024. Working Paper. [7](#)

- Bénabou, Roland**, “Tax and education policy in a heterogeneous-agent economy: What levels of redistribution maximize growth and efficiency?,” *Econometrica*, March 2002, 70 (2), 481–517. [86](#)
- Bergemann, Dirk and Philipp Strack**, “Dynamic revenue maximization: A continuous time approach,” *Journal of Economic Theory*, 2015, 159, 819–853. [64](#), [65](#)
- Blau, David M. and Donna B. Gilleskie**, “The Role of Retiree Health Insurance in the Employment Behavior of Older Men,” *International Economic Review*, May 2008, 49 (2), 475–514. [84](#)
- and **Ryan Goodstein**, “Can Social Security Explain Trends in Labor Force Participation of Older Men in the United States?,” *Journal of Human Resources*, 2010, 45 (2), 328–363. [6](#)
- Blundell, Richard, Eric French, and Gemma Tetlow**, “Retirement incentives and labor supply,” in “Handbook of the economics of population aging,” Vol. 1, Elsevier, 2016, pp. 457–566. [1](#), [6](#), [9](#), [26](#)
- , **Luigi Pistaferri, and Ian Preston**, “Consumption inequality and partial insurance,” *American Economic Review*, 2008, 98 (5), 1887–1921. [20](#)
- Board of Trustees, OASDI**, “The 2024 Annual Report of the Board of Trustees of the Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds,” Technical Report, Social Security Administration, Washington, DC 2024. [35](#)
- Boerma, Job, Aleh Tsyvinski, and Alexander P Zimin**, “Bunching and taxing multidimensional skills,” Technical Report, National Bureau of Economic Research 2022. [11](#)
- Borella, Margherita, Mariacristina De Nardi, and Fang Yang**, “The aggregate implications of gender and marriage,” *The Journal of the Economics of Ageing*, 2018, 11, 6–26. [82](#)
- , —, and —, “Are Marriage-Related Taxes and Social Security Benefits Holding Back Female Labor Supply?,” *The Review of Economic Studies*, 2023, 90 (1), 102–131. [7](#), [86](#), [89](#), [94](#)
- Braun, R. Anton, Karen A. Kopecky, and Tatyana Koreshkova**, “Old, Sick, Alone, and Poor: A Welfare Analysis of Old-Age Social Insurance Programmes,” *Review of Economic Studies*, 2017, 84 (2), 580–612. [86](#)
- Browning, Martin, Lars Peter Hansen, and James Heckman**, “Micro Data and General Equilibrium Models,” in John Taylor and Michael Woodford, eds., *Handbook of Macroeconomics*, Vol. 1A, Amsterdam: North Holland, 1999. [4](#)
- Cagetti, Marco**, “Wealth accumulation over the life cycle and precautionary savings,” *Journal of Business and Economic Statistics*, 2003, 21 (3), 339–353. [87](#)
- Choné, Philippe and Guy Laroque**, “Income Tax and Retirement Schemes,” Working paper, Center for Research in Economics and Statistics 2014. [7](#)
- Coile, Courtney C.**, “Working Longer in the United States: Trends and Explanations,” in Courtney C. Coile, Kevin Milligan, and David A. Wise, eds., *Social Security Programs and Retirement around the World: Working Longer*, University of Chicago Press, 2019. [6](#)
- Cremer, Helmuth, Jean-Marie Lozachmeur, and Pierre Pestieau**, “Social security, retirement age and optimal income taxation,” *Journal of Public Economics*, 2004, 88 (11), 2259–2281. [7](#)
- De Nardi, Mariacristina, Eric French, and John Bailey Jones**, “Why do the Elderly Save? The role of Medical Expenses,” *Journal of Political Economy*, February 2010, 118 (1), 39–75. [3](#), [7](#), [86](#), [90](#)
- , **Svetlana Pashchenko, and Ponpoje Porapakkarm**, “The Lifetime Costs of Bad Health,” *Review of Economic Studies*, 2024. [3](#), [79](#)



- Deaton, Angus and Christina Paxson**, “Intertemporal choice and inequality,” *Journal of political economy*, 1994, 102 (3), 437–467. [20](#), [22](#)
- Diamond, Peter Arthur and James A Mirrlees**, “A model of social insurance with variable retirement,” *Journal of Public Economics*, 1978, 10 (3), 295–336. [7](#)
- Dixit, Avinash**, “Entry and exit decisions under uncertainty,” *Journal of political Economy*, 1989, 97 (3), 620–638. [5](#)
- , “Art of Smooth Pasting,” *Fundamentals of Pure and Applied Economics*, 1993, 55. [63](#)
- Erosa, Andrés, Luisa Fuster, and Gueorgui Kambourov**, “Towards a Micro-Founded Theory of Aggregate Labour Supply,” *The Review of Economic Studies*, 2016, 83 (3), 1001–1039. [8](#)
- Fan, Xiaodong, Ananth Seshadri, and Christopher Taber**, “Estimation of a life-cycle model with human capital, labor supply, and retirement,” *Journal of Political Economy*, 2024, 132 (1), 48–95. [7](#)
- Farhi, Emmanuel and Iván Werning**, “Insurance and taxation over the life cycle,” *The Review of Economic Studies*, 2013, 80 (2), 596–635. [1](#), [2](#), [3](#), [6](#), [10](#), [11](#), [14](#), [16](#), [17](#), [19](#), [21](#), [22](#), [27](#), [35](#), [50](#), [69](#), [90](#), [99](#), [100](#)
- Feiveson, Laura and John Sabelhaus**, “How Does Intergenerational Wealth Transmission Affect Wealth Concentration?,” FEDS Notes, Board of Governors of the Federal Reserve System, Washington, DC 2018. [35](#)
- French, Eric**, “The effects of health, wealth, and wages on labour supply and retirement behaviour,” *The Review of Economic Studies*, 2005, 72 (2), 395–427. [1](#), [3](#), [7](#), [8](#), [28](#), [80](#), [82](#), [84](#), [87](#), [89](#), [90](#), [92](#), [94](#), [97](#)
- and **Jae Song**, “The Effect of Disability Insurance Receipt on Labor Supply,” *American Economic Journal: Economic Policy*, 2014, 6 (2). [90](#)
- and **John Bailey Jones**, “The effects of health insurance and self-insurance on retirement behavior,” *Econometrica*, 2011, 79 (3), 693–732. [1](#), [2](#), [3](#), [7](#), [8](#), [9](#), [17](#), [18](#), [21](#), [24](#), [25](#), [27](#), [49](#), [53](#), [80](#), [84](#), [87](#), [89](#), [93](#)
- Golosov, Mikhail, Aleh Tsyvinski, and Iván Werning**, “New Dynamic Public Finance: A User’s Guide,” *NBER Macroeconomics Annual*, 2006, 21, 317–363. [6](#)
- and —, “Designing optimal disability insurance: A case for asset testing,” *Journal of Political Economy*, 2006, 114 (2), 257–279. [7](#), [70](#), [105](#)
- and —, “Policy implications of dynamic public finance,” *Annual Review of Economics*, 2015, 7 (1), 147–171. [6](#)
- and **Ilia Krasikov**, “The Optimal Taxation of Couples,” Working Paper, National Bureau of Economic Research 2023. [11](#)
- , **Maxim Troshkin**, and **Aleh Tsyvinski**, “Redistribution and social insurance,” *The American Economic Review*, 2016, 106 (2), 359–386. [1](#), [3](#), [5](#), [6](#), [21](#), [90](#)
- , **Narayana Kocherlakota**, and **Aleh Tsyvinski**, “Optimal indirect and capital taxation,” *The Review of Economic Studies*, 2003, 70 (3), 569–587. [6](#)
- Gomes, Renato, Jean-Marie Lozachmeur, and Alessandro Pavan**, “Differential taxation and occupational choice,” *The Review of Economic Studies*, 2017, 85 (1), 511–557. [7](#)
- Gourinchas, Pierre-Olivier and Jonathan A. Parker**, “Consumption Over the Life Cycle,” *Econometrica*, January 2002, 70 (1), 47–89. [87](#)

- Graves, Sebastian, Victoria Gregory, Lars Ljungqvist, and Thomas J. Sargent**, “Time Averaging Meets Labor Supplies of Heckman, Lochner, and Taber,” Discussion Paper 18345, CEPR 2023. [4](#)
- Greenwood, Jeremy, Zvi Hercowitz, and Gregory W Huffman**, “Investment, capacity utilization, and the real business cycle,” *The American Economic Review*, 1988, pp. 402–417. [63](#)
- Grochulski, Borys and Narayana Kocherlakota**, “Nonseparable preferences and optimal social security systems,” *Journal of Economic Theory*, 2010, *145* (6), 2055–2077. [24](#), [25](#)
- **and Yuzhe Zhang**, “Optimal Contracts with Reflection,” Working Paper, Federal Reserve Bank of Richmond 2017. [24](#)
- Gruber, Jonathan and David A. Wise, eds**, *Social Security Programs and Retirement Around the World: Micro-Estimation* International Social Security, Chicago: University of Chicago Press, 2004. [6](#)
- **and David Wise**, “Social security and retirement: An international comparison,” *The American Economic Review*, 1998, *88* (2), 158–163. [6](#)
- Guner, Nezih, Remzi Kaygusuz, and Gustavo Ventura**, “Taxation and Household Labour Supply,” *The Review of Economic Studies*, July 2012, *79* (3), 1113–1149. [80](#)
- Gustman, Alan L. and Thomas L. Steinmeier**, “Effects of pensions on savings: analysis with data from the health and retirement study,” *Carnegie-Rochester Conference Series on Public Policy*, June 1999, *50*, 271–324. [94](#)
- **, Olivia S. Mitchell, Andrew A. Samwick, and Thomas L. Steinmeier**, “Evaluating Pension Entitlements,” in Olivia S. Mitchell, P. Brett Hammond, and Anna M. Rappaport, eds., *Forecasting Retirement Needs and Retirement Wealth*, University of Pennsylvania Press, 2000. [94](#)
- Haller, Andreas**, “Welfare Effects of Pension Reforms,” 2022. Working Paper, Norwegian School of Economics. [6](#)
- Heathcote, Jonathan, Kjetil Storesletten, and Giovanni L. Violante**, “Optimal Tax Progressivity: An Analytical Framework,” *The Quarterly Journal of Economics*, 2017, *132* (4), 1693–1754. [86](#)
- Heckman, James J.**, “The Common Structure of Statistical Models of Truncation, Sample Selection and Limited Dependent Variables and a Simple Estimator for Such Models,” in Sanford V. Berg, ed., *Annals of Economic and Social Measurement*, Vol. 5, National Bureau of Economic Research, October 1976, pp. 479–492. [80](#)
- Holter, Hans A., Lars Ljungqvist, Thomas J. Sargent, and Serhiy Stepanchuk**, “Singles, couples, time-averaging, and taxation,” *Journal of Monetary Economics*, 2025, *150*, 103702. [4](#)
- Hosseini, Roozbeh and Ali Shourideh**, “Retirement financing: An optimal reform approach,” *Econometrica*, 2019, *87* (4), 1205–1265. [7](#)
- **, Karen A. Kopecky, and Kai Zhao**, “The Evolution of Health over the Life Cycle,” *Review of Economic Dynamics*, 2022. [6](#), [78](#)
- Hubbard, R. Glenn, Jonathan Skinner, and Stephen P. Zeldes**, “Precautionary Saving and Social Insurance,” *Journal of Political Economy*, April 1995, *103* (2), 360–399. [86](#)
- Huggett, Mark and Juan Carlos Parra**, “How Well Does the US Social Insurance System Provide Social Insurance?,” *Journal of Political Economy*, 2010, *118* (1), 76–112. [7](#)
- Imrohoroglu, Selahattin and Sagiri Kitao**, “Social Security Reforms: Benefit Claiming, Labor Force Participation, and Long-Run Sustainability,” *American Economic Journal: Macroeconomics*, July 2012, *4* (3), 96–127. [84](#)

- Ippolito, Richard A.**, *Pension Plans and Employee Performance*, University of Chicago Press, 1997. [94](#)
- Jacka, Saul D. and J. R. Lynn**, “Finite-horizon optimal stopping, obstacle problems and the shape of the continuation region,” *Stochastics and Stochastics Reports*, 1992, *39*, 25–42. [59](#), [62](#), [64](#)
- Jacquet, Laurence, Etienne Lehmann, and Bruno Van der Linden**, “Optimal redistributive taxation with both extensive and intensive responses,” *Journal of Economic Theory*, 2013, *148* (5), 1770–1805. [7](#)
- Jones, John Bailey and Yue Li**, “Social Security Reform with Heterogeneous Mortality,” *Review of Economic Dynamics*, April 2023, *48*, 320–334. [7](#), [22](#)
- Judd, Kenneth L.**, *Numerical Methods in Economics*, 1 ed., Vol. 1, The MIT Press, September 1998. [95](#)
- Kapička, Marek**, “Efficient allocations in dynamic private information economies with persistent shocks: A first-order approach,” *Review of Economic Studies*, 2013, *80* (3), 1027–1054. [12](#)
- Kaplow, Louis**, “Optimal Income Taxation,” *Journal of Economic Literature*, June 2024, *62* (2), 637–738. [6](#)
- Karabarbounis, Marios**, “A road map for efficiently taxing heterogeneous agents,” *American Economic Journal: Macroeconomics*, 2016, *8* (2), 182–214. [7](#)
- Kruse, Thomas and Philipp Strack**, “Optimal stopping with private information,” *Journal of Economic Theory*, 2015, *159*, 702–727. [59](#), [70](#), [73](#)
- Kydland, Finn E and Edward C Prescott**, “Time to build and aggregate fluctuations,” *Econometrica: Journal of the Econometric Society*, 1982, pp. 1345–1370. [4](#)
- Lazear, Edward P and Robert L Moore**, “Pensions and turnover,” in “Pensions in the US Economy,” University of Chicago Press, 1988, pp. 163–190. [60](#)
- Le Monde**, “Réforme des retraites : environ 20 000 amendements déposés à l’Assemblée, dont plus de la moitié par le groupe LFI,” *Le Monde*, February 2023. Accessed: 2025-05-10. [1](#)
- Leland, Hayne E**, “Corporate debt value, bond covenants, and optimal capital structure,” *The journal of finance*, 1994, *49* (4), 1213–1252. [5](#), [62](#)
- Li, Yue**, “Paradoxical effects of increasing the normal retirement age: A prospective evaluation,” *European Economic Review*, January 2018, *101*, 512–527. [90](#)
- Liao, J. G. and Arthur Berg**, “Sharpening Jensen’s Inequality,” *The American Statistician*, 2019, *73* (3), 278–281. [21](#), [55](#)
- Liebman, Jeffrey B and Erzo FP Luttmer**, “Would people behave differently if they better understood social security? Evidence from a field experiment,” *American Economic Journal: Economic Policy*, 2015, *7* (1), 275–299. [6](#)
- Lockwood, Lee M**, “Incidental bequests and the choice to self-insure late-life risks,” *American Economic Review*, 2018, *108* (9), 2513–2550. [90](#)
- Low, Hamish and Luigi Pistaferri**, “Disability insurance and the dynamics of the incentive insurance trade-off,” *American Economic Review*, 2015, *105* (10), 2986–3029. [78](#), [90](#), [93](#), [94](#)
- Lucas, Robert E.**, *Models of Business Cycles*, Vol. 26, Oxford: Blackwell, 1987. [4](#)
- Maestas, Nicole and Julie Zissimopoulos**, “How Longer Work Lives Ease the Crunch of Population Aging,” *Journal of Economic Perspectives*, 2010, *24* (1). [6](#)

- , **Kathleen J. Mullen**, and **Alexander Strand**, “Does Disability Insurance Receipt Discourage Work? Using Examiner Assignment to Estimate Causal Effects of SSDI Receipt,” *American Economic Review*, 2013, 103 (5). [89](#)
- Makris, Miltiadis** and **Alessandro Pavan**, “Taxation under Learning by Doing,” *Journal of Political Economy*, 2021, 129 (6), 1878–1944. [6](#)
- McGarry, Kathleen**, “Health and Retirement: Do Changes in Health Affect Retirement Expectations?,” *Journal of Human Resources*, 2004, pp. 624–648. [6](#)
- Michau, Jean-Baptiste**, “Optimal redistribution: A life-cycle perspective,” *Journal of Public Economics*, 2014, 111, 1–16. [7](#), [60](#)
- Milgrom, Paul** and **Ilya Segal**, “Envelope theorems for arbitrary choice sets,” *Econometrica*, 2002, 70 (2), 583–601. [12](#), [24](#)
- Mirrlees, James A.**, “An exploration in the theory of optimum income taxation,” *The review of economic studies*, 1971, 38 (2), 175–208. [2](#), [8](#), [10](#), [12](#)
- Moser, Christian** and **Pedro Olea de Souza e Silva**, “Optimal paternalistic savings policies,” *Columbia Business School Research Paper*, 2019, (17–51). [7](#)
- Ndiaye, Abdoulaye**, “Essays on Optimal Taxation and Retirement Policies.” Ph.d. dissertation, Northwestern University, United States, Illinois 2018. [30](#)
- Nishiyama, Shinichi** and **Kent Smetters**, “Does social security privatization produce efficiency gains?,” *The Quarterly Journal of Economics*, 2007, 122 (4), 1677–1719. [7](#)
- Ozkan, Serdar**, “Preventive vs. curative medicine: A macroeconomic analysis of health care over the life cycle,” Technical Report, Federal Reserve Bank of St. Louis 2023. [84](#)
- Papageorge, Nicholas W.**, “Why medical innovation is valuable: Health, human capital, and the labor market,” *Quantitative Economics*, 2016, 7 (3), 671–725. [6](#)
- Pashchenko, Svetlana** and **Ponpoje Porapakkarm**, “Reducing Medical Spending of the Publicly Insured: The Case for a Cash-out Option,” *American Economic Journal: Economic Policy*, August 2019, 11 (3), 390–426. [78](#), [79](#)
- Pavan, Alessandro**, **Ilya Segal**, and **Juuso Toikka**, “Dynamic mechanism design: A myersonian approach,” *Econometrica*, 2014, 82 (2), 601–653. [12](#), [64](#), [65](#)
- Podor, Melinda** and **Timothy J Halliday**, “Health status and the allocation of time,” *Health Economics*, 2012, 21 (5), 514–527. [19](#)
- Prescott, Edward C.**, “Comment,” in Daron Acemoglu, Kenneth Rogoff, and Michael Woodford, eds., *NBER Macroeconomics Annual 2006*, Cambridge, Mass.: MIT Press, 2006. [4](#)
- , “Nobel Lecture: The Transformation of Macroeconomic Policy and Research,” *Journal of Political Economy*, 2006, 114 (2), 203–235. [4](#)
- Prescott, Edward C.**, **Richard Rogerson**, and **Johanna Wallenius**, “Lifetime aggregate labor supply with endogenous workweek length,” *Review of Economic Dynamics*, 2009, 12 (1), 23–36. [2](#), [9](#), [18](#), [53](#)
- Rogerson, Richard**, “Indivisible labor, lotteries and equilibrium,” *Journal of monetary Economics*, 1988, 21 (1), 3–16. [4](#)
- and **Johanna Wallenius**, “Nonconvexities, retirement, and the elasticity of labor supply,” *The American Economic Review*, 2013, 103 (4), 1445–1462. [1](#), [8](#), [18](#), [57](#)

- Rogerson, William P**, “Repeated moral hazard,” *Econometrica: Journal of the Econometric Society*, 1985, *53*, 69–76. [21](#), [55](#)
- Rothschild, Casey and Florian Scheuer**, “Redistributive taxation in the roy model,” *The Quarterly Journal of Economics*, 2013, *128* (2), 623–668. [7](#)
- Rust, John P**, “A dynamic programming model of retirement behavior,” in “The economics of aging,” University of Chicago Press, 1989, pp. 359–404. [60](#)
- Saez, Emmanuel**, “Using elasticities to derive optimal income tax rates,” *The review of economic studies*, 2001, *68* (1), 205–229. [16](#), [47](#), [49](#)
- , “Optimal income transfer programs: intensive versus extensive labor supply responses,” *The Quarterly Journal of Economics*, 2002, *117* (3), 1039–1073. [2](#), [7](#), [18](#), [30](#)
- **and Stefanie Stantcheva**, “Generalized social marginal welfare weights for optimal tax theory,” *The American Economic Review*, 2016, *106* (1), 24–45. [65](#)
- Seibold, Arthur**, “Reference Points for Retirement Behavior: Evidence from German Pension Discontinuities,” *American Economic Review*, 2021, *111* (4), 1126–1165. [6](#)
- Shourideh, Ali and Maxim Troshkin**, “Incentives and efficiency of pension systems,” Working Paper 2015. [7](#), [35](#), [60](#)
- Shoven, John B, Sita Nataraj Slavov, and David A Wise**, “Social Security Claiming Decisions: Survey Evidence,” Working Paper 23729, National Bureau of Economic Research August 2017. [6](#)
- Stantcheva, Stefanie**, “Optimal Taxation and Human Capital Policies over the Life Cycle,” *Journal of Political Economy*, 2017, *125* (6), 1931–1990. [1](#), [2](#), [6](#), [17](#), [20](#), [86](#), [90](#)
- , “Dynamic Taxation,” *Annual Review of Economics*, 2020, *12*, 801–831. [6](#)
- Stock, James H. and David A. Wise**, “Pensions, the Option Value of Work, and Retirement,” *Econometrica*, 1990, *58* (5), 1151–1180. [60](#)
- Storesletten, Kjetil, Christopher I Telmer, and Amir Yaron**, “Consumption and risk sharing over the life cycle,” *Journal of monetary Economics*, 2004, *51* (3), 609–633. [20](#)
- Werning, Iván**, “Nonlinear Capital Taxation,” 2011. Working Paper, MIT. [69](#)
- Yu, Zhixiu**, “Why are older men working more? The role of social security,” *Journal of Public Economics*, 2024, *231*. [7](#), [89](#)
- Zeldes, Stephen P.**, “Consumption and Liquidity Constraints: An Empirical Investigation,” *Journal of Political Economy*, April 1989, *97* (2), 305–346. [86](#)

## Part I

# Appendix

## A Theory Appendix

### A.1 Proof of Lemmas 2, 5, and 7

The Hamiltonian of the planner's problem is

$$\begin{aligned} & [C^t(y(\theta, h), \omega(\theta, h) - \beta v(\theta, h), \theta, h) - y(\theta, h)]s(h_-)\pi_h f^t(\theta|\theta_-) \\ & + \frac{1}{R}[K(v(\theta, h), \Delta(\theta, h), \theta, h, t+1)]s(h_-)\pi_h f^t(\theta|\theta_-) \\ & + \lambda[v - \omega(\theta, h)s(h_-)\pi_h f^t(\theta|\theta_-)] + \Gamma[\Delta - \omega(\theta, h)s(h_-)\pi_h \tilde{f}^t(\theta, \theta_-)] \\ & + p(\theta, h)[u_\theta^t(C^t(y(\theta, h), \omega(\theta, h) - \beta v(\theta, h), \theta, h), y(\theta, h), \theta, h) + \beta \Delta(\theta, h)] \end{aligned}$$

with boundary conditions for the costate  $p$

$$\lim_{\theta \rightarrow \bar{\theta}} p(\theta, h) = \lim_{\theta \rightarrow \underline{\theta}} p(\theta, h) = 0 \quad \forall h \in \mathcal{H}$$

Taking the first-order condition (FOC) with respect to  $y$ ,

$$\frac{\tau_L(\theta, h)}{1 - \tau_L(\theta, h)} = \frac{p(\theta, h)}{s(h_-)\pi_h f^t(\theta|\theta_-)} \frac{w_{\theta, t}}{w_t} u'(c(\theta, h)) \left( 1 + \frac{n(\theta, h)\psi_{nn}(n(\theta, h); h)}{\psi_n(n(\theta, h); h)} \right)$$

Define the uncompensated and compensated elasticities as in [Saez \(2001\)](#) at constant savings:

$$\varepsilon^u = \frac{\frac{\psi_n(n)}{n} + \frac{\psi_n(n)^2}{u'(c)^2} u''(c)}{\psi_{nn}(n) - \frac{\psi_n(n)^2}{u'(c)^2} u''(c)}, \quad \varepsilon^c = \frac{\frac{\psi_n(n)}{n}}{\psi_{nn}(n) - \frac{\psi_n(n)^2}{u'(c)^2} u''(c)}$$

With these elasticity definitions, we have

$$\frac{\tau_L(\theta, h)}{1 - \tau_L(\theta, h)} = \frac{p(\theta, h)}{s(h_-)\pi_h f^t(\theta|\theta_-)} \frac{\varepsilon_{w, \theta}}{\theta} u'(c(\theta, h)) \left( \frac{1 + \varepsilon_t^u(h)}{\varepsilon_t^c(h)} \right). \quad (18)$$

For the costate, the FOC with respect to  $\omega(\theta, h)$  yields:

$$\frac{dp(\theta, h)}{d\theta} = -H_\omega^*$$

where  $H^*$  is the Hamiltonian, that is,

$$\frac{dp(\theta, h)}{d\theta} = \left[ -\frac{1}{u'(c(\theta, h))} + \lambda + \Gamma \frac{\tilde{f}^t(\theta, \theta_-)}{f^t(\theta|\theta_-)} \right] s(h_-)\pi_h f^t(\theta|\theta_-)$$

Integrating over  $\theta$  and using both boundary conditions, we have for any  $h$

$$\lambda = \int_{\Theta} \frac{1}{u'(c(\theta, h))} f^t(\theta|\theta_-) d\theta \quad (19)$$

The FOCs for  $v(\theta, h)$  and  $\Delta(\theta, h)$  are, using the envelope theorem to replace  $K_v$  and  $K_\Delta$  with  $\lambda, \Gamma$ .

$$\frac{1}{u'(c(\theta, h))} = \frac{\lambda(\theta, h)}{\beta R} \quad (20)$$

and

$$-\frac{\Gamma(\theta, h)}{\beta R} = \frac{p(\theta, h)}{s(h_-)\pi_h f^t(\theta|\theta_-)}$$

Multiplying (18) with an arbitrary weighting function  $x(\theta)$ , and letting  $\mathcal{X}$  be the integral of  $x$ , we have

$$\begin{aligned} & \int \frac{\tau_L(\theta, h)}{1 - \tau_L(\theta, h)} s(h_-)\pi_h f^t(\theta|\theta_-) \frac{\theta}{\varepsilon_{w,\theta}} \frac{\varepsilon_t^c(h)}{1 + \varepsilon_t^u(h)} \frac{1}{u'(c(\theta, h))} x(\theta) d\theta = \int_{\Theta} p(\theta, h) x(\theta) d\theta \\ &= - \int \frac{dp(\theta, h)}{d\theta} \mathcal{X}(\theta) d\theta \\ &= \int \left[ \frac{1}{u'(c(\theta, h))} - \frac{\beta R}{u'_{t-1}} - \Gamma \frac{\tilde{f}^t(\theta, \theta_-)}{f^t(\theta|\theta_-)} \right] s(h_-)\pi_h f^t(\theta|\theta_-) \mathcal{X}(\theta) d\theta \\ &= \int \left[ \frac{1}{u'(c(\theta, h))} - \frac{\beta R}{u'_{t-1}} + \frac{\beta R p(\theta_-, h_-)}{s(h_-)\pi_h f^t(\theta_-|\theta_-)} \frac{\tilde{f}^t(\theta, \theta_-)}{f^t(\theta|\theta_-)} \right] s(h_-)\pi_h f^t(\theta|\theta_-) \mathcal{X}(\theta) d\theta \\ &= \int \left[ \frac{1}{u'} - \frac{\beta R}{u'_{t-1}} + \frac{\beta R \tau_L(\theta_-, h_-)}{1 - \tau_L(\theta_-, h_-)} \frac{\theta_-}{\varepsilon_{w,\theta_-}} \frac{1}{u'_{t-1}} \frac{\varepsilon_{t-1}^c(h_-)}{1 + \varepsilon_{t-1}^u(h_-)} \frac{\tilde{f}^t(\theta, \theta_-)}{f^t(\theta|\theta_-)} \right] s(h_-)\pi_h f^t(\theta|\theta_-) \mathcal{X}(\theta) d\theta \end{aligned}$$

so with  $\theta x(\theta) = 1$ ,  $\mathcal{X}(\theta) = \ln(\theta)$

$$\begin{aligned} & \int \frac{\tau_L}{1 - \tau_L} \frac{1}{\varepsilon_{w,\theta}} \frac{\varepsilon_t^c(h)}{1 + \varepsilon_t^u(h)} \frac{u'_{t-1}}{\beta R u'} f^t(\theta|\theta_-) d\theta \\ &= \int \left[ \frac{u'_{t-1}}{\beta R u'} - 1 + \frac{\tau_L(\theta_-, h_-)}{1 - \tau_L(\theta_-, h_-)} \frac{\theta_-}{\varepsilon_{w,\theta_-}} \frac{\varepsilon_{t-1}^c(h_-)}{1 + \varepsilon_{t-1}^u(h_-)} \frac{\tilde{f}^t(\theta, \theta_-)}{f^t(\theta|\theta_-)} \right] \ln(\theta) f^t(\theta|\theta_-) d\theta \\ &= \text{Cov} \left( \frac{u'_{t-1}}{\beta R u'}, \ln(\theta_t) \right) + \frac{\tau_L(\theta_-, h_-)}{1 - \tau_L(\theta_-, h_-)} \frac{\theta_-}{\varepsilon_{w,\theta_-}} \frac{\varepsilon_{t-1}^c(h_-)}{1 + \varepsilon_{t-1}^u(h_-)} \int_{\Theta} \tilde{f}^t(\theta, \theta_-) \ln(\theta) d\theta \\ &= \text{Cov} \left( \frac{u'_{t-1}}{\beta R u'}, \ln(\theta_t) \right) + \rho \frac{\tau_L(\theta_-, h_-)}{1 - \tau_L(\theta_-, h_-)} \frac{1}{\varepsilon_{w,\theta_-}} \frac{\varepsilon_{t-1}^c(h_-)}{1 + \varepsilon_{t-1}^u(h_-)} \end{aligned}$$

with AR(1) with persistence  $\rho$

$$\begin{aligned} & E_{t-1} \left[ \frac{\tau_{L,t}}{1 - \tau_{L,t}} \frac{1}{\varepsilon_{w,t}} \frac{\varepsilon_t^c(h)}{1 + \varepsilon_t^u(h)} \frac{1}{\beta R} \frac{u'(c_{t-1})}{u'(c_t)} | h \right] = \\ & \rho \frac{\tau_{L,t-1}}{1 - \tau_{L,t-1}} \frac{1}{\varepsilon_{w,\theta_-}} \frac{\varepsilon_{t-1}^c(h_-)}{1 + \varepsilon_{t-1}^u(h_-)} + \text{Cov} \left( \frac{1}{R\beta} \frac{u'(c_{t-1})}{u'(c_t)}, \ln(\theta_t) \right) \end{aligned} \quad (21)$$

This proves the general Lemma 7 with interactions. We can then focus on the nonconvexity



channel in Lemma 2, which corresponds to  $\varepsilon_{w,\theta_-} = \varepsilon_{w,\theta_t}$  and yields

$$E_{t-1} \left[ \frac{\tau_{L,t}}{1 - \tau_{L,t}} \frac{\varepsilon_t^c(h)}{1 + \varepsilon_t^u(h)} \frac{1}{\beta R} \frac{u'(c_{t-1})}{u'(c_t)} | h \right] = \\ \text{Cov} \left( \frac{1}{R\beta} \frac{u'(c_{t-1})}{u'(c_t)}, \ln(\theta_t) \right) + \rho \frac{\tau_{L,t-1}}{1 - \tau_{L,t-1}} \frac{\varepsilon_{t-1}^c(h_{-1})}{1 + \varepsilon_{t-1}^u(h_{-1})}$$

and the productive loss channel in Lemma 5, which corresponds to  $\frac{\varepsilon_t^c(h)}{1 + \varepsilon_t^u(h)} = \frac{\varepsilon_{t-1}^c(h_{-1})}{1 + \varepsilon_{t-1}^u(h_{-1})} = \frac{\varepsilon}{1 + \varepsilon}$ , where  $\varepsilon$  is the constant Frisch elasticity of the isoelastic disutility of labor  $\psi(n)$  in Economy 2. Equalizing the labor supply elasticities in (21) yields

$$E_{t-1} \left[ \frac{\tau_{L,t}}{1 - \tau_{L,t}} \frac{1}{\varepsilon_{w,\theta,t}(h)} \frac{1}{\beta R} \frac{u'(c_{t-1})}{u'(c_t)} | h \right] = \\ \rho \frac{\tau_{L,t-1}}{1 - \tau_{L,t-1}} \frac{1}{\varepsilon_{w,\theta-,t-1}(h_-)} + \left( 1 + \frac{1}{\varepsilon} \right) \text{Cov} \left( \frac{1}{R\beta} \frac{u'(c_{t-1})}{u'(c_t)}, \ln(\theta_t) \right)$$

## A.2 Proof of Lemma 3

**Part (i): Separable nonconvex preferences.** Consider the separable preferences (13),

$$u(c, n) = \tilde{u}(c) + \frac{l^{1-\gamma}}{1-\gamma}, \quad l = L - n - \phi,$$

where  $\phi$  collects the fixed costs. From the Saez (2001) constant-savings elasticity definitions used in the main text footnote,

$$\frac{\varepsilon^c}{1 + \varepsilon^u} = \frac{\frac{\psi_n(n)}{n}}{\frac{\psi_n(n)}{n} + \psi_{nn}(n)} = \frac{1}{1 + \frac{n\psi_{nn}(n)}{\psi_n(n)}},$$

because the terms involving  $u''$  cancel between the numerator and the denominator. With  $\psi(n) = -\frac{l^{1-\gamma}}{1-\gamma}$ , so that  $\psi_n(n) = l^{-\gamma}$  and  $\psi_{nn}(n) = \gamma l^{-\gamma-1}$ , we obtain

$$\frac{n\psi_{nn}(n)}{\psi_n(n)} = \frac{\gamma n}{l}, \quad \text{hence} \quad \frac{\varepsilon^c}{1 + \varepsilon^u} = \frac{l}{l + \gamma n}.$$

The ratio is strictly decreasing in  $n$ . As  $n \rightarrow 0$ , the ratio tends to 1. For the log-leisure case  $\gamma = 1$ , the ratio is  $\frac{l}{l+n}$ .

**Part (ii): French and Jones (2011) preferences.** Consider the preferences

$$u = \frac{(c^\gamma l^{1-\gamma})^{1-\nu}}{1-\nu},$$

and the agent's static labor supply problem at constant savings: the agent chooses hours  $n$  given the net wage  $w$  and unearned income  $I$ , with consumption  $c = wn + I$  and leisure  $l = \tilde{L} - n$ , where  $\tilde{L} = L - \phi$  is the time endowment net of the fixed costs. The tangency

condition of this static problem gives

$$\frac{c}{l} = \frac{\gamma}{1-\gamma} w,$$

so, with full income  $M = w\tilde{L} + I$ , the Marshallian demands are  $c = \gamma M$  and  $wl = (1-\gamma)M$ . Hence,

$$n = \gamma\tilde{L} - (1-\gamma)\frac{I}{w},$$

and the uncompensated elasticity is

$$\varepsilon^u = \frac{w}{n} \frac{dn}{dw} = (1-\gamma) \frac{I}{wn} = \gamma \frac{l}{n} - (1-\gamma).$$

For the compensated response, hold utility constant: utility is constant if and only if the composite  $c^\gamma l^{1-\gamma}$  is constant. Expenditure minimization with the same tangency condition gives

$$l = \bar{u} \left( \frac{1-\gamma}{\gamma w} \right)^\gamma,$$

so that  $\frac{d \ln l}{d \ln w} \Big|_u = -\gamma$ , hence  $\frac{dn}{dw} \Big|_u = \gamma \frac{l}{w}$  and  $\varepsilon^c = \gamma \frac{l}{n}$ . Therefore,

$$1 + \varepsilon^u = \gamma \frac{l+n}{n} \quad \text{and} \quad \frac{\varepsilon^c}{1 + \varepsilon^u} = \frac{l}{l+n},$$

independently of  $\gamma$  and  $\nu$ . □

*Remark:* for isoelastic convex disutility of labor  $\psi(n) = \frac{n^{1+1/\varepsilon}}{1+1/\varepsilon}$ , the same computation gives  $\frac{n\psi_{nn}(n)}{\psi_n(n)} = \frac{1}{\varepsilon}$  and the ratio collapses to the constant  $\frac{\varepsilon}{1+\varepsilon}$ , the [Farhi and Werning \(2013\)](#) benchmark.

### A.3 Proof of Proposition 4

The proof proceeds as follows. After showing the direction of movement of these elasticities, we obtain an *epsilon-of-a-room* for our inequality. For the innovation covariance term, we invoke the reduction in the savings wedge/dynamic incentives over time to fit it inside the room.

We start by observing that close to retirement, the intratemporal insurance term in covariance grows invariably small, so for any  $\zeta > 0$ , there exists  $t'$  so that for any  $t \geq t'$ ,  $\frac{1+\varepsilon_{t-1}^u(h_{-1})}{\varepsilon_{t-1}^c(h_{-1})} \text{Cov} \left( \frac{1}{R\beta} \frac{u'(c_{t-1})}{u'(c_t)}, \ln(\theta_t) \right) \leq \zeta$  and

$$E_{t-1} \left[ \frac{\tau_{L,t}}{1 - \tau_{L,t}} \frac{\varepsilon_t^c(h)}{1 + \varepsilon_t^u(h)} \frac{1}{\beta R} \frac{u'(c_{t-1})}{u'(c_t)} | h \right] \leq \rho \frac{\tau_{L,t-1}}{1 - \tau_{L,t-1}} (1 + \zeta) \frac{\varepsilon_{t-1}^c(h_{-1})}{1 + \varepsilon_{t-1}^u(h_{-1})}$$

We then characterize how the endogenous elasticities evolve with health.

**1. For general separable and nonconvex labor supply preferences defined by (13):** We calculate the compensated and uncompensated elasticities. First,  $u'(c) = c^{-\nu}$ ,  $u''(c) = -\nu c^{-\nu-1}$ , where  $u(c) = \frac{c^{1-\nu}}{1-\nu}$  is the CRRA utility function. Second,  $\psi_n(n) = (L-n)^{-\gamma}$  and  $\psi_{nn}(n) = \gamma(L-n)^{-\gamma-1}$ , where  $\psi(n) = -\frac{(L-n)^{1-\gamma}}{1-\gamma}$  represents the disutility of labor  $n$  and  $L$  is the available leisure time.

Substituting into the definitions of elasticities, we have:

$$\varepsilon^u = \frac{\frac{(L-n)^{-\gamma}}{n} - \nu(L-n)^{-2\gamma}c^{\nu-1}}{\gamma(L-n)^{-\gamma-1} + \nu(L-n)^{-2\gamma}c^{\nu-1}},$$

$$\varepsilon^c = \frac{\frac{(L-n)^{-\gamma}}{n}}{\gamma(L-n)^{-\gamma-1} + \nu(L-n)^{-2\gamma}c^{\nu-1}}.$$

The ratio of the elasticities  $\frac{\varepsilon^c}{1+\varepsilon^u}$  is calculated as follows, on the denominator  $1 + \varepsilon^u$ :

$$1 + \varepsilon^u = 1 + \frac{\frac{(L-n)^{-\gamma}}{n} - \nu(L-n)^{-2\gamma}c^{\nu-1}}{\gamma(L-n)^{-\gamma-1} + \nu(L-n)^{-2\gamma}c^{\nu-1}}$$

Combine terms over the common denominator:

$$1 + \varepsilon^u = \frac{\gamma(L-n)^{-\gamma-1} + \nu(L-n)^{-2\gamma}c^{\nu-1} + \frac{(L-n)^{-\gamma}}{n} - \nu(L-n)^{-2\gamma}c^{\nu-1}}{\gamma(L-n)^{-\gamma-1} + \nu(L-n)^{-2\gamma}c^{\nu-1}}$$

Simplify the numerator:

$$1 + \varepsilon^u = \frac{\gamma(L-n)^{-\gamma-1} + \frac{(L-n)^{-\gamma}}{n}}{\gamma(L-n)^{-\gamma-1} + \nu(L-n)^{-2\gamma}c^{\nu-1}}$$

The ratio becomes:

$$\frac{\varepsilon^c}{1 + \varepsilon^u} = \frac{\frac{(L-n)^{-\gamma}}{n}}{\gamma(L-n)^{-\gamma-1} + \frac{(L-n)^{-\gamma}}{n}}$$

Simplify further:

$$\frac{\varepsilon^c}{1 + \varepsilon^u} = \frac{1}{1 + \frac{\gamma n}{L-n}} = \frac{L-n}{L-n + \gamma n}$$

The available leisure time depends on health, so  $L = L(h)$ . We evaluate the elasticities at the endogenous hours worked, which depend on the whole history, but we emphasize its dependence on health and write  $n^* = n^*(h)$  for ease of notation. Then, writing  $q(h) \equiv \frac{n^*(h)}{L(h)-n^*(h)}$ ,

$$\frac{\varepsilon^c}{1 + \varepsilon^u} = \frac{1}{1 + \gamma q(h)}$$

Take the derivative with respect to  $h$ :

$$\frac{d}{dh} \left( \frac{\varepsilon^c}{1 + \varepsilon^u} \right) = - \frac{\gamma \frac{d}{dh} \left( \frac{n^*(h)}{L(h) - n^*(h)} \right)}{(1 + \gamma q(h))^2}$$

The derivative of  $\frac{n^*(h)}{L(h) - n^*(h)}$  using the quotient rule is:

$$\frac{d}{dh} \left( \frac{n^*(h)}{L(h) - n^*(h)} \right) = \frac{\frac{dn^*(h)}{dh}(L(h) - n^*(h)) - n^*(h) \left( \frac{dL(h)}{dh} - \frac{dn^*(h)}{dh} \right)}{(L(h) - n^*(h))^2}$$

If  $\frac{dL(h)}{dh} = 0$  (total available time does not depend on health), this simplifies to:

$$\frac{d}{dh} \left( \frac{n^*(h)}{L(h) - n^*(h)} \right) = \frac{\frac{dn^*(h)}{dh} L(h)}{(L(h) - n^*(h))^2}$$

Thus, the derivative becomes:

$$\frac{d}{dh} \left( \frac{\varepsilon^c}{1 + \varepsilon^u} \right) = - \frac{\gamma \frac{dn^*(h)}{dh} L(h)}{(1 + \gamma q(h))^2 \cdot (L(h) - n^*(h))^2}$$

The intuition is that if health worsens (poor  $h$ ),  $n^*(h)$  decreases ( $\frac{dn^*(h)}{dh} < 0$ ). The numerator  $\frac{dn^*(h)}{dh} L(h)$  is negative. The denominator is positive because  $\left( \frac{n^*(h)}{L(h) - n^*(h)} - \gamma \right)^2$  and  $(L(h) - n^*(h))^2$  are both positive. Thus,  $\frac{d}{dh} \left( \frac{\varepsilon^c}{1 + \varepsilon^u} \right) > 0$ , meaning the elasticity ratio increases as health worsens. As health worsens, labor supply  $n^*(h)$  decreases, making labor more sensitive to incentives relative to consumption. This raises the elasticity ratio  $\frac{\varepsilon^c}{1 + \varepsilon^u}$ .

If  $\frac{dL(h)}{dh} \neq 0$ ,

$$\frac{\varepsilon^c}{1 + \varepsilon^u} = \frac{1}{1 + \gamma \frac{n^*(h)/L(h)}{1 - n^*(h)/L(h)}} \quad (22)$$

and the function  $x \mapsto x/(1 - x)$  is increasing in  $[0, 1)$ . Hence, the elasticity ratio increases in poor health if the ratio of hours to available time  $n^*(h)/L(h)$  decreases in poor health. All the terms in the expectations are positive; applying the fact that if  $1 \leq X$  almost surely then  $E[Y] \leq E[XY]$ , we have

$$\begin{aligned} E_{t-1} \left[ \frac{\tau_{L,t}}{1 - \tau_{L,t}} \frac{1}{\beta R} \frac{u'(c_{t-1})}{u'(c_t)} | h \right] &\leq E_{t-1} \left[ \frac{\tau_{L,t}}{1 - \tau_{L,t}} \left( \frac{\varepsilon_{t-1}^c}{1 + \varepsilon_{t-1}^u} \right)^{-1} \frac{\varepsilon_t^c}{1 + \varepsilon_t^u} \frac{1}{\beta R} \frac{u'(c_{t-1})}{u'(c_t)} | h \right] \\ &\leq \rho \frac{\tau_{L,t-1}}{1 - \tau_{L,t-1}} (1 + \zeta) \end{aligned}$$

where the term  $\left( \frac{\varepsilon_{t-1}^c}{1 + \varepsilon_{t-1}^u} \right)^{-1}$  arises from dividing our starting inequality by the previous period

elasticity ratio. Taking  $\zeta$  small enough, we have the result for  $t > t'$

$$E_{t-1} \left[ \frac{\tau_{L,t}}{1 - \tau_{L,t}} \frac{1}{\beta R} \frac{u'(c_{t-1})}{u'(c_t)} | h \right] \leq \frac{\tau_{L,t-1}}{1 - \tau_{L,t-1}}$$

**2. For the separable limit of French and Jones (2011) preferences:** At the separable limit of French and Jones (2011) fixed time cost preferences (3) and (4) at  $\nu \rightarrow 1$ , we have

$$u(c_t, n_t; h_t) = \gamma \ln(c_t) + (1 - \gamma) \ln(L - n_t - \phi_n I\{n_t > 0\} - \phi_{RE} RE_t - \phi_h(h_t))$$

The elasticity ratio is the log-leisure case of (22): the multiplicative weights  $\gamma$  and  $1 - \gamma$  scale  $\psi_n$  and  $\psi_{nn}$  equally and cancel from the ratio, which equals  $\frac{l}{l+n}$  with leisure exponent equal to one, where the available leisure time is  $L(h) = L - \phi_n - \phi_{RE} - \phi_h(h)$ ; thus the result.

**3. For preferences with a fixed setup cost of work as in Prescott et al. (2009):** Hours devoted to market work are denoted by  $n \in [n_h, 1]$ , where

$$g(n) = \max\{0, n - n_h\}$$

maps hours into *labor services*.<sup>20</sup>

For analytical traceability, it helps to work with a smoothed version of the max function as in Prescott et al. (2009). That is, to differentiate at the kink, we replace  $g$  by the shifted hyperbolic tangent:

$$\tilde{g}(n) = \frac{1}{k} \tanh(k(n - n_h)), \quad k > 0.$$

Its inverse

$$\varphi(n) = \frac{1}{k} \operatorname{artanh}(k(n - n_h)) \tag{23}$$

is *strictly increasing and strictly convex* on  $(n_h, 1/k + n_h)$ .

The disutility of work is isoelastic in  $\varphi$ ,  $\psi(x) = \frac{x^{1+1/\epsilon}}{1+1/\epsilon}$ . For ease of notation, let  $d$  denote the disutility of hours and  $R$  the endogenous elasticity ratio  $\frac{\epsilon^c}{1+\epsilon^u}$

$$d(n) \equiv \psi(\varphi(n)), \quad R(n) \equiv \frac{\epsilon^c}{1 + \epsilon^u} = \frac{d'(n)/n}{d'(n)/n + d''(n)}$$

From (23),

$$\varphi'(n) = \frac{1}{1 - k^2(n - n_h)^2}, \quad \varphi''(n) = \frac{2k^2(n - n_h)}{[1 - k^2(n - n_h)^2]^2} > 0.$$

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<sup>20</sup>See Fig. 4, p. 27, of Prescott et al. (2009) for a graphical discussion of  $g$ .

Hence

$$\begin{aligned} d'(n) &= \psi'(\varphi(n)) \varphi'(n) = \varphi(n)^{1/\epsilon} \varphi'(n), \\ d''(n) &= \psi''(\varphi(n)) [\varphi'(n)]^2 + \psi'(\varphi(n)) \varphi''(n) \\ &= \frac{1}{\epsilon} \varphi(n)^{1/\epsilon-1} [\varphi'(n)]^2 + \varphi(n)^{1/\epsilon} \varphi''(n) \end{aligned}$$

We can now do Taylor approximations around the kink  $n_h$ . Let  $\tilde{n} \equiv n - n_h \downarrow 0^+$ . Using  $\text{artanh}(x) = x + \frac{x^3}{3} + O(x^5)$ ,

$$\varphi(n) = \tilde{n} + \frac{k^2 \tilde{n}^3}{3} + O(\tilde{n}^5), \quad \varphi'(n) = 1 + k^2 \tilde{n}^2 + O(\tilde{n}^4), \quad \varphi''(n) = 2k^2 \tilde{n} + O(\tilde{n}^3)$$

Substituting into  $d'(n)$  and  $d''(n)$  and retaining the lowest nonzero orders gives

$$\begin{aligned} \frac{d'(n)}{n} &= \frac{\tilde{n}^{1/\epsilon}}{\tilde{n}} + O(\tilde{n}^{1/\epsilon+1}) = \tilde{n}^{1/\epsilon-1} + O(\tilde{n}^{1/\epsilon+1}) \\ d''(n) &= \frac{1}{\epsilon} \tilde{n}^{1/\epsilon-1} + 2k^2 \tilde{n}^{1/\epsilon+1} + O(\tilde{n}^{1/\epsilon+2}) \end{aligned}$$

Insert into the elasticity ratio  $R(n)$ :

$$R(n) = \frac{1}{1 + \frac{d''(n)n}{d'(n)}} = \frac{1}{1 + \frac{1}{\epsilon} + 2k^2 \tilde{n}^2 + O(\tilde{n}^3)}$$

The final Taylor approximation yields

$$R(n) = \frac{1}{1 + \epsilon} \left[ 1 - \underbrace{\left( \frac{2k^2 \epsilon}{1 + \epsilon} \right) \tilde{n}^2}_{>0} + O(\tilde{n}^3) \right] \quad (24)$$

Equation (24) implies that

$$R'(n) = -\frac{4k^2 \epsilon}{1 + \epsilon} \tilde{n} + O(\tilde{n}^2) < 0 \quad \text{for } 0 < \tilde{n} \ll 1/k$$

As hours supplied  $n$  fall toward the fixed-cost kink  $n_h$ ,  $R(n)$  rises. A deterioration in health that reduces the optimal  $n$  therefore increases the effective Frisch elasticity, thus the result.

Our result goes beyond the tanh smoothing function. The proof relied only on two facts:

1. *Strict convexity* of the smoothing immediately right of  $n_h$  ( $\varphi''(n) > 0$ );
2. A *finite, positive* first derivative  $\varphi'(n)|_{n \downarrow n_h} = 1$ .

Any other smoothing (e.g., soft-plus, shifted logistic, spline) that shares these local properties generates

$$R(n) = \frac{1}{1 + \epsilon} - C (n - n_h)^2 + o((n - n_h)^2), \quad C > 0$$

and thus the same comparative static conclusion.

This highlights the importance of fixed costs in shaping labor supply decisions when work effort is near the participation threshold.

#### A.4 Proof of Proposition 6

Replacing (20) for  $\lambda$  in the previous period in (19), we obtain an inverse Euler equation (IEE) that applies by future health state and conditional on survival, that is,

$$\frac{\beta R}{u'(c(\theta_-, h_-))} = \int_{\Theta} \frac{1}{u'(c(\theta, h))} f^t(\theta|\theta_-) d\theta = \mathbb{E}_{t-1} \left[ \frac{1}{u'(c(\theta, h))} | h \right] \quad h \in \mathcal{H}_t$$

That is, the IEE (Rogerson 1985) that applies in a setting with dynamic moral hazard applies here not in expectation of future  $\{\theta, h, \text{survival}\}$  but, pathwise, conditional on each future health realization  $h$  and conditional on survival.

By definition, the savings wedge is constructed with the unconditional expectations at time  $t-1$ :

$$\tau_{K,t-1}(\theta_-, h_-) = 1 - \frac{1}{R\beta s_t(h_-)} \frac{u'(c(\theta_-, h_-))}{\int_{\Theta} \sum_{h \in \mathcal{H}} \pi_{h,h_-,t} u'(c(\theta, h)) f^t(\theta|\theta_-) d\theta}$$

We can apply Liao and Berg's (2019) sharpening of Jensen's inequality to the scalar random variable  $X = u'(c(\theta, h))$  and  $\varphi(x) = 1/x$  to obtain, taking expectations on  $(\theta, h)$ , and conditional on survival,

$$\begin{aligned} \frac{\sigma_{c,t}^2}{u_c(\underline{c}_t)^3} &\leq \int_{\Theta} \sum_{h \in \mathcal{H}} \pi_{h,h_-,t} \frac{1}{u'(c(\theta, h))} f^t(\theta|\theta_-) d\theta - \\ &\frac{1}{\int_{\Theta} \sum_{h \in \mathcal{H}} \pi_{h,h_-,t} u'(c(\theta, h)) f^t(\theta|\theta_-) d\theta} \leq \frac{\sigma_{c,t}^2}{u_c(\bar{c}_t)^3} \end{aligned}$$

where  $\sigma_{c,t}^2$  is the variance of marginal utility of consumption in the next period  $\sigma_{c,t}^2 \equiv \text{Var}_{t-1}[\tilde{u}_c(c_t) | \theta_-, h_-]$  and, respectively,  $\underline{c}_t, \bar{c}_t$  are the minimum and maximum values of future consumption:  $\underline{c}_t \equiv \min_{\{\theta \in \Theta, h \in \mathcal{H}_t\}} c_t(\theta^{t-1}, \theta, h^{t-1}, h)$ ,  $\bar{c}_t \equiv \max_{\{\theta \in \Theta, h \in \mathcal{H}_t\}} c_t(\theta^{t-1}, \theta, h^{t-1}, h)$ . Multiplying by  $u'(c(\theta_-, h_-)) / (\beta R)$ , the first integral will equal one due to the IEEs:

$$\frac{u'(c(\theta_-, h_-)) \sigma_{c,t+1}^2}{\beta R u_c(\underline{c}_{t+1})^3} \leq 1 - \frac{u'(c(\theta_-, h_-))}{\beta R \int_{\Theta} \sum_{h \in \mathcal{H}} \pi_{h,h_-,t} u'(c(\theta, h)) f^t(\theta|\theta_-) d\theta} \leq \frac{u'(c(\theta_-, h_-)) \sigma_{c,t+1}^2}{\beta R u_c(\bar{c}_{t+1})^3}$$

That is, subtracting 1, dividing by the survival probability  $s_t(h_-)$ , and adding 1, we obtain

$$\begin{aligned} 1 - \frac{1}{s_t(h_-)} \left( 1 - \frac{u'(c(\theta_-, h_-)) \sigma_{c,t+1}^2}{\beta R u_c(\underline{c}_{t+1})^3} \right) &\leq \underbrace{1 - \frac{u'(c(\theta_-, h_-))}{\beta R s_t(h_-) \int_{\Theta} \sum_{h \in \mathcal{H}} \pi_{h,h_-,t} u'(c(\theta, h)) f^t(\theta|\theta_-) d\theta}}_{\tau_{K,t-1}} \\ &\leq 1 - \frac{1}{s_t(h_-)} \left( 1 - \frac{u'(c(\theta_-, h_-)) \sigma_{c,t+1}^2}{\beta R u_c(\bar{c}_{t+1})^3} \right) \end{aligned}$$



The middle term of the inequality corresponds to the savings wedge. Thus, the result.

## A.5 Optimal Retirement in a Continuous-Time Model

This appendix develops a continuous-time version of the model of the main text in which the timing of retirement can be characterized analytically. Its implementation results (Appendix A.8) rely on risk neutrality in consumption and geometric Brownian wages; the quantitative implementation analysis is in Appendix C.8.

In this section, we describe a continuous-time economy in which workers are ex ante heterogeneous in productivity, experience idiosyncratic productivity shocks over their lifetime, and adjust their labor supply through flexible working hours and the timing of their retirement.

**Productivity, Technology, and Preferences:** Consider a continuous-time economy populated by a continuum of agents who live until age  $T$ . At each time  $t$ , each agent privately observes the realization of his current labor productivity  $\theta_t \in (0, +\infty)$ . Agents provide  $n_t \geq 0$  units of labor at time  $t$  at a wage rate equal to their productivity and earn gross income  $y_t = \theta_t n_t$ .

At time  $t = 0$ , initial productivity  $\theta_0 \in (0, +\infty)$  is drawn from a distribution  $F$  with density  $f$ . A standard Brownian motion  $B = \{B_t, \mathcal{F}_t; 0 \leq t \leq T\}$  on  $(\Omega, \mathcal{F}, \mathcal{P})$  drives the productivity shocks in future periods. A history of productivities  $(\theta^t) = \{\theta_s\}_{s \in [0, t]}$  is a sequence of realizations of the productivity process that evolves according to the law of motion

$$\frac{d\theta_t}{\theta_t} = \mu_t dt + \sigma dB_t \quad (25)$$

The real constants  $\mu_t - \frac{1}{2}\sigma^2$  and  $\sigma$  are, respectively, the drift and volatility of log productivity. When the drift and volatility are independent of time, productivity is a geometric Brownian motion (GBM), and log productivity is the continuous-time limit of a random walk. Health quality  $h_t \in \mathbb{R}$  is a diffusion process  $dh_t = \mu_t^h dt + \sigma_h dW_t$ , where  $W$  is in general a standard Brownian motion with its filtration. The correlation structure of  $B_t, W_t$  matters for a general treatment of our problem and would lead to a two-dimensional Hamilton–Jacobi–Bellman (HJB) equation that is not tractable with analytical methods. Instead, we assume a perfect correlation between high productivity and good health,  $W_t = -B_t$ . The interpretation of this assumption within the lens of our discrete-time model in the main text is that the wage (here,  $w_t = \theta_t$ ) deterministically decreases in poor health<sup>21</sup> and all available information on wages and health is the same.

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<sup>21</sup>Applying Ito’s lemma, we have  $h_t = h_0 + \int_0^t \mu_s^h ds - \frac{\sigma_h}{\sigma} \left( \log \left( \frac{\theta_t}{\theta_0} \right) - \int_0^t (\mu_s - \frac{1}{2}\sigma^2) ds \right)$ .

Agents have time-separable preferences over consumption  $\{c_t\}_{0 \leq t \leq T}$  and labor  $\{n_t\}_{0 \leq t \leq T}$  processes that are progressively measurable with respect to the filtration  $\mathcal{F}_t$ .<sup>22</sup> When an agent is working, ( $n_t > 0$ ), he incurs a health-dependent utility cost of staying in the labor market  $\phi_t(h_t)$ , and his current-period utility is  $u(c_t, n_t; h_t) - \phi_t(h_t)$ , where  $u$  is increasing in consumption, decreasing in labor, twice continuously differentiable, and concave and  $\phi_t$  is decreasing. Given our perfect correlation assumption between wages and health, the fixed cost of participation can be written directly in terms of productivity  $\phi_t(\theta_t)$ . Utility along the intensive margin is separable in consumption and labor and isoelastic in labor:

$$u(c_t, n_t) = u(c_t) - h(n_t) = u(c_t) - \kappa \frac{n_t^{1+\frac{1}{\varepsilon}}}{1+\frac{1}{\varepsilon}}$$

where  $\varepsilon > 0$  is the intensive Frisch elasticity of labor supply.

Retirement,  $n_t = 0$ , is an irreversible decision. Define a stopping time  $\mathcal{T}_R \in \mathcal{T}$ ,<sup>23</sup> the age after which a retired agent provides zero labor effort and does not incur the fixed utility cost. After retirement, an agent's utility in each period is  $u(c_t, 0)$ . We define the retirement age as the age at which an individual chooses to exit the labor force forever<sup>24</sup>—which the model allows to differ from the age at which an individual chooses to start claiming Old-Age, Survivors, and Disability Insurance (OASDI) benefits.<sup>25</sup>

**Planning Problem:** Preferences over consumption and labor  $\{c_t, n_t\}$  and retirement decisions  $\{\mathcal{T}_R\}$  are summarized by an agent's expected lifetime utility:

$$v_0(\{c_t, n_t, \mathcal{T}_R\}) \equiv \mathbb{E} \left\{ \int_0^{\mathcal{T}_R} e^{-\rho t} [u(c_t, n_t) - \phi_t(\theta_t)] dt + \int_{\mathcal{T}_R}^T e^{-\rho t} u(c_t, 0) dt \right\} \quad (26)$$

in which  $\rho$  is the rate of time preference. A utilitarian planner chooses incentive-compatible (IC) allocations to maximize social welfare:

$$\max_{\{c_t, n_t, \mathcal{T}_R\}} v_0(\{c_t, n_t, \mathcal{T}_R\})$$

<sup>22</sup>Consumption  $c_t(\theta^t)$  and labor  $n_t(\theta^t)$  depend on the whole history of productivities until time  $t$ . To simplify the notation, we drop the realizations  $\theta^t$  when referring to  $\mathcal{F}_t$ -measurable processes  $\{c_t, y_t\}$ .

<sup>23</sup>A random variable  $\mathcal{T}_R$  is a stopping time if  $\{\mathcal{T}_R \leq t\} \in \mathcal{F}_t, \forall t \geq 0$ . Intuitively, this definition means that at any time  $t$ , one must know whether retirement has occurred.

<sup>24</sup>The irreversible retirement assumption is motivated by empirical and theoretical considerations. [Rogerson and Wallenius \(2013\)](#) find empirical evidence in Current Population Survey data that retirement occurs as an abrupt transition from full-time to little or no work in the US. By age 70, the age at which individuals should start claiming Social Security benefits, 75% of men report working zero hours. In addition, this assumption is without loss of generality and can be relaxed. The main predictions of the model remain unchanged if we allow retirees to return to the labor market at a lower wage.

<sup>25</sup>In a decentralized economy, workers can actually claim SS benefits whenever they want, and their optimal retirement benefits are computed according to the history of their earnings. Because we work directly with allocations in this primal approach, the Social Security benefits are implicit in the model.

subject to the law of motion of productivity (25), the definition of indirect utility (26), and an intertemporal resource constraint. For simplicity, we work in partial equilibrium, and the planner can save aggregate resources in a small open economy and borrow at a net rate of return  $r$ . We study the planner's problem for a single cohort in isolation and abstract from intergenerational redistribution.<sup>26</sup> The planner's resource constraint is therefore

$$\mathbb{E}\left\{\int_0^T e^{-rt} c_t dt\right\} + G \leq \mathbb{E}\left\{\int_0^{\mathcal{T}_R} e^{-rt} \theta_t n_t dt\right\} \quad (27)$$

The left-hand side includes exogenous government spending  $G$ <sup>27</sup> and the cost of providing lifetime consumption to agents. The right-hand side is the sum of the net present value (NPV) of income  $y_t$  generated by workers until they retire. Because of the law of large numbers, the aggregate resource constraint is the expectation over the histories of productivities ( $\theta^t$ ).

### A.5.1 Optimal First-Best Retirement Choice

This section solves the planning problem with complete information. We highlight features of the optimal retirement decision that are absent in existing models with no endogenous retirement choice but that have important implications for optimal policy.

Let the rate of time preference equal the rate of return of government savings,  $\rho = r$ . From the IEE, productivity shocks are fully insured, and consumption is the same across different histories:  $u'(c_t(\theta^t)) = \lambda$ , where  $\lambda$  is the marginal social cost of public funds.<sup>28</sup> When it is optimal to work, the marginal rate of transformation of labor into consumption is the wage rate,  $\theta_t$ . Therefore, labor supply satisfies  $\kappa n_t^{\frac{1}{\varepsilon}} = \lambda \theta_t$ . With full information, the planner maximizes social welfare by maximizing total resources available in the economy. Consumption is smoothed, and more productive agents work more hours and produce more output. It is therefore only natural that, as long as the fixed cost of staying in the labor market for highly productive workers is not too high compared with that of low-productivity workers (Technical Assumption 3), the planner makes highly productive workers retire later than low-productivity workers.

**Assumption 3.** For some constant  $\psi$ ,  $\phi'(\theta) \leq \psi \theta^\varepsilon$ ,  $\forall \theta$ .

**Proposition 9.** (First-best retirement decision) Suppose that Assumption 1 holds. Then, there exists a time-dependent productivity threshold  $\theta_R^{fb}(t)$  such that retirement occurs if and only if productivity falls below it:  $\mathcal{T}_R^{fb} = \inf\{t; \theta_t \leq \theta_R^{fb}(t)\}$ .

<sup>26</sup>Given that we study insurance and redistribution across one cohort, time is equivalent to age for the cohort.

<sup>27</sup> $G$  can capture many sources of exogenous government revenues and expenses as well as intergenerational transfers to or from another cohort and so on.

<sup>28</sup> $\lambda$  is the multiplier on the planner's resource constraint (27).

*Proof.* The planner's problem is

$$\max_{\{\lambda, c_t, n_t, \mathcal{T}_R\}} \mathbb{E} \left\{ \int_0^T e^{-\rho s} [u(c_t) - \lambda c_t] dt + \int_0^{\mathcal{T}_R} e^{-\rho s} [\lambda \theta_t n_t - \kappa \frac{(n_t)^{1+\frac{1}{\varepsilon}}}{1+\frac{1}{\varepsilon}} - \phi_t(\theta_t)] dt \right\}$$

subject to the law of motion of productivity (25). From the optimal allocations  $u'(c) = \lambda$  and  $\kappa n_t^{\frac{1}{\varepsilon}} = \lambda \theta_t$ , denote  $\mathbb{E} \left\{ \int_0^T e^{-\rho s} [u(c_t) - \lambda c_t] dt \right\} = h(\lambda)$ . Then, the above objective is rewritten as

$$\max_{\{\lambda, \mathcal{T}_R\}} h(\lambda) + \mathbb{E} \left\{ \int_0^{\mathcal{T}_R} e^{-\rho t} [\lambda^{1+\varepsilon} \frac{(\theta_t)^{1+\varepsilon}}{\kappa^\varepsilon (1+\varepsilon)} - \phi_t(\theta_t)] dt \right\}$$

Denote a maximizer by  $\lambda^*$ . By an envelope condition, the expected change in the payoff if retirement is delayed by an infinitesimally short time is  $\lambda^{*1+\varepsilon} \frac{(\theta_t)^{1+\varepsilon}}{\kappa^\varepsilon (1+\varepsilon)} - \phi_t(\theta_t)$ . Taking  $\psi < \frac{\lambda^{*1+\varepsilon}}{\kappa^\varepsilon}$  in the condition of growth bounded from above of  $\phi_t(\theta)$  in Proposition 9 or assuming that  $G$  is high enough that the marginal utility of consumption  $\lambda^{*1+\varepsilon}$  is high, and the inequality holds, then the expected change in payoff is increasing in productivity. The dynamic single crossing condition in Kruse and Strack (2015) holds, and Theorem 4.3 of Jacka and Lynn (1992) implies that a time-varying threshold determines the shape of the stopping region (retirement rule).

Note that when  $\phi_t$  is independent of productivity, or nonincreasing in productivity, the “bounded growth from above” condition in the proposition holds, implying Proposition 9.  $\square$

This proposition means that the planner balances the need to induce the highly productive (high-wage) agents to continue working with the need to avoid the fixed utility cost for less productive (low-earning) workers. In the first-best case, it is therefore optimal to set productivity cutoffs below which retirement occurs.

To understand the determinants and lifetime evolution of these retirement cutoffs, we consider the case in which agents are risk neutral. In this tractable case, we analytically show that there is an option value of waiting for higher productivity shocks before retirement. In addition, this option value decreases over time. Therefore, the implicit labor supply elasticity over the retirement margin increases over time. The following corollary summarizes this result in terms of the retirement thresholds  $\theta_R^{fb}(t)$ .

**Corollary 10.** (*Option value of continued work vs. retirement*) Suppose that Assumption 3 holds and productivity is a GBM. Denote by  $\theta_S$  the static participation threshold.

1. For all  $t < T$ ,  $\theta_R^{fb}(t) \leq \theta_S$ , and the marginal social value of continued work is negative at retirement, that is,  $\theta_R^{fb}(t) n^{fb}(\theta_R^{fb}(t)) - h(n^{fb}(\theta_R^{fb}(t))) - \phi(\theta_R^{fb}(t)) \leq 0$ .
2. The retirement thresholds  $\theta_R^{fb}(t)$  are increasing in  $t$ . In addition,  $\lim_{t \rightarrow T} \theta_R^{fb}(t) = \theta_S$ .

Point 1 of the corollary states that retirement occurs below a level of productivity at which it would be inefficient to work in a static environment. This creates an option value of waiting for higher productivity shocks and higher earnings before retirement, which is not present in models with permanent productivity shocks such as those in [Michau \(2014\)](#) and [Shourideh and Troshkin \(2015\)](#). Working today instead of retiring preserves the option of retiring later at a higher wage, hence the term “option value of work.” Indeed, when there is no uncertainty on future earnings, the marginal value of labor is equal to the fixed utility cost of work at retirement, and the option value is zero. In practice, this option value is negative at retirement. [Lazear and Moore \(1988\)](#), [Rust \(1989\)](#) and [Stock and Wise \(1990\)](#) estimate structural models of retirement with uncertain earnings and find that people continue to work at any age, as long as the expected present utility value of continuing work is greater than or equal to the expected present value of immediate retirement.

Point 2 of the corollary states that the option value of continued work decreases over time as the horizon shortens. The option value of continued work vanishes at the end of the horizon, and only then is the irreversible retirement decision similar to a static participation decision, and the marginal value of labor is equal to the fixed utility cost of work.

*Proof.* To qualify results further, we now consider agents who are risk-neutral in consumption, so that  $u(c_t) = c_t$ . Consumption is not pinned down by the Euler equation. We eliminate consumption from the planner’s problem by replacing the resource constraint in the planner’s social welfare function:

$$w \equiv \max_{\mathcal{T}_R} \mathbb{E} \left\{ \int_0^{\mathcal{T}_R} e^{-\rho t} [\theta_t n_t^{fb} - \kappa \frac{(n_t^{fb})^{1+\frac{1}{\varepsilon}}}{1+\frac{1}{\varepsilon}} - \phi(\theta_t)] dt \right\} - G$$

subject to the law of motion of productivity (25). Normalizing government spending to zero,  $G = 0$ , and replacing the first-best labor allocations using the optimality condition  $\kappa(n_t^{fb})^{\frac{1}{\varepsilon}} = \theta_t$ , the social welfare function  $w(\theta_t, t)$  satisfies the following HJB equation:

$$0 = \max \left\{ -w(\theta, t), -\rho w(\theta, t) + \frac{\theta^{1+\varepsilon}}{\kappa^\varepsilon(1+\varepsilon)} - \phi(\theta_t) + (\mu_t \theta) \partial_\theta w(\theta, t) + \frac{\sigma^2 \theta^2}{2} \partial_{\theta\theta} w(\theta, t) + \partial_t w(\theta, t) \right\}$$

The terms to the right of  $-\rho w(\theta, t)$  consist of the marginal social value of labor minus the fixed cost and derivatives of social welfare with respect to time and productivity.

Now, consider the case of productivity that evolves according to a GBM; that is,  $\mu_t$  and  $\sigma$  are, respectively, constants  $\mu$  and  $\sigma$ . We show that even when the fixed cost is a constant  $\phi(t) = \phi$ , there is an option value of waiting for higher productivity shocks before retirement. In addition, this option value decreases over time. Therefore, even when the fixed cost is constant over time, the elasticity of the retirement margin increases over time. Hence, the extensive margin elasticity of labor supply increases over time, despite the intensive Frisch

elasticity and the fixed cost being time independent.

Consider the infinite horizon model,  $T = +\infty$ . To ensure convergence of social welfare, we assume

$$\rho > (1 + \varepsilon)(\mu + \frac{1}{2}\sigma^2\varepsilon). \quad (28)$$

Social welfare is now time-independent, and replacing the HJB equation in this setting is

$$\max\{0 - w(\theta), -\rho w(\theta) + \mu\theta w_\theta + \frac{\sigma^2\theta^2}{2}w_{\theta\theta} + \frac{\theta^{1+\varepsilon}}{\kappa^\varepsilon(1+\varepsilon)} - \phi(\theta)\} \quad (29)$$

We conjecture that the solution is of the following form: There is a threshold  $\theta_R^{fb}$  such that an agent retires if and only if his productivity falls below the threshold  $\theta_t \leq \theta_R^{fb}$ . This implies that  $w(\theta) = 0$  for all  $\theta \leq \theta_R^{fb}$  and for  $\theta > \theta_R^{fb}$ ,  $w$  is a nonnegative solution to the equation

$$-\rho w(\theta) + \mu\theta w_\theta + \frac{\sigma^2\theta^2}{2}w_{\theta\theta} = -\frac{\theta^{1+\varepsilon}}{\kappa^\varepsilon(1+\varepsilon)} + \phi(\theta)$$

Moreover,  $w$  must be  $C^1$  on its entire domain. This implies  $w(\theta_R^{fb}) = 0$ , a value-matching condition, and  $w_\theta(\theta_R^{fb}) = 0$ , a smooth-pasting condition. Finally, observe that, for  $\theta \leq \theta_R^{fb}$ , the second term on the right-hand side of (29) implies that  $\frac{\theta^{1+\varepsilon}}{\kappa^\varepsilon(1+\varepsilon)} \leq \phi(\theta)$ ; that is, at retirement and afterward, the marginal social value of continued work is negative. In particular,  $\hat{\theta}_R^{fb} \leq \theta_S$ .

Set  $\phi_1\theta^{1+\varepsilon_\phi} + \phi_0$  with  $\varepsilon_\phi < \varepsilon$ . Define the quadratic polynomial  $P(x) = -\rho + \mu x + \frac{\sigma^2}{2}x(x-1)$ . The homogeneous equation

$$-\rho w(\theta) + \mu\theta w_\theta + \frac{\sigma^2\theta^2}{2}w_{\theta\theta} = 0$$

admits the general solution

$$w(\theta) = C_- \theta^{x_-} + C_+ \theta^{x_+}$$

in which  $x_-$  and  $x_+$  are the negative and positive roots of  $P$ . We find a particular solution for each nonhomogeneous term, denoted  $A\theta^{1+\varepsilon}$ ,  $A'\theta^{1+\varepsilon_\phi}$ , and  $B$  in which  $A = -\frac{1}{\kappa^\varepsilon(1+\varepsilon)P(1+\varepsilon)}$ ,  $A' = \frac{\phi_1}{P(1+\varepsilon_\phi)}$  and  $B = -\frac{\phi}{\rho}$ . By the assumption in (28),  $P(1+\varepsilon) < 0$ . The sum of these particular solutions  $A\theta^{1+\varepsilon} + A'\theta^{1+\varepsilon_\phi} + B$  is the value of social welfare if agents never retire.

By the superposition principle of linear homogeneous ordinary differential equations (ODEs), the solution takes the form

$$w(\theta) = A\theta^{1+\varepsilon} + A'\theta^{1+\varepsilon_\phi} + B + C_- \theta^{x_-} + C_+ \theta^{x_+}$$

for  $\theta > \theta_R^{fb}$  and  $w(\theta) = 0$  for  $\theta \leq \theta_R^{fb}$ . From (28), we ensure that  $x_+ > 1 + \varepsilon$ . Since

$n^{fb} - \kappa \frac{(n^{fb})^{1+\frac{1}{\varepsilon}}}{1+\frac{1}{\varepsilon}} = \frac{\theta^{1+\varepsilon}}{\kappa^\varepsilon(1+\varepsilon)}$ , we can conjecture that  $w(\theta) =_{\theta \rightarrow +\infty} \mathcal{O}(\theta^{1+\varepsilon})$ . Therefore,  $C_+ = 0$ .

By the value-matching and smooth-pasting conditions:

$$A(\theta_R^{fb})^{1+\varepsilon} + A'(\theta_R^{fb})^{1+\varepsilon_\phi} + B + C_-(\theta_R^{fb})^{x_-} = 0 \quad (30)$$

$$(1+\varepsilon)A \frac{(\theta_R^{fb})^{1+\varepsilon}}{\theta_R^{fb}} + (1+\varepsilon_\phi)A' \frac{(\theta_R^{fb})^{1+\varepsilon_\phi}}{\theta_R^{fb}} + x_- C_- \frac{(\theta_R^{fb})^{x_-}}{\theta_R^{fb}} = 0 \quad (31)$$

Multiplying (30) by  $x_-$  and (31) by  $\theta_R^{fb}$  and subtracting the two yields

$$(1+\varepsilon-x_-)A(\theta_R^{fb})^{1+\varepsilon} + (1+\varepsilon_\phi-x_-)A'(\theta_R^{fb})^{1+\varepsilon_\phi} = x_-B$$

When  $\varepsilon_\phi = \varepsilon$ , the equation becomes simply

$$(1+\varepsilon-x_-)(A+A')(\theta_R^{fb})^{1+\varepsilon} = x_-B$$

Setting  $A(\phi_1) = A + A'$ , the expression of  $\theta_R^{fb}$  and  $w$  in Corollary 10 follows by replacing the values of  $A(\phi_1)$  and  $B$ .

$$\theta_R^{fb} = \left( \frac{\phi_0}{\rho} \frac{x}{A(\phi_1)(1+\varepsilon+x)} \right)^{\frac{1}{\varepsilon}}$$

and the static participation threshold is

$$\theta_S = \left( \frac{\phi_0}{[\kappa^\varepsilon(1+\varepsilon)]^{-1} - \phi_1} \right)^{\frac{1}{\varepsilon}}$$

Both  $\theta_R^{fb}$  and  $\theta_S$  are increasing in  $\phi_0$  and in  $\phi_1$ . In addition, since  $\frac{\rho-(1+\varepsilon)(\mu+\frac{\sigma^2}{2}\varepsilon)}{\rho} < 1$  and  $\frac{(x)}{(1+\varepsilon+x)} < 1$ , we obtain  $\theta_R^{fb} < \theta_S$ . Now, in a finite horizon, the problem is time-dependent, and the thresholds are time-dependent. When time goes to  $T$ , the value of waiting for productivity to improve decreases, and the thresholds converge to  $\theta^*$ . Only the dynamic single-crossing property of the derivative operator is needed in the finite horizon for this to hold. This is again an application of [Jacka and Lynn \(1992\)](#).  $\square$

To develop some intuition for the proof, when we set  $\phi(\theta) = \phi_0 + \phi_1\theta^{1+\varepsilon}$  and consider the infinite horizon limit  $T \rightarrow \infty$ , the retirement threshold is independent of time,  $\theta_R^{fb}$ . We proceed similarly to [Leland \(1994\)](#) by decomposing the value of social welfare into two terms:

$$w(\theta) = \underbrace{A(\phi_1)\theta^{1+\varepsilon} - \frac{\phi_0}{\rho}}_{\text{social value of working forever (SVWF)}} - \underbrace{\left(\frac{\theta_R^{fb}}{\theta}\right)^x}_{\text{discounting at retirement } E[e^{-\rho \mathcal{T}_R^{fb}}|\theta]} \underbrace{\left[A(\phi_1)(\theta_R^{fb})^{1+\varepsilon} - \frac{\phi_0}{\rho}\right]}_{\text{SVWF starting at retirement threshold}}$$

with positive constant  $x$  and nonincreasing function  $A(\phi_1)$ . The value of social welfare  $w(\theta)$  is the value of the lifetime utility of output if the agent were to work forever, minus



the value of lifetime utility of output if he were to work forever at the optimal retirement threshold, discounted by the expected value of the discount factor at retirement. This value is zero at retirement. From a smooth-pasting argument as in [Dixit \(1993\)](#), the value of his marginal social welfare is also zero at retirement. This gives an explicit value of the retirement threshold

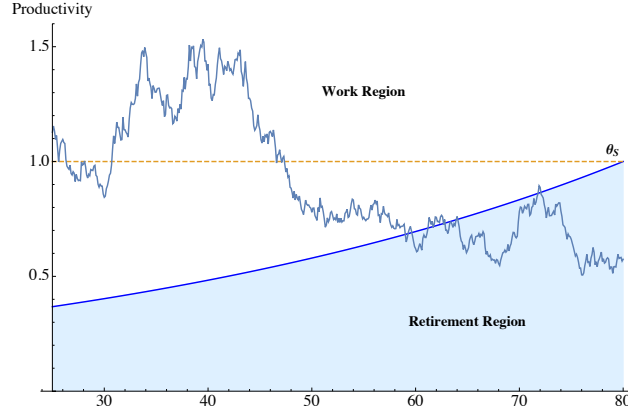
$$\theta_R^{fb} = \left( \frac{\phi_0}{\rho} \frac{x}{A(\phi_1)(1 + \varepsilon + x)} \right)^{\frac{1}{\varepsilon}}$$

and the static participation threshold is

$$\theta_S = \left( \frac{\phi_0}{[\kappa^\varepsilon(1 + \varepsilon)]^{-1} - \phi_1} \right)^{\frac{1}{\varepsilon}}$$

Note that both  $\theta_R^{fb}$  and  $\theta_S$  are increasing in  $\phi_0$  and in  $\phi_1$ , meaning that workers retire earlier when their fixed costs are large. In addition, the marginal social value of continued work is negative at retirement  $\theta_R^{fb} < \theta_S$ .

Figure 10: First-Best Retirement Decision



Notes: Example of productivity history. Horizontal axis  $t$ , vertical axis  $\theta_t$ . Retirement region shaded.  $\theta_s$ : static participation cutoff. The retirement region expands with age.

In summary, the solution to the first-best planning problem generates the following insights about the implications of optimal retirement: First, low-productivity agents retire earlier than highly productive agents. Second, there is an option value of waiting for higher earnings before retiring. Therefore, the implicit labor supply elasticity increases over time.

**Nonseparable preferences.** The following lemma characterizes the first-best retirement decision with preferences that are nonseparable in consumption and labor.

**Lemma 11.** *Suppose  $u$  is a [Greenwood et al. \(1988\)](#)-type utility function. The optimal retirement rule in the first-best is a cutoff rule  $\mathcal{T}_R^{fb} = \inf\{t; \theta_t \leq \theta_R^{fb}(t)\}$ .*

*Proof.* Denote by  $\lambda$  the Lagrangian multiplier on the government's resource constraint. The

FOC on  $c_t$  when an agent works is  $\left(c_t - \frac{n_t^{\frac{1+\varepsilon}{1+\varepsilon}}}{1+\varepsilon}\right)^{-\nu} = \lambda$  and  $c_t^{-\nu} = \lambda$  when an agent is retired. The FOC for the labor supply of workers is  $n_t^{\frac{1}{\varepsilon}} \lambda = \lambda \theta_t$  so that  $n_t = \theta_t^\varepsilon$ . After rearranging and simplifying, the terms in  $\lambda$  cancel out, and the planner's retirement problem is rewritten as:

$$\max_{\{\lambda, \mathcal{T}_R\}} \mathbb{E} \left\{ \int_0^{\mathcal{T}_R} e^{-\rho t} \left[ \lambda \frac{(\theta_t)^{1+\varepsilon}}{(1+\varepsilon)} - \phi_t(\theta_t) \right] dt \right\}$$

The proof ends as in the proof of Proposition 9 applying Theorem 4.3 in [Jacka and Lynn \(1992\)](#).  $\square$

When the planner cannot observe productivity, first-best allocations with constant consumption are not achievable, as any agent would be better off retiring immediately. Nevertheless, history-dependent versions of these intuitions carry through in the second-best retirement policies.

## A.6 Optimal Second-Best Retirement

### A.6.1 Second-Best Problem Under Risk Neutrality

We introduce the first-order approach (FOA) in the simpler setting in which agents are risk neutral in consumption and productivity is a GBM. We relax incentive compatibility by considering a family of deviations that [Bergemann and Strack \(2015\)](#) call *consistent deviations*. The effect of these deviations on promised utility can be summarized by what [Pavan et al. \(2014\)](#) call the *impulse response function*. This FOA is standard in the dynamic contracting literature with persistent shocks.

The value of the agent's productivity if he reports his productivity truthfully is

$$\theta_t = \theta_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma B_t\right)$$

We define  $\Phi$  by  $\theta_t \equiv \Phi(t, \theta_0, B_t)$  and set the following definition, which is motivated by [Bergemann and Strack \(2015\)](#).

**Definition 12.** (*Consistent deviations*). A deviation is called consistent if an agent, with real productivity  $\theta_t = \Phi(t, \theta_0, B_t)$  and associated initial shock  $\theta_0$ , misreports his initial shock by announcing  $\tilde{\theta}_0 \in \Theta_0$  at  $t = 0$  and continues to misreport  $\tilde{\theta}_t = \Phi(t, \tilde{\theta}_0, B_t)$  instead of his true productivity  $\theta_t$  at all future dates  $t \leq T$ .

With this definition, an agent who follows a consistent deviation misreports his true type in all future periods. An agent's reported productivity  $\tilde{\theta}_t = \Phi(t, \tilde{\theta}_0, B_t)$  would be equal to the productivity he would have had if his initial shock had been  $\tilde{\theta}_0$  instead of  $\theta_0$ . From these misreports, the planner can infer the true realized path of Brownian shocks  $B_t$ . However,

since the allocations depend on the history of productivities instead of the Brownian shocks, the inference on the Brownian shocks is not of immediate use for the principal. [Bergemann and Strack \(2015\)](#) show that incentive compatibility with respect to consistent deviations—a one-dimensional class of deviations—is sufficient for full incentive compatibility in the risk-neutral and GBM case. This result allows us to derive the IC optimal allocations and shadow fixed costs.

Consider the ex ante utility at time 0 of an agent with initial productivity  $\theta_0$  who announces  $\tilde{\theta}_0$  and follows consistent deviations; denote it  $v(\theta_0, \tilde{\theta}_0)$ . Then,

$$v(\theta_0, \tilde{\theta}_0) = \mathbb{E}^{\{\tilde{\theta}\}} \left\{ \int_0^T e^{-\rho t} c_t(\tilde{\theta}_0) dt - \int_0^{\mathcal{T}_R(\tilde{\theta}_0)} e^{-\rho t} \left[ \kappa \frac{\left( \frac{y_t(\tilde{\theta}_0)}{\Phi(t, \theta_0, B_t)} \right)^{1+\frac{1}{\varepsilon}}}{1 + \frac{1}{\varepsilon}} + \phi_t \left( \Phi(t, \theta_0, B_t) \right) \right] dt \middle| \tilde{\theta}_0 \right\}$$

Restricting attention to consistent deviations alone, the incentive problem turns into a static one. Truthful reports at time zero are necessary for incentive compatibility; that is,  $v(\theta_0) = \max_{\tilde{\theta}_0} v(\theta_0, \tilde{\theta}_0)$ , and an envelope condition allows us to obtain the derivative of ex ante utility. The sensitivity of ex ante utility with respect to initial reports satisfies:

$$v_\theta(\theta_0) = \mathbb{E} \left\{ \int_0^{\mathcal{T}_R} e^{-\rho t} \left[ \left( 1 + \frac{1}{\varepsilon} \right) \left( \frac{\Phi_\theta(t, \theta_0, B_t)}{\theta_t} \right) \kappa \frac{\left( \frac{y_t}{\theta_t} \right)^{1+\frac{1}{\varepsilon}}}{1 + \frac{1}{\varepsilon}} - \Phi_\theta(t, \theta_0, B_t) \phi'_t(\theta_t) \right] dt \middle| \theta_0 \right\}$$

$\Phi_\theta(t, \theta_0, B_t)$  is what [Pavan et al. \(2014\)](#) call the *impulse response function* and [Bergemann and Strack \(2015\)](#) call the *stochastic flow* in continuous time. Here, with GBM productivity, the stochastic flow is the ratio of current productivity to initial productivity, that is,

$$\Phi_\theta(t, \theta_0, B_t) = \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma B_t\right) = \theta_t / \theta_0$$

Then, the IC constraint simplifies to

$$v_\theta(\theta_0) = \frac{1}{\theta_0} \mathbb{E} \left\{ \int_0^{\mathcal{T}_R} e^{-\rho t} \left[ \left( 1 + \frac{1}{\varepsilon} \right) \kappa \frac{\left( \frac{y_t}{\theta_t} \right)^{1+\frac{1}{\varepsilon}}}{1 + \frac{1}{\varepsilon}} - \theta_t \phi'_t(\theta_t) \right] dt \middle| \theta_0 \right\} \quad (32)$$

### A.6.2 Optimal Second-Best Retirement Choice

Consider the case of agents who are risk neutral in consumption and productivity that evolves according to a GBM. Risk neutrality in consumption implies that consumption need not be distorted. Because of the strict concavity of  $u(c)$  in the case of risk-averse agents with a utilitarian planner, the equivalent generalized social marginal welfare weights ([Saez and Stantcheva 2016](#)) reflect decreasing marginal utility of consumption. Low-productivity agents have lower consumption and higher marginal utility and therefore higher social welfare weights. To ensure comparability between the risk-averse utilitarian and the risk-neutral cases, we assume that the planner puts Pareto welfare weights  $\alpha(\theta_0)$  on each agent with initial type  $\theta_0$ . Since, with concave utility, the marginal utility of consumption is nonincreasing, we

assume the function  $\alpha : \Theta_0 \mapsto (0; +\infty)$  is nonincreasing. We normalize the sum of Pareto weights to one  $\int_0^\infty \alpha(\theta_0) dF(\theta_0) = 1$  and call the summand of weights  $\Lambda(\theta) = \int_0^\theta \alpha(\theta_0) dF(\theta_0)$ .

The following lemma formulates the second-best retirement decision problem by substituting the optimal allocations in the planner's problem.

**Lemma 13.** (*Allocations and wedges*) *The labor wedges are time-invariant and depend only on the initial heterogeneity and the welfare weights:*

$$\frac{\tau_t^L}{1 - \tau_t^L} = \frac{\tau(\theta_0)}{1 - \tau(\theta_0)} = \left(1 + \frac{1}{\varepsilon}\right) \frac{1}{\theta_0} \frac{\Lambda(\theta_0) - F(\theta_0)}{f(\theta_0)}$$

*In addition, the planner's problem is to choose the retirement rule so as to solve:*

$$\max_{\mathcal{T}_R} \int_0^\infty \mathbb{E} \left\{ \int_0^{\mathcal{T}_R} e^{-\rho t} \left[ (1 - \tau(\theta_0))^\varepsilon [y_t^{fb} - \kappa \frac{(y_t^{fb})^{1+\frac{1}{\varepsilon}}}{1 + \frac{1}{\varepsilon}}] - [\phi_t - \frac{\tau(\theta_0)}{1 - \tau(\theta_0)} \frac{\varepsilon}{1 + \varepsilon} \theta_t \phi'_t(\theta_t)] \right] dt \right\} dF(\theta_0)$$

*Proof.* The problem of the planner is to choose allocations  $\{c, y\}$  and a retirement rule  $\mathcal{T}_R$  to maximize social welfare subject to the definition of ex ante utility, the resource constraint (27), the relaxed IC constraint (32) and the law of motion of productivity (25). We rewrite the problem from the FOA with risk neutrality below for reading convenience.

$$\begin{aligned} & \max_{\{c, y, v, \mathcal{T}_R\}} \int_0^\infty \alpha(\theta_0) v(\theta_0) dF(\theta_0) \\ & \text{s.to } \frac{d\theta_t}{\theta_t} = \mu dt + \sigma dB_t \\ & v(\theta_0) = \mathbb{E}_0 \left\{ \int_0^T e^{-\rho t} c_t dt - \int_0^{\mathcal{T}_R} e^{-\rho t} \left[ \kappa \frac{(y_t)^{1+\frac{1}{\varepsilon}}}{1 + \frac{1}{\varepsilon}} + \phi_t \right] dt \middle| \theta_0 \right\} \\ & 0 \leq \mathbb{E} \left\{ \int_0^{\mathcal{T}_R} e^{-\rho t} y_t dt \right\} - \mathbb{E} \left\{ \int_0^T e^{-\rho t} c_t dt \right\} \\ & v_\theta(\theta_0) = \frac{1}{\theta_0} \mathbb{E}_0 \left\{ \int_0^{\mathcal{T}_R} e^{-\rho t} \left[ \left(1 + \frac{1}{\varepsilon}\right) \kappa \frac{(y_t)^{1+\frac{1}{\varepsilon}}}{1 + \frac{1}{\varepsilon}} - \theta_t \phi'_t(\theta_t) \right] dt \middle| \theta_0 \right\} \quad (\text{FOA}) \end{aligned}$$

Eliminate consumption from the problem by plugging the definition of ex ante utility at time zero into the feasibility constraint (27). The feasibility constraint then becomes:

$$\int_0^\infty \left( v(\theta_0) + \mathbb{E}_0 \left\{ \int_0^{\mathcal{T}_R} e^{-\rho t} \left[ \kappa \frac{(y_t)^{1+\frac{1}{\varepsilon}}}{1 + \frac{1}{\varepsilon}} + \phi_t \right] dt \middle| \theta_0 \right\} \right) dF(\theta_0) \leq \int_0^\infty \mathbb{E}_0 \left\{ \int_0^{\mathcal{T}_R} e^{-\rho t} y_t dt \middle| \theta_0 \right\} dF(\theta_0) \quad (33)$$

Denote by  $\lambda$  the multiplier on the new feasibility constraint (33). If  $v(\theta_0)$  is interior, the FOC on  $v$ :  $\alpha(\theta_0) f(\theta_0) - \lambda f(\theta_0) = 0$  integrated over  $\Theta_0$  yields  $\lambda = 1$ . The problem is then to

maximize the Lagrangian

$$\int_0^\infty \alpha(\theta_0) v(\theta_0) dF(\theta_0) - \left[ \int_0^\infty \left( v(\theta_0) + E_0 \left\{ \int_0^{\mathcal{T}_R} e^{-\rho t} \left[ \kappa \frac{\left( \frac{y_t}{\theta_t} \right)^{1+\frac{1}{\varepsilon}}}{1+\frac{1}{\varepsilon}} + \phi_t \right] dt \middle| \theta_0 \right\} \right) dF(\theta_0) \right. \\ \left. - \int_0^\infty E_0 \left\{ \int_0^\nu e^{-\rho t} y_t dt \middle| \theta_0 \right\} dF(\theta_0) \right]$$

subject to the incentive constraints from the FOA (32) and the law of motion of productivity (25). By partial integration,

$$\int_0^\infty v(\theta_0) dF(\theta_0) = \int_0^\infty \frac{1 - F(\theta_0)}{f(\theta_0)} v_\theta(\theta_0) dF(\theta_0) + \lim_{\theta \rightarrow 0} v(\theta) \\ \int_0^\infty \alpha(\theta_0) v(\theta_0) dF(\theta_0) = \int_0^\infty \frac{1 - \Lambda(\theta_0)}{f(\theta_0)} v_\theta(\theta_0) dF(\theta_0) + \lim_{\theta \rightarrow 0} v(\theta)$$

Eliminating  $v$  from the Lagrangian using partial integration and the expression of  $v_\theta$  from the IC constraint, the planner's problem becomes

$$\int_0^\infty E_0 \left\{ \int_0^{\mathcal{T}_R} e^{-\rho t} \left[ y_t - \kappa \frac{\left( \frac{y_t}{\theta_t} \right)^{1+\frac{1}{\varepsilon}}}{1+\frac{1}{\varepsilon}} \left[ 1 + \left( 1 + \frac{1}{\varepsilon} \right) \frac{\Lambda(\theta_0) - F(\theta_0)}{f(\theta_0)} \frac{1}{\theta_0} \right] - \left[ \phi_t - \frac{\Lambda(\theta_0) - F(\theta_0)}{f(\theta_0)} \frac{\theta_t}{\theta_0} \phi'_t(\theta_t) \right] \right] dt \middle| \theta_0 \right\} dF(\theta_0) \quad (34)$$

The FOC for  $y_t$  implies that the labor wedge is time-invariant and depends only on initial heterogeneity and the welfare weights.

$$\frac{\tau_t^L}{1 - \tau_t^L} = \frac{\tau(\theta_0)}{1 - \tau(\theta_0)} = \left( 1 + \frac{1}{\varepsilon} \right) \frac{1}{\theta_0} \frac{\Lambda(\theta_0) - F(\theta_0)}{f(\theta_0)}$$

Since  $y_t^{fb} - \kappa \frac{\left( \frac{y_t^{fb}}{\theta_t} \right)^{1+\frac{1}{\varepsilon}}}{1+\frac{1}{\varepsilon}} = \frac{\theta_t^{1+\varepsilon}}{\kappa^\varepsilon(1+\varepsilon)}$  and  $y_t^{sb} - \kappa \frac{\left( \frac{y_t^{sb}}{\theta_t} \right)^{1+\frac{1}{\varepsilon}}}{1+\frac{1}{\varepsilon}} \left[ 1 + \left( 1 + \frac{1}{\varepsilon} \right) \frac{\Lambda(\theta_0) - F(\theta_0)}{f(\theta_0)} \frac{1}{\theta_0} \right] = (1 - \tau(\theta_0))^\varepsilon \frac{\theta_t^{1+\varepsilon}}{\kappa^\varepsilon(1+\varepsilon)}$ , we can replace  $y_t^{sb}$  in the planner's objective (34) to obtain

$$\max_\nu \int_{\underline{\theta}}^\infty E \left\{ \int_0^{\mathcal{T}_R} e^{-\rho t} \left[ (1 - \tau(\theta_0))^\varepsilon [y_t^{fb} - \kappa \frac{\left( \frac{y_t^{fb}}{\theta_t} \right)^{1+\frac{1}{\varepsilon}}}{1+\frac{1}{\varepsilon}}] - \left[ \phi_t - \frac{\tau(\theta_0)}{1 - \tau(\theta_0)} \frac{\varepsilon}{1 + \varepsilon} \theta_t \phi'_t(\theta_t) \right] \right] dt \right\} dF(\theta_0)$$

□

The normalization of Pareto weights and the assumption of nonincreasing weights implies that  $\Lambda(\theta_0) - F(\theta_0)$  is always nonnegative. The labor wedges are therefore nonnegative. In the risk-neutral case, with GBM productivity, the labor wedges depend only on the inverse intensive Frisch elasticity of labor supply, initial heterogeneity, and the welfare weights of the planner. Because there is no income effect, consumption can be allocated freely over time without distorting the labor margin.

In the context of private information, labor distortions are such that the flow utility of consumption and disutility of labor are lower than in the first best. This is captured by the factor  $(1 - \tau(\theta_0))^\varepsilon < 1$  in front of  $[y_t^{fb} - \kappa \frac{\left( \frac{y_t^{fb}}{\theta_t} \right)^{1+\frac{1}{\varepsilon}}}{1+\frac{1}{\varepsilon}}]$  in the planner's objective. These labor distortions create incentives for the agents to retire early. However, the virtual fixed cost

either increases or decreases depending on the sign of  $\phi'_t(\theta_t)$ .

If  $\phi'_t$  is negative, the virtual fixed cost increases compared to the first best. Its effect goes in the same direction as the decrease in output  $y$ , and agents retire earlier than in the first best. Therefore, if  $\phi'_t$  is negative, all agents retire earlier in the second best than in the first best. In addition, retirement is a cutoff rule. If  $\phi'_t$  is positive, the virtual fixed cost decreases compared to the first best and depends negatively on the intensive Frisch elasticity of labor and the labor wedge. Its effect goes in the opposite direction as the decrease in  $y$ . Once we have solved the retirement decision problem in the first-best case, the derivation of the analogous rule for the second-best scenario is relatively simple. Dividing the planner's objective by  $(1 - \tau(\theta_0))^\varepsilon$ , one can observe that the choice of the retirement rule in the second best is equivalent to the choice of the retirement rule in the first best when the fixed utility cost is replaced by a virtual cost  $\tilde{\phi}$ , defined as  $\tilde{\phi}(t, \theta_t) = \frac{\phi(t, \theta_t)}{(1 - \tau(\theta_0))^\varepsilon} (1 - \frac{\tau(\theta_0)}{1 - \tau(\theta_0)} \frac{\varepsilon}{1 + \varepsilon} \varepsilon_{\phi, \theta}(\theta_t, t))$ . In contrast to the first-best case, the retirement rule depends on initial productivity. Define  $S(\tau(\theta_0), t) \equiv \tilde{\phi}(t, \theta_t)/\phi(t, \theta_t)$ , and  $S(\tau(\theta_0)) \equiv \tilde{\phi}(\theta_t)/\phi(\theta_t)$  when  $\phi$  is time-invariant. The following proposition summarizes the results on the second-best retirement decision.

**Proposition 14.** *(Second-best retirement decision)*

1. *There exists a time-dependent and initial productivity-dependent deterministic retirement threshold  $\theta_R^{sb}(t, \theta_0)$  such that  $\mathcal{T}_R^{sb} = \inf\{t; \theta_t \leq \theta_R^{sb}(t, \theta_0)\}$ .*
2. *Set  $\phi(\theta) = \phi_1 \theta^{1+1/\varepsilon_\phi} + \phi_0$ . In the infinite horizon limit,  $T = +\infty$ , the retirement thresholds are time invariant  $\hat{\theta}_R^{sb} : \Theta_0 \mapsto \mathbb{R}^{+*}$ ,  $\mathcal{T}_R^{sb} = \inf\{t; \theta_t \leq \theta_R^{sb}(\theta_0)\}$  and*

$$\theta_R^{sb}(\theta_0) = \theta_R^{fb} S(\tau(\theta_0))^{\frac{1}{\varepsilon}}$$

3. *If  $\phi_1 \leq 0$ , retirement occurs earlier in the second best than in the first best for all agents  $\theta_R^{sb}(t, \theta_0) \geq \theta_R^{fb}(t)$ . If  $\phi_1 > 0$ , a criterion for whether retirement happens early or is delayed compared to the first best is*

$$S(\theta_0) = \frac{1}{(1 - \tau(\theta_0))^\varepsilon} (1 - \frac{\tau(\theta_0)}{1 - \tau(\theta_0)} \frac{1 + 1/\varepsilon_\phi}{1 + 1/\varepsilon})$$

*For a given  $T < +\infty$ , retirement occurs earlier in the second best than in the first best:  $\theta_R^{sb}(t, \theta_0) \geq \theta_R^{fb}(t)$  for all  $t \leq T$  if and only if  $S(\theta_0) \geq 1$ .*

Point 1 of the proposition highlights that the retirement thresholds depend on the initial productivity of the agents. Again, the option of continued work compared to retiring is negative at retirement. The second point gives an explicit formula for the optimal retirement

threshold at the infinite horizon as in the discussion after Corollary 10.<sup>29</sup> Point 2 gives an explicit expression for the retirement thresholds at the infinite horizon.

Point 3 of the proposition states that if the fixed utility cost is increasing in productivity, there is a force that pushes for delayed retirement. High types have a high fixed cost and lower information rents than in the case when the fixed cost is independent of productivity. This creates an effect that goes in the opposite direction of the income tax. Depending on the strength of this effect, retirement may occur early or be delayed with respect to the first best. The proposition shows that the relative weight of the two forces depends on the criterion  $S$  that, in turn, depends on the intensive Frisch elasticity of labor supply, the elasticity of the fixed cost with respect to the wage, and the welfare weights of the planner. This criterion allows one to determine which productivity types should be induced to retire before  $S(\theta_0) \geq 1$  or after  $S(\theta_0) < 1$  the first best.

## A.7 Implementation of the Discrete-Time Optimum: Discussion

The wedges characterized in Section 4 are properties of the constrained-efficient allocation, not tax instruments. This subsection discusses what a tax system implementing them looks like.

First, the optimum is history dependent. The recursive formulation of Section 3 makes the dependence explicit: allocations are functions of promised utility and the marginal promise, which encode the full history of income reports. Standard decentralizations of dynamic Mirrlees optima—the insurance-annuity implementation of Farhi and Werning (2013) or the nonlinear capital taxes of Werning (2011)—inherit this history dependence in both the retirement benefit and the labor income tax. Appendix C.8 measures how much of the optimal wedge variation is spanned by progressively richer, less history-dependent tax schedules, from statutory age-and-health schedules to earnings-tested ones.

Second, the extensive margin is priced by retirement consumption. In our environment, the planner provides participation incentives through the schedule of consumption promised upon exit as a function of the exit age—the model counterpart of claiming-age-dependent retirement benefits. This instrument, rather than negative marginal tax rates, sustains positive labor wedges alongside voluntary retirement.

Third, the savings margin interacts with mortality. The negative-wedge region of Proposition 6 maps into savings instruments that become more generous with age and with poorer health—in practice, age- and income-conditioned preferential treatment of retirement savings accounts, with asset or earnings tests playing the role that consumption observability plays in

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<sup>29</sup>There is no concern over immiseration at the infinite horizon here since, with risk neutrality in consumption, consumption is not pinned down by FOCs.



the mechanism, as in Golosov and Tsyvinski's (2006) asset-tested disability insurance. In the quantification of Appendix C.8, an earnings-conditioned savings instrument captures nearly two-thirds of the optimal savings wedge variation, more than any other simple instrument.

Appendix A.8 develops a complete implementation in a continuous-time version of the model with risk neutrality in consumption, in which retirement benefits depend only on lifetime income and the retirement age (using Kruse and Strack 2015).

## A.8 Implementation with a Simple Social Security Program

When can one reduce the history dependence of the optimal policies? In this subsection, we show that in the limit case of workers who are risk neutral in consumption, the optimal policies can be implemented by a retirement benefit system that looks similar to the US Social Security system (in that it depends on lifetime income and retirement age) and by a history-independent labor income tax. To construct this implementation, we proceed in two steps. First, we construct retirement-age-dependent post-retirement transfers that replicate the effects of the shadow fixed cost. Given optimal hours and said transfers, the agent's private retirement decision would coincide with the optimal retirement decision. Second, using these post-retirement transfers and a labor wedge, we construct a social security system and a history-independent income tax that implement the optimum.

### A.8.1 Shadow Fixed Costs

**Definition 15.** We define the shadow fixed cost as the change in fixed cost  $\tilde{\phi}_t = (1 + \tau_t^\phi)\phi(\theta_t)$  that makes the agent privately choose the second-best retirement decision  $\mathcal{T}_R^*$  given  $\{c_t^*, y_t^*, \tilde{\phi}_t\}$ , that is,

$$\mathcal{T}_R^* = \arg \max_{\mathcal{T}_R} v_t^{lf}(\mathcal{T}_R; \{c_t^*, y_t^*, (1 + \tau_t^\phi)\phi(\theta_t)\})$$

The change in fixed utility cost that would make the agent privately choose the second-best retirement decision given  $\{y_t^*\}$  is:

$$\tau_t^\phi \phi(\theta_t) = \underbrace{\tau_0^L \frac{\varepsilon}{1 + \varepsilon} y_t^*}_{\text{increased shadow fixed cost from labor wedge}} - \underbrace{\frac{\tau_0^L}{1 - \tau_0^L} \frac{\varepsilon}{1 + \varepsilon} \varepsilon_{\phi, \theta}(\theta_t)}_{\text{net shadow fixed cost}} \phi(\theta_t), \quad (35)$$

where  $\varepsilon_{\phi, \theta}(\theta_t)$  is the elasticity of the fixed utility cost with respect to productivity. The first term is a positive fixed cost and comes from the fact that the labor wedge distorts retirement downward. The net shadow fixed cost  $\tau_t^R$  corrects for this effect;  $(\tau_t^R - \tau_t^\phi)\phi(\theta_t) = -\tau_0^L \frac{\varepsilon}{1 + \varepsilon} y_t^*$  and is equal to the second term of (35) in equilibrium.

### A.8.2 Post-Retirement Transfers

Recall from A.6.2 that if agents are risk neutral in consumption, then consumption is undistorted and the labor wedge at age  $t$  is equal to the time-zero labor wedge  $\tau_L^t(\{\theta^t\}) = \tau_L^0(\theta_0)$ , where  $\tau_L^0(\theta_0)$  is determined by the government's redistributive motive in the initial period. Lemma 16 below gives general conditions on the distribution of initial heterogeneity such that there exist government Pareto weights that rationalize a constant optimal labor wedge,  $\tau_L^t(\{\theta^t\}) = \tau_L$ . More specifically, these conditions are satisfied if initial productivity is Pareto distributed for a range of social welfare functions, from utilitarian (labor wedge equal to zero) to Rawlsian (largest labor wedge) and a Rawlsian-utilitarian mixture (intermediate levels of labor wedge).<sup>30</sup>

**Lemma 16.** *For any smooth distribution  $f$  such that  $\theta_0 f(\theta_0) \rightarrow_{\theta_0 \rightarrow \infty} 0$  (which is satisfied by all densities that have a mean), for all  $\tau$ , there exist Pareto weights that are smooth on the support of  $f$  but put weight on the min of the support of  $f$ , such that the optimal tax is constant,  $\tau(\theta_0) = \tau$ .*

*Proof.*  $\Lambda(\theta) = \int_0^\theta \alpha(\theta_0) dF(\theta_0)$ . We want in the interior  $(1 + \frac{1}{\varepsilon})(\Lambda(\theta) - F(\theta)) = \frac{\tau}{1-\tau} \theta f(\theta)$ . The limit condition in the lemma comes from the fact that  $\Lambda(\infty) - F(\infty) = 0$ . Now, the derivative is a necessary condition in the interior:

$$(1 + \frac{1}{\varepsilon})[\alpha(\theta) - 1] = \frac{\tau}{1-\tau}[1 + \underline{\theta}].$$

Thus,

$$\alpha(\theta) = \left( \frac{\varepsilon}{1+\varepsilon} \right) \left( 1 + \frac{\tau}{1-\tau} [1 + \frac{\theta f'(\theta)}{f(\theta)}] \right)$$

with a mass at the bottom support  $\underline{\theta}$  such that the sum of weights adds up to 1, that is,

$$\alpha_{\underline{\theta}} = \left( \frac{\varepsilon}{1+\varepsilon} \right) \left( 1 - \int_{\underline{\theta}}^\infty \alpha(\theta_0) dF(\theta_0) \right) = \left( \frac{\varepsilon}{1+\varepsilon} \right) \left( -\frac{\tau}{1-\tau} (1 + \int_{\underline{\theta}}^\infty \theta f'(\theta) d\theta) \right) = \frac{\varepsilon}{1+\varepsilon} \frac{\tau}{1-\tau} \underline{\theta} f(\underline{\theta})$$

and, of course, the condition that the weights are positive everywhere, that is,

$$\frac{\theta f'(\theta)}{f(\theta)} \geq -\frac{1}{\tau}$$

In particular, if the support of the distribution starts at zero, there is no mass at zero. If the distribution is Pareto  $(\underline{\theta}, \alpha)$  such that  $f(\theta) = \frac{a\theta^a}{\theta^{a+1}}$ , then  $\frac{\theta f'(\theta)}{f(\theta)} = -1 - a$ , and the weights are constant:

$$\alpha(\theta) = \left( \frac{\varepsilon}{1+\varepsilon} \right) \left( 1 - \frac{\tau}{1-\tau} a \right)$$

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<sup>30</sup>If the distribution is Pareto with shape parameter  $a$  on  $[\underline{\theta}, \infty)$  and the government puts weight  $\alpha_{\underline{\theta}}$  at  $\underline{\theta}$  and equal weights on  $(\underline{\theta}, \infty)$ , then the labor wedge is  $\tau_L = (1 + \frac{1}{\varepsilon}) \frac{1}{a} \frac{\alpha_{\underline{\theta}}}{1+\alpha_{\underline{\theta}}}$ . The labor wedge is  $\tau_L = 0$  if  $\alpha_{\underline{\theta}} = 0$  (utilitarian) and  $\tau = (1 + \frac{1}{\varepsilon}) \frac{1}{2a}$  if  $\alpha_{\underline{\theta}} = 1$  (Rawlsian) and is increasing in  $\alpha_{\underline{\theta}}$ .

and

$$\alpha_{\underline{\theta}} = \frac{\varepsilon}{1 + \varepsilon} \frac{\tau}{1 - \tau} a$$

One can also invert this to show that with a Pareto distribution and the types of Pareto weights that we call Rawlsian-utilitarian, the tax is given by:

$$\tau = \left(1 + \frac{1}{\varepsilon}\right) \frac{1}{a} \frac{\alpha_{\underline{\theta}}}{1 + \alpha_{\underline{\theta}}}$$

which is a zero marginal tax  $\tau = 0$  if  $\alpha_{\underline{\theta}} = 0$  and utilitarian and is  $\tau = \left(1 + \frac{1}{\varepsilon}\right) \frac{1}{2a}$  if Rawlsian.  $\square$

If the government sets a flat labor income tax equal to  $\tau$  and a post-retirement transfer  $\pi$  is a function of retirement age, then the agent chooses hours conditional on work optimally,  $y_t = y_t^*$ , and his private retirement decision satisfies

$$\max_{\nu} \mathbb{E} \left\{ \int_0^{\nu} e^{-\rho t} \left[ (1 - \tau) y_t^* - h\left(\frac{y_t^*}{\theta_t}\right) - \phi(\theta_t) \right] dt + e^{-\rho \nu} \pi(\nu) \right\} \quad (36)$$

The planner's choice of the optimal retirement decision is different from the agent's private choice in two respects. First, because of labor income taxes, the government values output relative to the fixed cost more than the agent does. Second, the government wants to distort the fixed cost faced by the agent because of the redistributive value of the net shadow fixed cost. The transfer  $\pi$  implements the optimal retirement decision if  $\mathcal{T}_R^*$  is a solution to the agent's private retirement decision problem (36).

Under Assumption 3, we construct  $\pi$  by evaluating the agent's expected utility at the productivity process reflected at the second-best retirement cutoff  $\theta_R^*(t)$ . Intuitively, the reflected productivity is a process that equals productivity as long as it stays above the cutoff. Once productivity falls below the cutoff and the planner would want the agent to retire, the reflected process follows its own dynamics and is defined so that it stays above the cutoff at all times.

We started with the definition of a reflected process above the second-best retirement boundary.

**Definition 17.** (*Reflected process*) Let  $\theta_R^* : [0, T] \rightarrow \mathbb{R}$  be a càdlàg function. A process  $\{\tilde{\theta}_t\}_t$  on  $(\Omega, \mathcal{F}, \mathcal{P})$  is called a reflected version of  $\{\theta_t\}_t$  with barrier  $\theta_R^*$  if it satisfies the following conditions:

1.  $\tilde{\theta}_t$  is constrained to stay above  $\theta_R^*$ : For every  $t \in [0, T]$ , we have  $\tilde{\theta}_t \geq \theta_R^*(t)$  a.s.
2. Until  $\{\theta_t\}$  hits the barrier, both processes coincide: For every  $0 \leq t < \mathcal{T}_R^*$ , we have  $\theta_t = \tilde{\theta}_t$  a.s.

3.  $\tilde{\theta}$  is always higher than  $\theta$ : For every  $t \in [0, T]$ , we have  $\tilde{\theta}_t \geq \theta_t$  a.s.

4. When  $\{\theta_t\}$  hits the barrier,  $\{\tilde{\theta}_t\}_t$  is at  $\theta_R^*$ :  $\tilde{\theta}_{\mathcal{T}_R^*} = \theta_R^*(\mathcal{T}_R^*)$  a.s.

The next proposition ensures the existence of a reflected version of the productivity process at the second-best retirement boundary for our GBM diffusion process.

**Proposition 18.** *Let  $\theta_R^* : [0, T] \rightarrow \mathbb{R}$  be a càdlàg function. If  $\{\theta_t\}_t$  has no jumps, then there exists a version of  $\{\theta_t\}_t$  reflected at  $\theta_R^*$ .*

The conditions of this proposition are just the necessary ones to obtain our result. In general, there exist reflected versions at càdlàg boundaries for an extensive class of processes, including those with downward jumps.

**Proposition 19.** *Suppose Assumption 3 holds. Define as  $\{\tilde{\theta}_t\}_t$  the reflected process above  $\theta_R^*(t)$ ; then,*

$$\pi(t) = \mathbb{E}_t \left\{ \int_t^T e^{-\rho(s-t)} \left[ (1-\tau)\tilde{y}_s^* - h\left(\frac{\tilde{y}_s^*}{\tilde{\theta}_s}\right) - \phi(\tilde{\theta}_s) \right] ds \right\}$$

*implements the second-best retirement decision, where  $\tilde{y}_t^* = (1-\tau)^\varepsilon \frac{\tilde{\theta}_t^{1+\varepsilon}}{\kappa^\varepsilon(1+\varepsilon)}$ .*

*Proof.* If the government sets a flat labor income tax equal to  $\tau$  from Lemma 16 and a post-retirement transfer  $\pi$  is a function of the retirement age, then the agent chooses his hours conditional on work optimally  $y_t = y_t^*$ , and his private retirement decision satisfies:

$$\max_\nu \mathbb{E} \left\{ \int_0^\nu e^{-\rho t} \left[ (1-\tau)y_t^* - h\left(\frac{y_t^*}{\theta_t}\right) - \phi(\theta_t) \right] dt + e^{-\rho\nu} \pi(\nu) \right\}$$

The occurrence of  $\pi$  is here to implement the wedges  $\tau_\phi$  with post-retirement transfers. From Proposition 14, we know that the second-best retirement decision is a cutoff rule,  $\theta_R^*$ . Then, Theorem 5 of Kruse and Strack (2015) applies and implies that

$$\pi(t) = \mathbb{E}_t \left\{ \int_t^T e^{-\rho(s-t)} \left[ (1-\tau)\tilde{y}_s^* - h\left(\frac{\tilde{y}_s^*}{\tilde{\theta}_s}\right) - \phi(\tilde{\theta}_s) \right] ds \right\}$$

implements the second-best retirement decision, where  $\tilde{y}_t^* = (1-\tau)^\varepsilon \frac{\tilde{\theta}_t^{1+\varepsilon}}{\kappa^\varepsilon(1+\varepsilon)}$ . □

The transfer achieves the implementation of the second-best retirement decision by doing the following. First, when the net shadow fixed cost and labor wedge result in distortions for delayed (resp., early) retirement, the planner provides a marginal change in the transfer that increases (resp., decreases) the option value of continued work of the agent until (resp., after) productivity falls to  $\theta_R^*(t)$ . Proposition 19 states that the marginal change in the optimal transfer is the agent's private value of work at a level of labor income that is constrained to

stay above the level of labor income that triggers retirement in the second best. In particular, if  $\pi$  implements  $\mathcal{T}_R^*$ , then a lump-sum transfer added to  $\pi$  implements  $\mathcal{T}_R^*$ . This will allow us to complement any smooth history-independent labor income tax with a history-dependent retirement benefit and a lump-sum transfer to implement the optimum.

**Proposition 20.** *Let  $T(y_t)$  be a differentiable history-independent labor income tax. There exist retirement benefits  $b$  and a lump-sum transfer  $t_0$  such that  $(T, b, t_0)$  implements the optimum. In addition,*

$$b(\nu, \{y_t\}) = \underbrace{\delta(\nu) \mathbb{E} \left\{ \int_0^{\mathcal{T}_R^*} e^{-\rho t} \tau y_t^* \right\}}_{\text{level around second best}} + \underbrace{\pi(\nu) - \delta(\nu) \mathbb{E}[e^{-\rho \mathcal{T}_R^*} \pi(\mathcal{T}_R^*)]}_{\text{deferral rate}} + \underbrace{f(\{y_s\})}_{\text{function of past earnings}}$$

for any retirement age, where  $e^{-\rho \mathcal{T}_R^*} f(\{y_s\}) = \int_0^{\mathcal{T}_R^*} e^{-\rho t} [T(y_t) - \tau y_t] dt$  and  $\delta(t) \equiv \frac{1 - e^{-\rho(T-t)}}{1 - e^{-\rho T}}$  is the lifetime equivalent of a stream of units of consumption from time  $t$  until death.

*Proof.* We start by setting the savings tax such that the agents are not willing to save privately. Because of risk neutrality in consumption, the savings tax can be set to zero. Given a history-independent income tax  $T(y_t)$  and a history-dependent lifetime retirement benefit  $b(\{y_t\}, \mathcal{T}_R)$ , the agent's consumption before retirement is  $c_t = y_t - T(y_t)$ , and after retirement, the NPV of consumption is  $c_{\mathcal{T}_R} g(\mathcal{T}_R) = b(\{y_t\}, \mathcal{T}_R)$ . The agent's private optimization problem is:

$$\max_{y_t, \nu} \int_{\underline{\theta}}^{\infty} \mathbb{E} \left\{ \int_0^{\nu} e^{-\rho t} [y_t - T(y_t) - h(\frac{y_t}{\theta_t})] - \phi_t(\theta) \right\} dt + e^{-\rho \nu} b(\{y_s\}, \nu) \Big\} dF(\theta_0)$$

We search for affordable benefits of the form

$$b(\{y_t\}, \nu) = -T(0)g(\nu) + \pi(\nu) + f(\{y_t\}, \mathcal{T}_R^*)$$

that guarantees that  $y_t = y_t^*$ , after which we will know that  $\nu = \mathcal{T}_R^*$  by construction of the transfer  $\pi$ . The necessary condition for optimal hours is, given  $\mathcal{T}_R^*$ ,

$$(1 - T'(y_t) + e^{-\rho(\mathcal{T}_R^* - t)} \frac{\partial b(\{y_s\}, \mathcal{T}_R^*)}{y_t}) = h(\frac{y_t^*}{\theta_t}) = 1 - \tau$$

Setting

$$e^{-\rho(\mathcal{T}_R^* - t)} \frac{\partial b(\{y_s\}, \mathcal{T}_R^*)}{y_t} = T'(y) - \tau$$

and integrating pathwise over  $y$ ,

$$e^{-\rho \mathcal{T}_R^*} f(\{y_s\}, \mathcal{T}_R^*) = \int_0^{\mathcal{T}_R^*} e^{-\rho t} [T(y_t) - \tau y_t] dt$$

guarantees that  $y_t = y_t^*$ . The transfer  $\pi(\mathcal{T}_R)$  guarantees that  $\nu = \mathcal{T}_R^*$  as long as it is affordable by the aggregate resource constraint. Now observe that if  $\pi$  implements the second-best retirement decision, any lump-sum transfer added to  $\pi$  also implements the second-best retirement decision. This will allow us to adjust the lump-sum transfer  $-T(0)$  until the aggregate resource constraint is satisfied in equilibrium.

$$T(0)g(0) + E\left\{ \int_0^{\mathcal{T}_R^*} e^{-\rho t} T(y_t^*) \right\} = E[e^{-\rho \mathcal{T}_R^*} \pi(\mathcal{T}_R^*)] + E[e^{-\rho \mathcal{T}_R^*} f(\{y_s^*\}, \mathcal{T}_R^*)]$$

Replacing the expression of  $-T(0)$  in  $b$  yields that, for any  $\mathcal{T}_R$ ,

$$\begin{aligned} b(\{y_t\}, \nu) = & \underbrace{\frac{g(\nu)}{g(0)} \left( E\left\{ \int_0^{\mathcal{T}_R^*} e^{-\rho t} T(y_t^*) \right\} - E[e^{-\rho \mathcal{T}_R^*} f(\{y_s^*\}, \mathcal{T}_R^*)] \right)}_{\text{level around second best corrected for tax distortion}} \\ & + \underbrace{\pi(\nu) - \frac{g(\nu)E[e^{-\rho \mathcal{T}_R^*} \pi(\mathcal{T}_R^*)]}{g(0)}}_{\text{actuarial adjustment}} \\ & + \underbrace{f(\{y_s\}, \mathcal{T}_R^*)}_{\text{function of past earnings}} \end{aligned}$$

which simplifies from the expression of  $f$  to:

$$\begin{aligned} b(\{y_t\}, \nu) = & \underbrace{\frac{g(\nu)}{g(0)} \left( E\left\{ \int_0^{\mathcal{T}_R^*} e^{-\rho t} \tau y_t^* \right\} \right)}_{\text{level around second best corrected for tax distortion}} \\ & + \underbrace{\pi(\nu) - \frac{g(\nu)E[e^{-\rho \mathcal{T}_R^*} \pi(\mathcal{T}_R^*)]}{g(0)}}_{\text{actuarial adjustment}} \\ & + \underbrace{f(\{y_s\}, \mathcal{T}_R^*)}_{\text{function of past earnings}} \end{aligned}$$

We rename without loss of generality the income tax  $T$  to a tax on labor income without the lump-sum transfer. The benefits  $b(\{y_t\}, \nu)$ , combined with labor income tax and the lump-sum transfer, implement the planner's optimum.  $\square$

## B Data Appendix

### B.1 Data

Our analysis primarily relies on data from the Panel Study of Income Dynamics (PSID), supplemented with data from the Medical Expenditure Panel Survey (MEPS) for analyses related to medical spending. This section describes the construction of the sample from these sources and the methods used to define health and disability status across the two datasets.

**Panel Study of Income Dynamics:** The PSID is a longitudinal survey of a representative sample of the US population, collecting detailed information on labor market outcomes, income, health, and wealth. Initiated in 1968 with interviews of approximately 4,800 families, the PSID has continuously followed the original respondents and their descendants, with interviews conducted annually until 1997 and biennially thereafter. The original sample comprises two components: 1,872 low-income families drawn from the Survey of Economic Opportunity (SEO) and 2,930 families selected by the Survey Research Center (SRC) at the University of Michigan to represent the general US population.

In constructing our sample, we restrict to individuals from the SRC component who participated in interviews between 1968 and 2015. To make the sample more representative of the US population, we exclude the SEO sample, as most of these families were dropped from the survey after 1997. Our analysis focuses on individuals born between 1920 and 1940, a cohort that experienced a similar set of Social Security rules and for whom the PSID provides sufficient data observations across the life cycle.

Table 6: PSID Sample Selection

| Selection                              | Individuals | Observations |
|----------------------------------------|-------------|--------------|
| Initial sample                         | 76,880      | 3,075,200    |
| Non-SEO sample                         | 36,430      | 1,457,200    |
| Household heads and spouse if present  | 21,225      | 355,314      |
| Age 20 to 90                           | 21,031      | 350,617      |
| No missing labor or health information | 19,433      | 222,319      |
| Born between 1920 and 1940             | 2,248       | 44,404       |

*Notes:* Description of our Panel Study of Income Dynamics subsample selection. We focus on the 1920 to 1940 birth cohort over the age of 20.

Table 6 reports the sample selection and resulting sample size. The final sample includes 2,248 individuals and 44,404 person–year observations. We use data from the PSID to estimate health status transitions, wage profiles and dynamics, survival probabilities, the initial distribution over state variables, and the moment conditions that the model is estimated to match.

**Medical Expenditure Panel Survey:** The MEPS is a nationally representative survey of families, individuals, medical providers, and employers across the United States. The survey covers individuals of all ages, although age is top-coded at 85. Medical spending is reported at the individual level and cross-validated with information from medical providers and insurance companies, which enhances data accuracy.

We use data from the 1999–2012 waves of the MEPS, restricting the sample to household heads aged 20 and older. We exclude observations with missing information on age, medical spending, health insurance coverage during working years, or health status. The final sample includes 120,731 individuals and 211,709 person–year observations, as summarized in Table 7. We use the MEPS data to estimate out-of-pocket medical expenditures for a representative population.

Table 7: MEPS Sample Selection

| Selection                           | Individuals | Observations |
|-------------------------------------|-------------|--------------|
| Initial sample                      | 259,263     | 491,795      |
| Household heads                     | 130,978     | 231,866      |
| Age above 20                        | 125,587     | 222,495      |
| With insurance information at 25–65 | 121,385     | 213,005      |
| Nonmissing health status            | 120,731     | 211,709      |

*Notes:* Description of our Medical Expenditure Panel Survey subsample selection. We keep household heads aged 20 and older and exclude observations with missing information.

## B.2 Health and Disability Measurement

In the PSID, health and disability status are constructed based on responses to a series of self-reported work limitation questions. In each wave, respondents are asked the following: (1) *Do you have any physical or nervous condition that limits the type of work or the amount of work that you can do?* (2) *Does this condition keep you from doing some types of work?* (3) *For work you can do, how much does it limit the amount of work you can do—a lot, somewhat, or just a little?*

The first question allows Yes or No responses. The second question, available beginning in 1986, offers Yes, No, or *Can do nothing* as response options. The third question has been coded differently over time: Since 1976, the response options have been *Not at all*, *Somewhat*, *Just a little*, or *A lot*; before 1976, they were *We can’t work*, *It limits me a lot*, *Some, not much*, or *Limitation but not on work*. Since the second question became available in 1986, for observations from before that year, health status is defined based only on the responses to the first and third questions.



Following [Low and Pistaferri \(2015\)](#), we classify individuals into three health categories: good health ( $h_t = 0$ ) if they respond “No” to the first question or “Not at all” to the third question; poor health ( $h_t = 1$ ) if they answer “Yes” to the first question and “Somewhat” or “Just a little” to the third question; and with a disability ( $h_t = 2$ ) if they answer “Yes” to the first question, “Can do nothing” to the second question, and “A lot” to the third question, in accordance with the Social Security Administration (SSA) criterion for disability insurance qualification. Alternatively, [Hosseini et al. \(2022\)](#) construct a frailty index to measure health status using PSID survey questions available since 2003, while the PSID has also included self-reported health rankings since 1984. However, these measures are not suitable for our analysis, as we require consistent information on health dynamics over the entire life cycle. The restricted availability of these measures precludes their use in our analysis.

In the MEPS, health status is based on self-reported health rankings, following [Pashchenko and Porapakarm \(2019\)](#). Each year  $t$  includes three survey waves, and in each wave, respondents rate their health on a five-point scale: (1) *excellent*, (2) *very good*, (3) *good*, (4) *fair*, or (5) *poor*. We define individuals as being in good health ( $h_t = 0$ ) if their health rank falls in categories 1–3 across all three waves; as with a disability ( $h_t = 2$ ) if they report “poor” in any wave; and as being in poor health ( $h_t = 1$ ) otherwise. [Table 8](#) below compares the health measures across the two data sources and demonstrates their consistency.

Table 8: Health Measures Over Ages – Comparison of Two Datasets

| Health Status         | Ages 50–60 |      | Ages 60–70 |      | Ages 70 and older |      |
|-----------------------|------------|------|------------|------|-------------------|------|
|                       | MEPS       | PSID | MEPS       | PSID | MEPS              | PSID |
| With a disability (%) | 10         | 9.3  | 11.7       | 12.9 | 15                | 19.7 |
| Poor health (%)       | 21         | 10.6 | 23.6       | 17.0 | 28.7              | 19.3 |
| Good health (%)       | 69         | 80.1 | 64.7       | 70.1 | 56.3              | 60.0 |

*Data Source:* Panel Study of Income Dynamics and Medical Expenditure Panel Survey.

### B.3 First Step of Estimation

This section describes the procedures used to estimate the first-step inputs, including health transitions, survival probabilities, hourly wages, and out-of-pocket medical expenses directly from raw data.

**Health Transitions:** We estimate the evolution of health status,  $Pr(h_t = j|h_{t-1} = i)$  for  $i, j \in \{0, 1, 2\}$ , by running probit regressions of the indicator variable  $I(h_t = j|h_{t-1} = i)$  on a set of age dummies, using the sample with  $(h_{t-1} = i)$ . The predicted values from these

regressions provide estimates of the transition probabilities. We then linearly interpolate the estimated transition probabilities across ages so that the transition matrices change smoothly over the life cycle.

**Survival Probabilities:** Using data from the PSID, we estimate age- and health-dependent survival probabilities by running a logistic regression of an indicator for survival on a polynomial in age, previous health status, and their interactions whenever they are statistically significant. The estimated coefficients are reported in

[Table 9](#).

Table 9: Estimated Coefficients for the Survival Probabilities

|                                          | Coefficients | S.E.     |
|------------------------------------------|--------------|----------|
| Age <sup>2</sup>                         | -.0006***    | (0.0000) |
| Health <sub>t-1</sub> = 1                | -1.3198***   | (0.2288) |
| Health <sub>t-1</sub> = 2                | -3.1419***   | (0.4189) |
| Health <sub>t-1</sub> × Age <sup>2</sup> | 0.0109***    | (0.0032) |
| Constant                                 | 7.8801***    | (0.1373) |
| Observations                             | 132,070      |          |

*Notes:* The table shows estimated coefficients of a logistic regression using the Panel Study of Income Dynamics. Dependent Variable: indicator of survival. Robust standard errors in parentheses, clustered at the individual level.

**Out-of-Pocket Medical Expenses:** Out-of-pocket medical expenses are computed as total medical expenditures net of the amount covered by health insurance, based on data from the MEPS. Following the procedure of [Pashchenko and Porapakarm \(2019\)](#), we estimate the profiles of total medical expenditures by running weighted regressions on age and year dummies separately for each health group, applying both the cross-sectional and longitudinal weights. We similarly estimate the profiles of insurance coverage by regressing them on age and year dummies, separately by health status. For working-age individuals, the medical coverage is computed as the amount paid by insurance programs, while for individuals aged 65 and older, coverage reflects expenses paid by Medicare. Out-of-pocket expenditures are then computed as the product of estimated coinsurance rates and total medical expenses at each age and health status. Following [De Nardi et al. \(2024\)](#), we adjust the estimated medical expenditures by multiplying them by 1.60 for individuals younger than age 65 and by 1.90 for those aged 65 and older, to make medical spending consistent with the aggregate spending patterns reported in the National Health Expenditure Account. Finally, we smooth the resulting profiles by regressing them on a quadratic function of age.

**Hourly Wages:** Hourly wages are calculated as annual earnings divided by annual hours worked, based on data from the PSID. Respondents report their annual earnings and hours

worked for the previous year. To account for extreme wage values, observations with hourly wages below \$6.50 or above \$250 (in 2016 dollars) are excluded. Since wages are observed only for labor market participants, we adjust for selection bias in the observed wages by estimating age- and health-specific hourly wage profiles using the two-stage Heckman selection model (Heckman 1976), as adapted in recent studies, such as Guner et al. (2012). Some previous studies address this selection bias by matching profiles of labor market participants using the model to estimate the parameters of wage equations (French 2005; French and Jones 2011). We choose the Heckman selection model as it provides a good fit to the targeted moments while also being computationally efficient.

Specifically, in the first stage, we estimate the selection equation (labor force participation) using a probit regression on all PSID observations, generating an inverse Mill’s ratio. In the second stage, we estimate hourly wages by regressing them on age polynomials, the interactions of health and age polynomials, and the inverse Mill’s ratio obtained in the first stage. The coefficients from this two-step procedure are reported in Table 10.

To estimate the stochastic components of hourly wages  $(\rho, \sigma_\rho^2)$ , we use the wage residuals from the previous steps. We restrict the age range to between 25 and 65 and drop the top 0.5% of the residuals to mitigate the effect of extreme outliers on the variance. The estimation is performed with maximum likelihood and standard Kalman filter recursions. The estimated values for  $(\rho, \sigma_\rho^2)$  are (0.981, 0.016), which are consistent with the estimates from the standard literature.

**Initial Distribution:** To construct the initial distribution of the relevant state variables at age 30, we draw randomly from the empirical joint distribution of wages, health, assets, and labor force participation for individuals aged 28–32 in the target cohort, using data from the PSID. We adjust the mean of log wages to match the estimated wage profiles within each health group. For the initial Social Security wealth, we assume that all individuals have worked for five years, working 2,000 hours per year at the hourly wage rate at age 30, following the procedure used by French (2005), to impute initial values of average indexed monthly earnings. Table 11 summarizes the initial distribution of assets, wages, previous labor force participation, and health status. It shows that individuals in good health have higher wages and assets, on average, than those in poor health and with disabilities.

## B.4 Second Step of Estimation

**Target Profiles:** The target profiles include labor force participation by health status, hours worked conditional on participation by health status, mean and median assets, and mean rates of labor market reentry over the life cycle. We construct these profiles with the data from the PSID by estimating fixed-effects regressions, following the procedure outlined

Table 10: Estimated Coefficients for the Logarithm of Hourly Wages

|                           | Coefficients | S.E.      |
|---------------------------|--------------|-----------|
| <i>Wage Equation</i>      |              |           |
| Age                       | 0.0582***    | (0.00500) |
| Age <sup>2</sup>          | -0.00067***  | (0.00006) |
| Health×Age                | -0.0113***   | (0.00127) |
| Health×Age <sup>2</sup>   | 0.00013***   | (0.00002) |
| Constant                  | 1.447***     | (0.104)   |
| <i>Selection Equation</i> |              |           |
| Age                       | 0.0977***    | (0.00453) |
| Age <sup>2</sup>          | -0.00162***  | (0.00004) |
| Health                    | -1.371***    | (0.0682)  |
| Health×Age                | 0.0122***    | (0.00119) |
| Family Size               | 0.0225***    | (0.00369) |
| Birth Year                | -0.0123***   | (0.00080) |
| Constant                  | 23.79***     | (1.581)   |
| Inverse Mill's Ratio      | 0.182***     | (0.0482)  |
| Observations              | 64,941       |           |

*Notes:* Table shows estimated coefficients of two-stage Heckman selection model based on the Panel Study of Income Dynamics. Dependent variable of the wage equation: logarithm of wages. Dependent variable of selection equation: indicator of labor force participation. Robust standard errors in parentheses, clustered at the individual level. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 11: Summary Statistics for the Initial Conditions

|                          | Overall | Health Status |             |                   |
|--------------------------|---------|---------------|-------------|-------------------|
|                          |         | Good health   | Poor health | With a disability |
| Assets (in 2016 dollars) |         |               |             |                   |
| Mean                     | 60,241  | 61,388        | 42,022      | 30,289            |
| Standard Deviation       | 146,975 | 149,670       | 91,894      | 19,310            |
| Wages (in 2016 dollars)  |         |               |             |                   |
| Mean                     | 18.95   | 19.19         | 15.13       | 12.50             |
| Standard Deviation       | 9.32    | 9.32          | 8.29        | 6.98              |
| Participation            |         |               |             |                   |
| Mean                     | 0.96    | 0.97          | 0.89        | 0.45              |
| Standard Deviation       | 0.20    | 0.18          | 0.31        | 0.50              |
| Percentage               |         | 94.74         | 4.17        | 1.09              |
| Observations             | 4,849   | 4,594         | 202         | 53                |

*Data Source:* Panel Study of Income Dynamics.

in French (2005).

Labor force participation is measured as a dummy variable, taking the value of one if an individual's reported annual hours worked exceed 800 hours, thereby capturing both full-time and part-time employment. Hours worked are based on self-reports from the PSID, with respondents in survey year  $t$  reporting their hours worked during year  $t - 1$ .

Asset measurement includes real estate, farm or business equity, vehicles, stocks, mutual funds, individual retirement accounts (IRAs), Keogh plans, liquid assets, bonds, and investment trusts, net of mortgages and other debts. Pension wealth and Social Security wealth are excluded. We define assets as the sum of all these categories plus home equity net of any outstanding debts. Following Borella et al. (2018), we assume equal division of asset ownership between partners for aggregation purposes.

Asset data are available only for selected survey years starting in 1984. To impute asset values for years without direct observations, we estimate a fixed-effects regression of assets on age polynomials, log wages, family size, education, unemployment rates, health status indicators, interactions between log wages and the age polynomials, and interactions between health status and the age polynomials, education, family size, unemployment rates, and wages. We use both imputed and observed values to construct asset profiles.

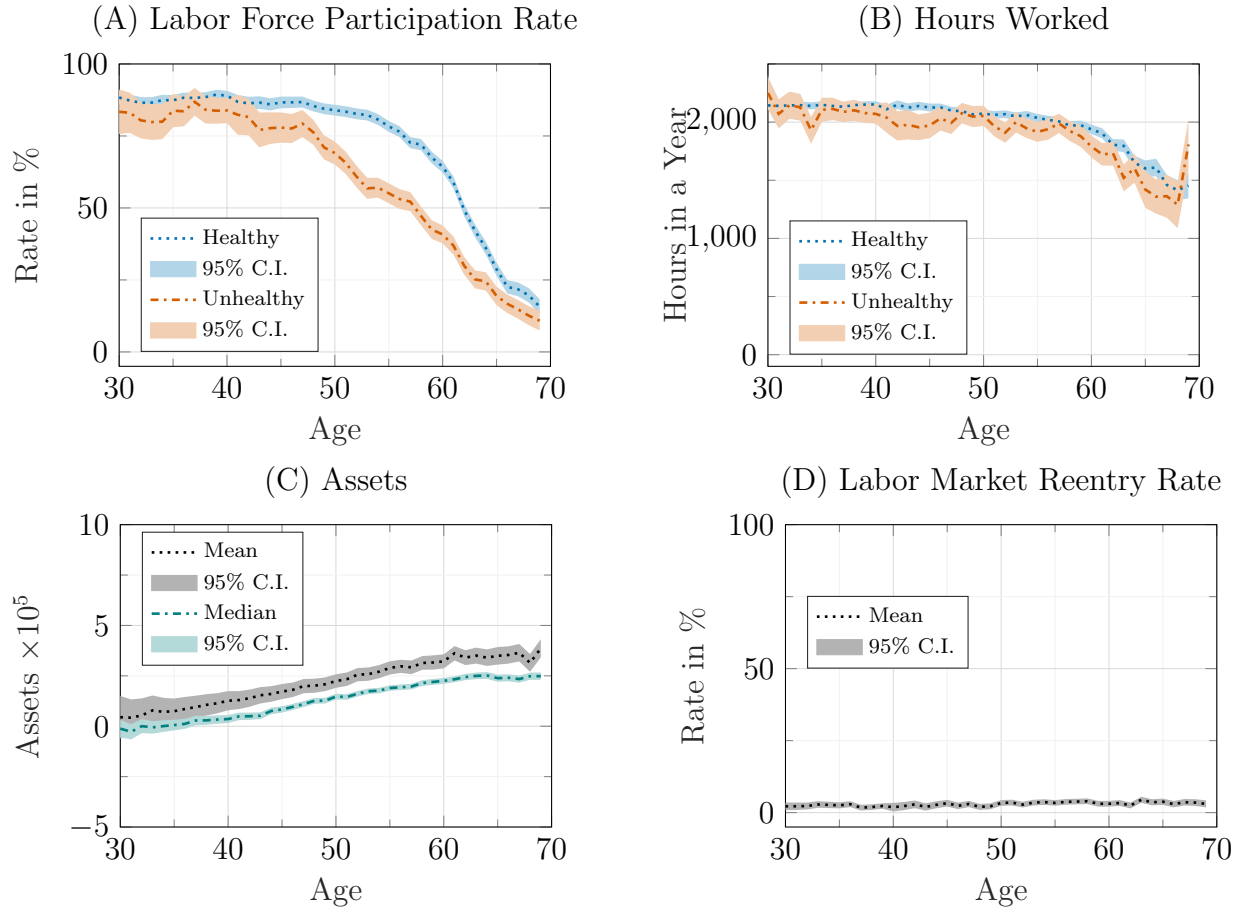
We estimate the target profiles by running the following fixed-effects regression model:

$$Z_{it} = f_i + \sum_{k=1}^T B_{gk} I\{age_{it} = k\} \times I\{h_{it} = 0\} + \sum_{k=1}^T B_{bk} I\{age_{it} = k\} \times I\{h_{it} \neq 0\} \\ + \sum_{f=1}^F B_f \text{familysize}_{it} + B_u U_t + u_{it}$$

where  $Z_{it}$  represents the outcome of interest, either hours worked or labor force participation, for individual  $i$  at age  $t$ ,  $f_i$  denotes an individual-specific effect,  $\text{familysize}_{it}$  is family size dummies,  $U_t$  is the unemployment rate, and  $\{\{B_{gk}\}_{k=1}^T, \{B_{bk}\}_{k=1}^T, \{B_f\}_{f=1}^F, B_u\}$  are parameters. To ensure the target profiles are not contaminated by family size, year (business cycle) effects, or individual-specific effects, we set family size to 3 and the unemployment rate to 6.5%. To control for birth year (cohort) and the correlation between the person-specific effect and health status, we use the mean individual-specific effect for individuals who are aged 50, in average health at age 50, and born in 1935. For the asset and labor market reentry profiles, we do not condition on health status.

The resulting estimated profiles are displayed in Figure 11. Healthy individuals work significantly more than unhealthy individuals over the life cycle, particularly after age 50. Labor market reentry rates are low in the data, remaining below 5% throughout the observed period.

Figure 11: Data Profiles



*Notes:* Panel A shows the labor force participation by health status. Panel B shows the hours worked conditional on participation by health status. The blue and red lines represent healthy and unhealthy individuals. Panel C shows the mean and median assets. Panel D shows the mean labor market reentry rate. Data profiles are estimated with the Panel Study of Income Dynamics. The shaded region represents the 95% confidence intervals.

## C Quantitative Appendix

### C.1 The Estimated Economy: Setup, Estimation, and Fit

This appendix estimates the primitives of the economy of Section 2—the nonseparable preferences with health-dependent fixed participation costs and a reentry cost, the AR(1) ability process, and the health-transition, wage, and mortality profiles—in the status quo US policy environment: Social Security (SS), Medicare, disability insurance, pensions, taxes, and incomplete insurance markets. Section 5 of the main text uses the estimated model to quantify the impact of health on retirement, and section 6 computes the constrained-efficient optimum of planning problem at these estimates.

#### C.1.1 Setup

Our modeling of retirement—labor force participation and SS claiming—builds on [French \(2005\)](#) and [French and Jones \(2011\)](#). In each period  $t$ , individuals face shocks to health (good, poor, or disability), mortality risk, wages, and medical expenditure, and decide how much to consume, how much to work (participation and hours), and whether to claim SS retirement benefits (if eligible).

**Preferences:** Individuals enter the model at age 30 and live up to age 85, with nonseparable preferences over consumption and leisure as specified in equation (3). They also incur health-dependent time costs for labor force participation and reentry, so that

$$l_t = L - n_t - \phi_n(h_t)I\{n_t > 0\} - \phi_{RE}RE_t$$

**Health Transitions and Mortality:** Individuals face uncertainty in their health status,  $h_t \in \{0, 1, 2\}$ , which evolves according to a Markov process with probabilities of transitioning from state  $i$  to  $j$  of  $\pi_{j,i,t+1}$ . Mortality risk at age  $t + 1$  is captured by the probability  $s_{t+1} = S(h_t, t + 1)$ , which depends on both the individual’s current health status and age.

**Medical Expenses:** Out-of-pocket medical expenditures  $m_t$  vary by age, health status, and participation and are given by

$$m_t = M(h_t, t, P_t) \tag{37}$$

defined as the individual’s total medical expenditure net of coverage provided by insurance prior to age 65 and Medicare thereafter.<sup>31</sup>

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<sup>31</sup>[Blau and Gilleskie \(2008\)](#), [Imrohoroglu and Kitao \(2012\)](#) and [French and Jones \(2011\)](#) explicitly model employer-provided health insurance, while [Ozkan \(2023\)](#) models self-insurance in the form of preventative medical expenditures. Our model abstracts from heterogeneity in health insurance to incorporate other dimensions of heterogeneity.

**Hourly Wages:** Hourly wages have two components: a deterministic function of health and age, and an autoregressive process for ability that evolves according to:

$$\begin{aligned}\ln w_t(\theta_t, h_t) &= W(h_t, t) + \ln(\theta_t), \\ \ln(\theta_t) &= \rho \ln(\theta_{t-1}) + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma_\rho^2)\end{aligned}\tag{38}$$

**SS Retirement Benefits:** We model SS benefits in detail to match the current US system closely. Benefit levels are determined by the primary insurance amount (PIA), a progressive piecewise-linear function of a worker’s average indexed monthly earnings (AIME)—approximately the average of a worker’s highest 35 years of earnings—so that low earners receive a more generous replacement rate. Eligibility to claim begins at the early retirement age (ERA) of 62, and claiming is an irreversible, one-time decision. Benefits claimed before the normal retirement age (NRA) are reduced by an actuarial penalty for early claiming, while delaying claims beyond the NRA, up to age 70, increases benefits through the delayed retirement credit (DRC); [Table 12](#) summarizes these adjustments for claiming ages 62–70, as percentages of the unreduced benefit at the NRA, under an NRA of 65 and a DRC of 4.5%. In addition, beneficiaries under age 70 who continue to earn labor income are subject to a retirement earnings test (RET) that withholds benefits above an earnings threshold, with withheld benefits credited through increased future payments. Details of the AIME/PIA formulas, the actuarial adjustment ([Table 12](#)), the earnings test, disability insurance, pensions, and taxes are in [Appendix C.2](#).

Table 12: Effects of Early or Delayed Social Security Claiming

| NRA | DRC | Benefit, as a percentage of PIA, payable at age |       |       |     |       |     |       |     |       |
|-----|-----|-------------------------------------------------|-------|-------|-----|-------|-----|-------|-----|-------|
|     |     | 62                                              | 63    | 64    | 65  | 66    | 67  | 68    | 69  | 70    |
| 65  | 4.5 | 80                                              | 86.67 | 93.33 | 100 | 104.5 | 109 | 113.5 | 118 | 122.5 |

*Notes:* Social Security retirement benefits, expressed as a percentage of the primary insurance amount, that an individual can receive if he claims at ages 62–70. Abbreviations: NRA = normal retirement age; DRC = delayed retirement credit (percentage points); PIA = primary insurance amount.

**Disability Insurance and Pension Benefits:** Disability insurance (DI) benefits  $db_t$  are modeled as the standard SS benefits for individuals who meet disability criteria, adjusted by a probability factor reflecting typical DI qualification rates. Pension income  $pb_t$  is modeled as a function of SS benefits to reflect historical earnings and age-specific pension accruals. [Appendix C.2](#) presents more details on the disability insurance and pension benefits.



**Taxes:** We model income taxation following [Bénabou \(2002\)](#), [Heathcote et al. \(2017\)](#), and [Borella et al. \(2023\)](#), reflecting the US system. Individuals face effective tax rates dependent on age (and thus time). Taxes paid on annual total income  $y_t$  are given by  $T_t(y_t, \tau_t) = y_t - \lambda_t y_t^{1-\tau_t}$ , where  $\tau_t$  denotes the degree of progressivity and  $\lambda_t$  captures the average level of taxation. In addition, the government uses a proportional payroll tax  $\tau_t^p$  on labor income to help finance SS and Medicare. This tax applies up to a threshold  $\bar{y}_t^{ss}$ , with the payroll tax paid at age  $t$  given by  $T_t^p(w_t(\theta_t, h_t)n_t, \tau_t^p) = \tau_t^p \min[w_t(\theta_t, h_t)n_t, \bar{y}_t^{ss}]$ .

**Assets, Borrowing Constraints, and Government Transfers:** In each period, individuals receive from several sources: interest on assets  $ra_t$ , labor income  $w_t(\theta_t, h_t)n_t$ , pension benefits  $pb_t$ , SS retirement benefits net of the RET  $ss_t$  (if applicable), DI benefits  $db_t$  (if applicable), and government transfers  $tr_t$  (if applicable). The budget constraint is given by:

$$\begin{aligned} a_{t+1} &= a_t + Y_t(y_t, \tau_t, \tau_t^p) + (b_t * ss_t) + db_t I\{h_t = 2\} + tr_t - m_t - c_t \\ y_t &= ra_t + w_t(\theta_t, h_t)n_t + pb_t \\ Y_t(y_t, \tau_t, \tau_t^p) &= y_t - T_t(y_t, \tau_t) - T_t^p(w_t(\theta_t, h_t)n_t, \tau_t^p) \\ a_{t+1} &\geq \underline{a} \end{aligned} \tag{39}$$

The variable  $b_t \in \{0, 1\}$  is an indicator that equals one if the individual has claimed retirement benefits,  $y_t$  is annual taxable income, and  $r$  is the net return on assets.<sup>32</sup> Income taxes  $T_t(\cdot)$  depend on total taxable income, payroll taxes  $T_t^p$  on labor income, and after-tax income is  $Y_t(\cdot)$ . Since individuals cannot borrow against future SS benefits and can rarely borrow against most forms of pension wealth, we set the borrowing limit  $\underline{a} = 0$ .<sup>33</sup> To ease comparison with the literature on life cycle taxation with exogenous retirement, we allow for accidental bequests but assume no planned bequests at the terminal age,  $a_{T+1} = 0$ .<sup>34</sup> Government transfers,  $tr_t$ , ensure a minimum consumption level  $\underline{c}$ , as in [De Nardi et al. \(2010\)](#) and [Hubbard et al. \(1995\)](#), capturing US safety net programs such as SNAP and Supplemental Security Income.<sup>35</sup>

**Choices:** In each period, individuals choose consumption  $c_t(\cdot)$ , labor force participation  $p_t(\cdot)$ , hours worked  $n_t(\cdot)$ , savings  $a_{t+1}(\cdot)$ , and SS application  $b_t(\cdot)$  (if eligible) to maximize discounted expected utility (discount rate  $\beta$ ), subject to (39). The recursive program is

<sup>32</sup>That is,  $r = (1 - \tau_k)r_f$  with the pretax interest rate  $r_f$ , and  $\tau_k$  is an unmodeled corporate tax and other direct capital taxes as in [Stantcheva \(2017\)](#).

<sup>33</sup>This no-borrowing assumption is standard in the literature (e.g., [Hubbard et al. 1995](#); [Zeldes 1989](#)). In our setting, since the lower bound of the support of the distribution of wages is zero, the natural borrowing limit is zero.

<sup>34</sup>We target the time discount factor to match mean and median assets over the life cycle.

<sup>35</sup>That is,  $tr_t = \max\{0, \underline{c} + m_t - (a_t + Y_t + ss_t + db_t)\}$ . Papers such as [Braun et al. \(2017\)](#) explicitly model these safety net programs.

detailed in Appendix C.2.1.

**Summary:** Our framework captures essential features of French (2005) and French and Jones (2011) but differs with important innovations. First, we allow people to choose to work until death, rather than restricting endogenous retirement decisions to end at age 70. Second, in contrast to prior studies that focus on male household heads, our model targets aggregate variables and outcomes. Third, we explicitly model the disability state and DI benefits, which are important components of retirement policies.

### C.1.2 Estimation

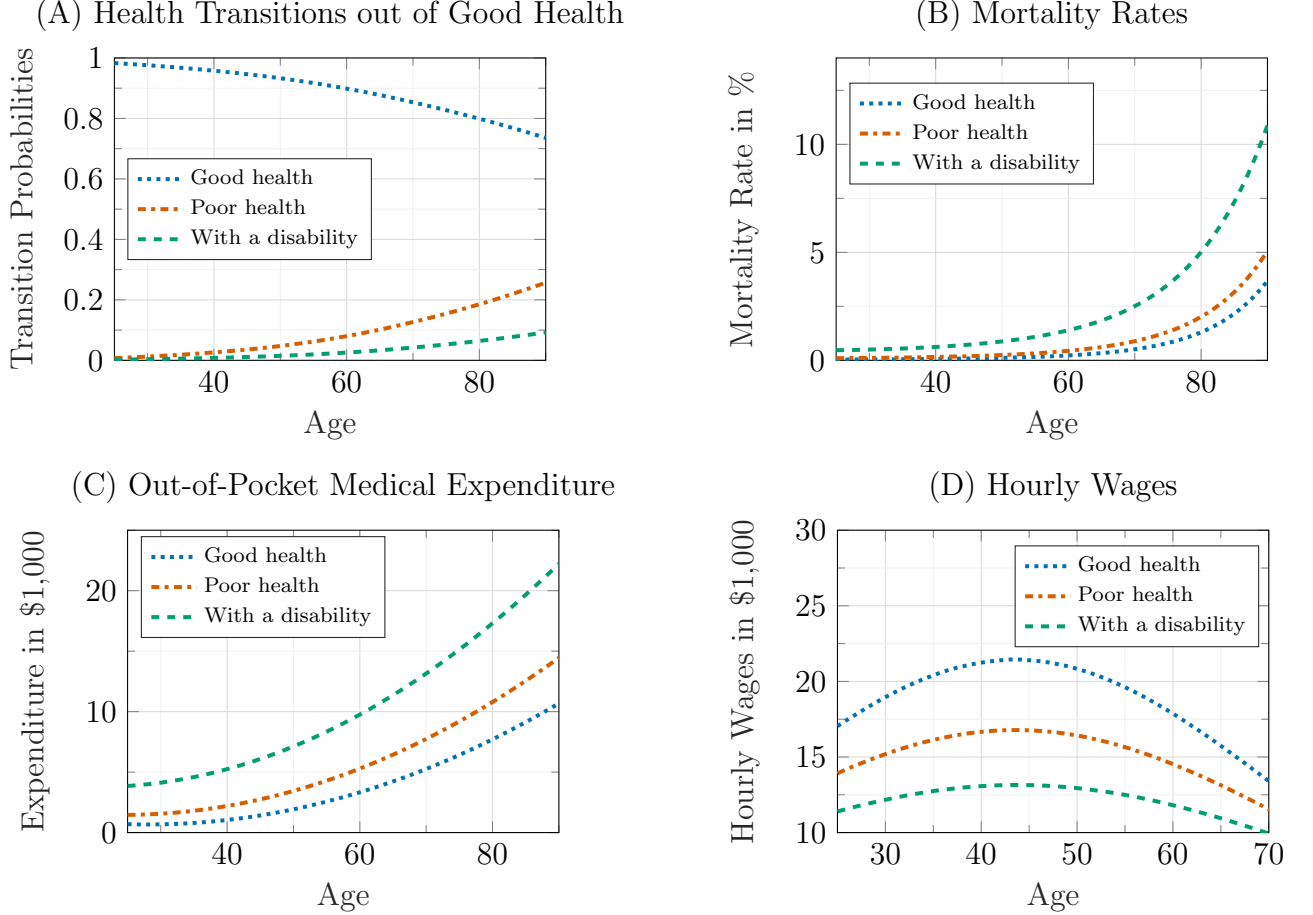
We estimate the model using a two-step strategy, as is standard in the literature (e.g., Cagetti 2003; Gourinchas and Parker 2002): parameters that can be cleanly identified without the model are estimated or calibrated first, and the remaining preference parameters are then estimated by the method of simulated moments (MSM), taking the first-step parameters as given.

**First-Step Estimation:** The first step estimates health transitions, mortality rates, out-of-pocket medical spending, and wage profiles directly from raw data. We primarily use the Panel Study of Income Dynamics (PSID), 1968–2015 waves, focusing on individuals born between 1920 and 1940—a cohort subject to similar SS rules whose outcomes we observe throughout the life cycle. We supplement it with the Medical Expenditure Panel Survey (MEPS), 1999–2012 waves, to estimate medical expenditure profiles. Health transitions and survival probabilities are estimated from the PSID with probit and logistic regressions; out-of-pocket medical expenses are constructed from the MEPS by combining total medical expenditures and insurance coverage, controlling for age and health status; and hourly wages are derived from PSID earnings and hours, adjusted for selection bias by means of the Heckman selection model. The estimated ability process from the PSID is highly persistent, with autoregressive coefficient and innovation variance  $(\rho, \sigma_\rho^2) = (0.981, 0.016)$ . We also calibrate parameters based on values used in the existing literature: the gross interest rate is set to 4% (French 2005), the consumption floor is set to \$3,500 (French and Jones 2011), and SS policy parameters are taken from official SSA rules. Appendix B.1 presents the data and sample selection, and Appendix B.3 presents the estimation procedures.

**Estimated Profiles by Health Status:** Figure 12 displays the resulting life cycle profiles. The probability of staying in good health declines with age, absorbed primarily by transitions into poor health (panel A). Mortality rates rise with age for all health categories, rapidly so for people with a disability at older ages (panel B). Out-of-pocket medical expenses are considerably higher in poorer health (panel C). Healthy individuals earn consistently higher

hourly wages than workers in poor health, who, in turn, outearn individuals with a disability throughout the life cycle (panel D).

Figure 12: First-Step Estimation, Profiles by Health Status



*Notes:* Probabilities of transitioning out of good health (panel A), mortality rates (panel B), out-of-pocket medical expenditure (panel C), and hourly wages (panel D), as functions of age by health status. The blue/red/green (dotted/dash-dotted/dashed) lines represent people in good health/people in poor health/people with a disability. Monetary values are expressed in 2016 dollars.

*Data Source:* Panel Study of Income Dynamics, Medical Expenditure Panel Survey.

**Second-Step Estimation:** In the second step, we use the MSM to estimate the preference parameters  $\mathcal{P} = (\gamma, \nu, \phi_n(h = 0), \phi_n(h = 1, 2), \phi_{RE}, L, \beta)$ . Our MSM estimator  $\hat{\mathcal{P}}$  minimizes the weighted distance, measured by a generalized method of moments (GMM) criterion, between the estimated target profiles from the PSID and the simulated profiles generated by the model. We require the model to match, for individuals from age 30 to 69, average labor force participation by health status, hours worked conditional on participation by health status, mean and median nonpension assets, and mean labor market reentry rates—a total of 280 moment conditions. Because of the limited number of observations of individuals with a disability in the PSID, individuals in poor health and those with disabilities ( $h = 1, 2$ )

are assigned the same fixed cost of working,  $\phi_n(h = 1, 2)$ . Further details are provided in Appendix C.3.

Table 13 reports the estimated structural parameters.<sup>36</sup> The estimated discount factor is 0.98, identified by intertemporal substitution and thus by labor supply and asset profiles, especially for the age group accumulating wealth for retirement. The annual time endowment is estimated at 4,491 hours, with fixed costs of working of 791 hours for healthy individuals and 868 hours for those in poor health or with a disability, identified by employment rates and hours worked by health status and within the range of earlier estimates (French and Jones 2011; Borella et al. 2023). The reentry cost is estimated at 3,312 hours, identified by the low reentry rates in the data. The estimated CRRA for flow utility and consumption weight are 1.70 and 0.54, implying a CRRA for consumption,  $-(\partial^2 u / \partial c_t^2) c_t / (\partial u / \partial c_t)$ , of approximately 1.38, identified by the asset and labor supply profiles: higher risk aversion contributes to greater precautionary savings and encourages more work earlier in life.

Table 13: Estimated Structural Parameters

| Parameter                                     | Definition                    | Estimates | S.E.   |
|-----------------------------------------------|-------------------------------|-----------|--------|
| $\gamma$                                      | Consumption weight            | 0.5354    | 0.0047 |
| $\nu$                                         | CRRA for flow utility         | 1.7029    | 0.0681 |
| $\beta$                                       | Time discount factor          | 0.9797    | 0.0023 |
| $L$                                           | Time endowment                | 4490.5    | 21.65  |
| $\phi_{RE}$                                   | Labor market reentry cost     | 3311.5    | 82.02  |
| $\phi_n(h = 0)$                               | Fixed cost of work, healthy   | 791.07    | 15.65  |
| $\phi_n(h = 1, 2)$                            | Fixed cost of work, unhealthy | 867.54    | 22.68  |
| $\chi^2$ statistic (degrees of freedom = 273) |                               | 1463      |        |

Notes: Estimated structural parameters and standard errors from the method of simulated moments procedures.

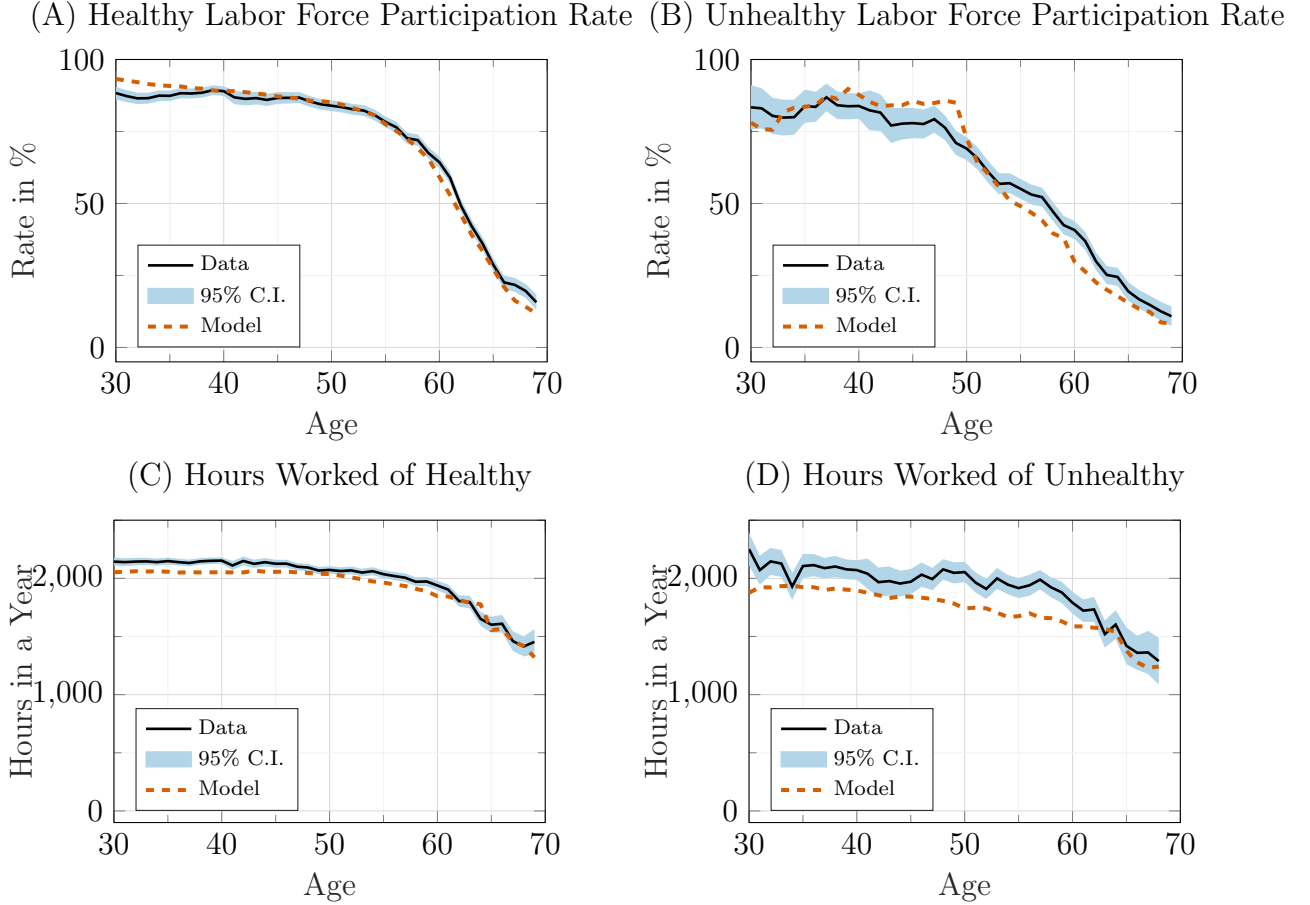
### C.1.3 Match of Moments

Figure 13 compares the simulated life cycle profiles to the empirical PSID profiles, with 95% confidence intervals. The model reproduces the key patterns of labor supply over the life cycle: high and stable participation in the 30s and 40s, earlier declines beginning in the 50s for those in poor health or with a disability, and sharp drops during the 60s across all health categories, coinciding with falling wages and growing retirement incentives through pensions and SS. As highlighted by Yu (2024), modeling disability status and DI benefits is essential for explaining the participation declines of unhealthy individuals starting in their 50s, as DI provides an alternative pathway to retirement for people with disabilities (Maestas

<sup>36</sup>For a more in-depth discussion of identification of preference parameters, see French (2005) and French and Jones (2011). As in previous studies, despite the overidentification test results reported in Table 13, the model successfully reproduces life cycle profiles that closely match those observed in the data.

et al. 2013; French and Song 2014; Low and Pistaferri 2015; Li 2018); this yields a closer fit for this group than in earlier work, such as French (2005).

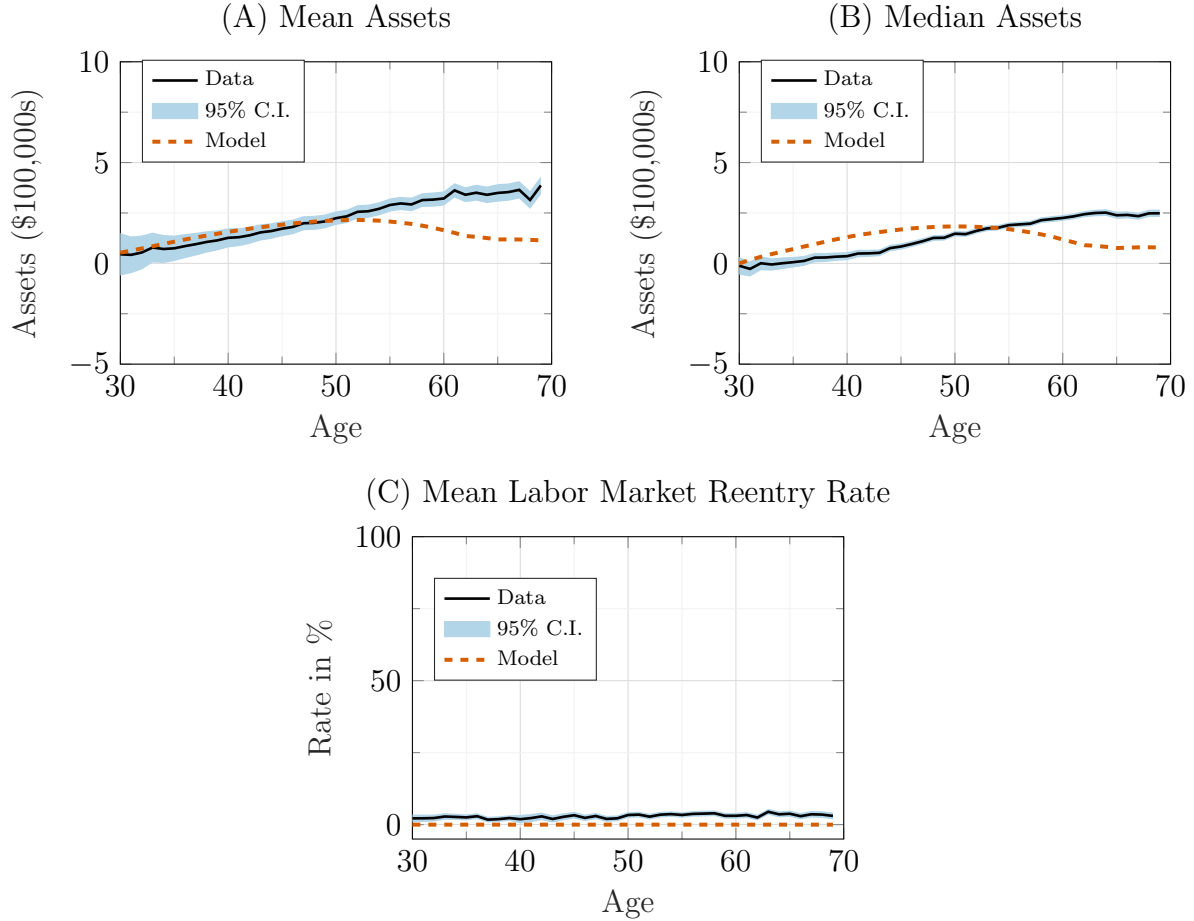
Figure 13: Data and Estimated Profiles



*Notes:* Panels A and B show the model fit for labor force participation by health status; panels C and D for hours worked conditional on participation by health status. Data profiles are estimated with the Panel Study of Income Dynamics; the shaded region represents the 95% confidence intervals, and the orange dashed lines represent model profiles. Monetary values are expressed in 2016 dollars.

The model also matches mean and median assets well until the mid-50s. The divergence beyond the 50s likely arises because, to make valid comparisons with the literature on optimal taxation with exogenous retirement (Farhi and Werning 2013; Golosov et al. 2016; Stantcheva 2017), we model accidental bequests but abstract from incidental bequests, which typically increase savings in older age groups (e.g., De Nardi et al. 2010; Lockwood 2018). Labor market reentry rates are already low in the data, below 5%, and our model generates no reentry in equilibrium; the large estimated reentry cost justifies modeling reentry as costly and treating final exits as effectively irreversible in the normative analysis. Figure 14 presents these additional matched profiles.

Figure 14: Additional Data and Estimated Profiles



*Notes:* Panels A and B show the model fit for mean and median assets, and panel C shows mean reentry. Data profiles are estimated with the Panel Study of Income Dynamics. The shaded region represents the 95% confidence intervals. The orange lines represent model profiles. Monetary values are expressed in 2016 dollars. The model matches well the mean and median assets until age 55. Reentry is already low in the data (below 5%), and our model generates no reentry in equilibrium.

## C.2 Additional Specifications of the Quantitative Model

**Social Security Policy Rules:** According to the Social Security Administration (SSA), individuals born between 1920 and 1940 faced an early retirement age (ERA) of 62 and, on average, a normal retirement age (NRA) of 65. Their delayed retirement credit (DRC) ranged from 3% to 6%, depending on their year of birth. In our analysis, we use the cohort-average DRC of 4.5%.

The primary insurance amount (PIA) formula included bend points at \$3,720 and \$22,392, with maximum average indexed monthly earnings (AIME) capped at \$43,800. The retirement earnings test imposed different tax rates and thresholds based on age: Beneficiaries under 65 lost \$1 of benefits for every \$2 of earnings above \$6,000, while those aged 65 and older lost \$1 for every \$3 above \$8,186. These parameters, summarized in Table 14, are incorporated into the model.

Table 14: Social Security Parameters Summary

| Parameter        | Description                          | Value             | Source      |
|------------------|--------------------------------------|-------------------|-------------|
| $\tau_t^{ss}$    | Payroll tax rate                     | 3%–7.65%          | SSA         |
| $\bar{y}_t^{ss}$ | Threshold, payroll tax               | \$14,459–\$64,974 | SSA         |
| $y_{db}$         | Disability benefits income threshold | \$3,600           | SSA         |
| $aime_0$         | 1st bend point in PIA formula        | \$3,720           | SSA         |
| $aime_1$         | 2nd bend point in PIA formula        | \$22,392          | SSA         |
| $aime_{max}$     | Maximum Social Security wealth       | \$43,800          | SSA         |
| $ERA$            | Early retirement age                 | 62                | SSA         |
| $NRA$            | Normal retirement age                | 65                | SSA         |
| $DRC$            | Delayed retirement credit            | 4.5%              | SSA         |
| $RET$            | Retirement earnings test             | Under NRA         | NRA-69      |
| $\tau_{ret}$     | Tax rate                             | 50%               | 33% SSA     |
| $y_{ret}$        | Threshold                            | \$6,000           | \$8,186 SSA |

Notes: Monetary values are expressed in 1987 dollars.

Source: Social Security Administration.

**AIME and PIA Formulas:** Lifetime career earnings enter benefits through the AIME, calculated from a worker’s highest 35 years of earnings. To reflect this in the model, we construct an annualized version of AIME,  $aime_t$ , that evolves according to:

$$aime_{t+1} = \begin{cases} \min\{aime_t + \frac{w_t(\theta_t, h_t)n_t}{35}, aime_{max}\} & \text{if age} < 60 \\ \min\{aime_t + \max\{0, \frac{w_t(\theta_t, h_t)n_t - aime_t}{35}\}, aime_{max}\} & \text{if age} \geq 60 \end{cases}$$

which is capped at a maximum threshold  $aime_{max}$ .<sup>37</sup> This limits the incentives to continue

<sup>37</sup>Individuals are assumed to enter the labor market at age 25. That is, by age 30, the starting age of the model, they have accumulated 5 years of labor earnings. This is a common assumption (French 2005).

working after a worker has accumulated 35 years of earnings, unless the income from an additional year exceeds that in earlier lower-earnings years.

$PIA_t$ , the base benefit amount, is a piecewise linear function of  $aime_t$ :

$$\begin{aligned} PIA_t = & 0.9 * \min\{aime_t, aime_0\} \\ & + 0.32 * \min\{\max\{aime_t - aime_0, 0\}, aime_1 - aime_0\} \\ & + 0.15 * \max\{aime_t - aime_1, 0\} \end{aligned}$$

where  $aime_0$  and  $aime_1$  denote the bend points in the formula. This structure results in a more generous replacement rate for low earners.

**Retirement Earnings Test:** Individuals under age 70 who work while receiving retirement benefits are subject to the retirement earnings test (RET). If their labor income exceeds a specified threshold  $y_{ret}$ , each dollar earned above that threshold reduces their benefit payments by  $\tau_{ret}$  dollars, until the entire benefit is withheld. For example, before 2000, individuals under age 65 had their benefits reduced by \$1 for every \$2 of earnings, while those aged 65 and older faced a \$1 reduction for every \$3 of earnings. Thus, the actual benefit received at age  $t$ ,  $ss_t$ , reflects this reduction:

$$ss_t = \max\{0, ssb_t - \tau_{ret} * \max\{0, (w_t(\theta_t, h_t)n_t - y_{ret})\}\} \quad \text{if age} < 70$$

Here,  $ssb_t$  is the  $PIA_t$  amount adjusted by early or delayed claiming. Benefits withheld due to the RET are credited through increased future payments.<sup>38</sup>

**Disability Insurance:** Individuals with a disability are eligible to receive disability insurance (DI) benefits prior to the NRA if their labor income falls below a specified threshold. In our model, the DI application is not treated as an endogenous decision because it is not the primary focus of the paper and obtaining relevant data is challenging. In studies that explicitly model DI applications, such as [Low and Pistaferri \(2015\)](#), the probability of a successful application for DI depends on age and health status. To keep our model tractable, we assume that DI benefits are granted exclusively to individuals who have a disability and meet the income requirement. Specifically, we assume that individuals with a disability are eligible for DI benefits if they are older than 50 and their labor income is below \$3,600, corresponding to the monthly substantial gainful activity threshold of \$300, as defined by

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In addition, according to the SSA, an individual's earnings up to two years before retirement eligibility age (currently age 60) are indexed to average wage growth to account for increases in living standards over time. In our model, to capture the general rise in the standard of living during working periods, we adjust  $AIME$  for inflation by indexing wages to the personal consumption expenditure price index.

<sup>38</sup>Following [French and Jones \(2011\)](#), if a year's worth of benefits is withheld because of the RET before the NRA, future benefits are increased by the actuarial reduction factor for early claiming. If benefits are withheld after the NRA, future benefits are increased by the DRC.



the SSA for nonblind individuals with a disability.

$$db_t = \pi^{db} PIA_t I_{\{h_t=2\}} I_{\{w_t(\theta_t, h_t) n_t \leq y_{ab}\}} \quad \text{if age} < NRA$$

The DI benefit amount  $db_t$  is set equal to the individual's PIA, without the early claiming penalty that would normally reduce retirement benefits received before the NRA. To reflect the fact that not all individuals with a disability receive benefits successfully, the benefit amount is discounted by  $\pi^{db}$ , the probability of receiving DI benefits for old-age groups with severe work limitations, as estimated by [Low and Pistaferri \(2015\)](#). This approach captures the average DI benefit amount received by individuals with a disability.

**Pension Benefits:** Pension benefits are modeled to capture the basic features of defined benefit (DB) plans. Similarly to Social Security benefits, pensions depend on an individual's work history and are illiquid before a specified age, which we assume to be 62. In contrast to Social Security, DB pension plans often create strong incentives for workers to exit the labor force at particular ages through the structure of accrual rates ([Gustman et al. 2000](#); [Ippolito 1997](#)). To reflect these features, we impute  $pb_t$  as a function of Social Security benefits, following the approach of [French \(2005\)](#), and model it as a regressive function of  $pb(PIA_t)$ , adjusted by a term  $\epsilon_t$  to reflect age-specific pension accrual.

$$pb_t = pb(PIA_t) + \epsilon_t$$

The parameters used in  $pb(PIA_t)$  are based on estimates by [Gustman and Steinmeier \(1999\)](#), and the age-specific pension accrual rates are taken from [Gustman et al. \(2000\)](#). The adjustment  $\epsilon_t$  is made to account for the difference in accrual between Social Security and pension wealth over the life cycle, which tends to be positive at older ages. [French \(2005\)](#) provides a detailed description of this procedure.

**Taxes:** Individuals in our model face the effective time-varying tax rates on their total income, estimated from PSID data as in [Borella et al. \(2023\)](#). For tractability, we assume that individuals anticipate changes in the tax rates on total income. Payroll tax rates and thresholds are based on historical data from the SSA for Old-Age, Survivors, and Disability Insurance (OASDI) and Medicare's Hospital Insurance (HI) programs. Since 1960, the combined OASDI and HI tax rates for employees and employers have ranged from 3% to 7.65%.

### C.2.1 Dynamic Program of Positive Economy

The life cycle in the model is divided into three distinct stages. The first stage, from ages 30 to 61, represents the working years during which individuals are not eligible for pension or Social Security retirement benefits. In this period, they make decisions only

about consumption and labor supply. The second stage, from ages 62 to 69, is a transitional period when individuals continue to choose consumption and labor supply but also decide whether to claim Social Security retirement benefits. The third stage, from ages 70 to 85, reflects a post-benefits-claiming period during which individuals have already claimed retirement benefits and continue to make decisions about consumption and labor supply.

The individual's problem at age  $t$  is summarized by a value function in state  $(X_t)$ :

$$\begin{aligned}
V_t(X_t) = \max_{c_t, n_t, b_t} \{ & \frac{1}{1-\nu} (c_t^\gamma [L - n_t - \phi_n(h_t)I\{n_t > 0\} - \phi_{RE}RE_t]^{1-\gamma})^{1-\nu} \\
& + \beta s_{t+1} \int V_{t+1}(X_{t+1}) dF(X_{t+1}|X_t, t, c_t, n_t, b_t) \} \\
X_t = & (a_t, w_t, h_t, b_{t-1}, P_{t-1}, aime_t)
\end{aligned} \tag{40}$$

subject to equations (1)–(2), (5), (37)–(38), and (39). The function  $F(\cdot|\cdot)$  defines the conditional distribution of states, subject to the laws of motion of shocks and the budget constraint. Variables such as  $m_t$  and  $pb_t$  are not explicitly included in  $X_t$  since they depend on other state variables.

The model is solved by backward induction with value function iteration. An individual's decisions in period  $t$  depend on the state  $X_t$ , preference parameters  $\mathcal{P} = (\gamma, \nu, \phi_n(h=0), \phi_n(h=1,2), \phi_{RE}, L, \beta)$ , and structural parameters that determine the data-generating process for the state variables  $\chi = (r, \sigma_\rho^2, \rho, \{W(h_t, t)\}_{t=1}^T, \{\pi_{h_{t+1}, h_t, t}\}_{t=1}^{T-1}, \{s_{t+1}\}_{t=1}^{T-1}, \{m_t\}_{t=1}^T, \{pb_t\}_{t=1}^T, \{ss_t\}_{t=1}^T, \{db_t\}_{t=1}^T, Y_t(\cdot))$ . At time  $T$ , the consumption decision  $c_T$  is made by solving the above problem  $V_T(X_T)$  with the terminal value. Our model accounts for savings used for self-insurance against retirement and risks such as medical expenses and income fluctuations, and accidental bequests, while abstracting from any legacy bequests after age 85. Given the decision rules and value function at time  $T$ , we then solve the decision rules at time  $T-1$ ,  $T-2$ , ..., 0. The solution to the problem (40) consists of sequences of consumption rules  $\{c_t(X_t, \mathcal{P}, \chi)\}_{1 \leq t \leq T}$ , hours worked rules  $\{n_t(X_t, \mathcal{P}, \chi)\}_{1 \leq t \leq T}$ , and Social Security benefit application rules  $\{b_t(X_t, \mathcal{P}, \chi)\}_{1 \leq t \leq T}$ . The labor force participation rules at  $t$ ,  $p_t(X_t, \mathcal{P}, \chi)$ , are determined by the hours worked decisions. Assets in the next period,  $a_{t+1}(X_t, \mathcal{P}, \chi)$ , are obtained by applying these decision rules to the asset accumulation equation.

To numerically solve these decision rules, we discretize the state variables into a finite number of points within a grid that includes 30 asset states, 10 wage states, 10 AIME states, 2 Social Security application states, three health states, and two previous labor force participation states. We directly compute the value function at these points and integrate the value function with respect to the innovation of wages using five-node Gauss–Hermite quadrature (Judd 1998). We use linear interpolation within the grid and linear extrapolation outside of the grid to evaluate the value function at points that cannot be directly computed.

After solving the model, we use the decision rules to simulate individual life cycle patterns using forward induction. Starting from an initial realized state variable  $X_0$ , decisions are made with the decision rules at  $t = 0$ . Then, given time-zero decisions, the state  $X_1$  is generated by means of transition probabilities and shock realizations. This process is repeated through age  $T$ , generating complete simulated histories.

### C.3 Estimation Strategy

We estimate the remaining preference parameters,  $\mathcal{P} = (\gamma, \nu, \phi_n(h = 0), \phi_n(h = 1, 2), \phi_{RE}, L, \beta)$ , using the method of simulated moments (MSM). Because of the limited number of observations on individuals with a disability in the PSID, individuals in poor health and those with disabilities are assigned the same fixed cost of working to simplify the model. The objective is to find a vector of preference parameters  $\hat{\mathcal{P}}$  that minimizes the weighted distance between the estimated target profiles from the PSID and the corresponding simulated profiles generated by the model. The MSM estimator  $\hat{\mathcal{P}}$  minimizes the GMM criterion function, which evaluates the distance between the model and data moments:

$$\hat{\mathcal{P}} = \arg \min_{\mathcal{P}} \frac{I}{1 + \tau} \hat{\varphi}(\mathcal{P}; \chi)' \hat{\mathbf{W}}_I \hat{\varphi}(\mathcal{P}; \chi)$$

The parameter  $\tau$  denotes the ratio of the number of empirical observations to the number of simulated observations, and  $\hat{\varphi}(\mathcal{P}; \chi)$  is a  $7\mathbb{T}$ -element vector of moment conditions, such that

$$\hat{\varphi}(\mathcal{P}; \chi) = \begin{bmatrix} E[p_{iht}|h, t] - \int p_t(X, \mathcal{P}, \chi) dF_{h,t}(X|h, t) \\ E[n_{iht}|h, t] - \int n_t(X, \mathcal{P}, \chi) dF_{h,t}(X|h, t) \\ E[a_{iht}|h, t] - \int a_t(X, \mathcal{P}, \chi) dF_{h,t-1}(X|h, t) \\ E[re_{it}|t] - \int re_t(X, \mathcal{P}, \chi) dF_t(X|t) \end{bmatrix}_{t \in \{30, \dots, 69\}, h \in \{healthy, unhealthy\}}$$

where  $E[p_{iht}|\cdot]$ ,  $E[n_{iht}|\cdot]$ ,  $E[a_{iht}|\cdot]$ , and  $E[re_{iht}|\cdot]$  are the moments that our model is estimated to match:

1. Labor force participation by health status (healthy, unhealthy) over ages 30–69 (  $2\mathbb{T}$  moments),
2. Hours worked conditional on participation by health status (healthy and unhealthy) over ages 30–69 (  $2 \times \mathbb{T}$  moments),
3. Mean and median assets over ages 30–69 (  $2 \times \mathbb{T}$  moments),
4. Mean labor market reentry rates over ages 30–69 ( $\mathbb{T}$  moment).

This gives a total of  $7 \times \mathbb{T} = 280$  moment conditions. The model generates  $\int p_t(X, \mathcal{P}, \chi) dF(\cdot)$ ,  $\int n_t(X, \mathcal{P}, \chi) dF(\cdot)$ ,  $\int a_t(X, \mathcal{P}, \chi) dF(\cdot)$ , and  $\int re_t(X, \mathcal{P}, \chi) dF(\cdot)$ , where  $F_t(\cdot)$  is the cumulative distribution function (CDF) of the state variables at time  $t$  and  $F_{ht}(\cdot)$  indicates the

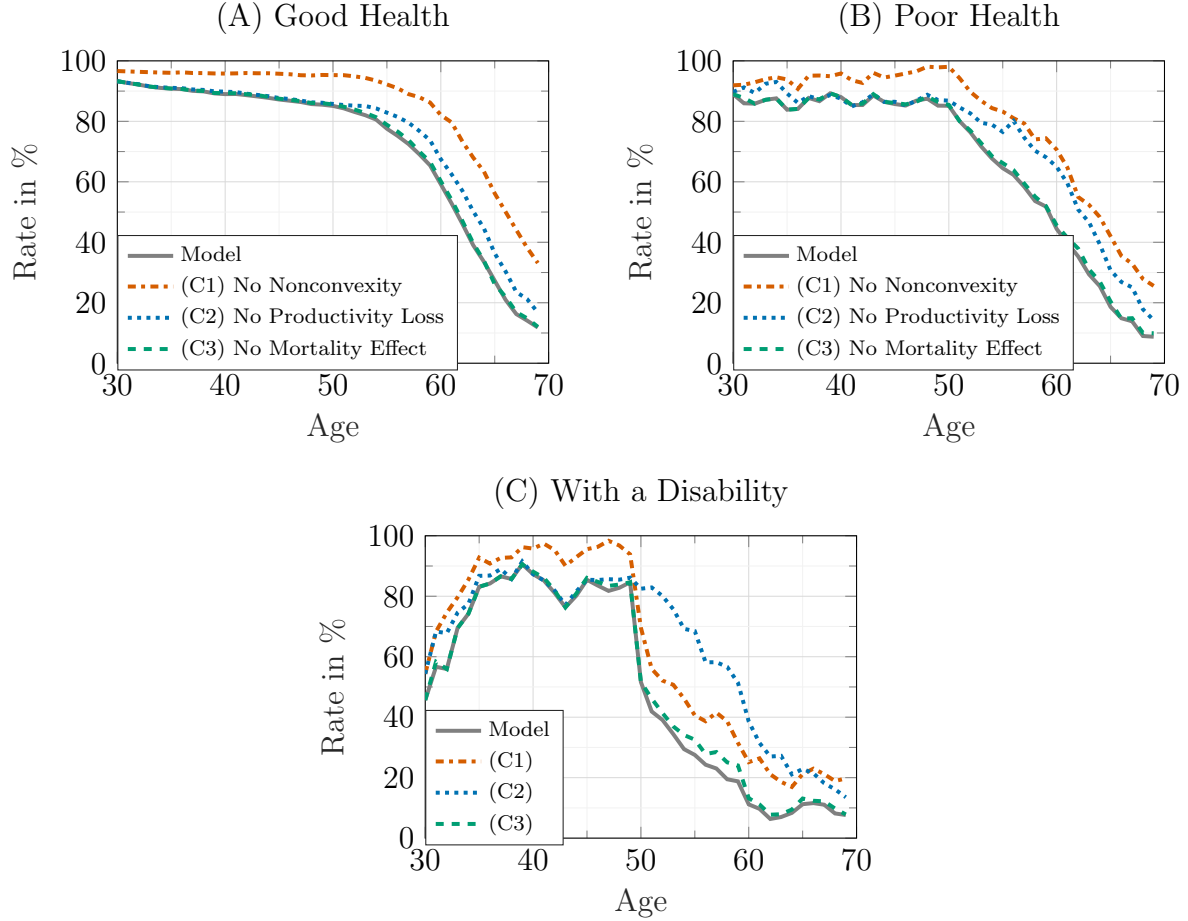
CDF of the state variables at period  $t$  given health status  $h$ . The weighting matrix,  $\hat{\mathbf{W}}_I$ , is set to the inverse of a diagonal  $7T \times 7T$  matrix, which consists of the elements of the variance–covariance matrix from the data along its main diagonal. To identify the preference parameters in  $\mathcal{P}$ , we follow the approach in French (2005) and impose two assumptions: that individuals’ work decisions do not affect their health status or wage and that age changes work and savings incentives but not preferences.

The estimation proceeds in the following steps: First, we estimate the empirical life cycle profiles for labor force participation, hours worked, and assets by health status from the data. Second, given the initial distribution of relevant state variables and a set of parameters  $\chi$  that determine the data-generating process for the state variables, we generate matrices of health and wage shocks. Third, given the values of  $\mathcal{P}$ , we solve the model numerically using value function iteration and generate decision rules that allow us to simulate life cycle profiles for participation, hours worked, wealth accumulation, and labor market reentry. Fourth, we construct moment conditions and calculate the distance between the simulated profiles and data profiles using the GMM criterion. Fifth, we pick a new vector of preference parameters,  $\mathcal{P}_{new}$ , and iterate the above two steps until the parameter vector that minimizes the criterion function is found.

## C.4 Decomposition

### C.4.1 Additional Figures: Decomposition of Allocations

Figure 15: Effects of Preferences and Technological Factors on Labor Force Participation by Health Status



*Notes:* Labor force participation rates as a function of age by health status for the estimated quantitative model and counterfactual simulations that shut down different channels. Panels A/B/C represent people in good health/people in poor health/people with a disability. The *No Nonconvexity* benchmark (C1) sets fixed costs of participation to zero. The *No Productivity Loss* benchmark (C2) sets wages equal to the wages of healthy individuals. The *No Mortality Effect* benchmark (C3) sets survival probabilities equal to those of healthy individuals. The *No Nonconvexity* benchmark features the largest increase in labor force participation for people in good and poor health, followed by the *No Productivity Loss* benchmark. For people with a disability, however, the *No Productivity Loss* benchmark features the largest increase in labor force participation in old age.

## C.5 Computation of the Mirrlees Optimum

**Algorithm:** We extend the computational approach of [Farhi and Werning \(2013\)](#). The recursive dual state is  $(\lambda, \Gamma, \theta_-, h_-, P_-, t)$ . Within a period, the allocation over the ability shock  $\varepsilon$  solves an optimal-control problem integrated as an ODE in  $\varepsilon$ , with a shooting step on the costate’s transversality condition. Health enters as  $n_h$  independent branch problems per grid point, aggregated with transition-and-survival weights; survival-inclusive promises keep every first-order condition free of survival terms. Participation is a pointwise minimum over two concentrated Hamiltonians (work, with fixed time costs; not work, with the continuation indexed by lagged nonparticipation), so exit and reentry are endogenous at every age up to 84, after which the closed-form retirement continuation applies. The AR(1) ability process  $\ln \theta_t = \rho \ln \theta_{t-1} + \ln \varepsilon_t$  enters through the type map  $m(\theta_-, \varepsilon) = \theta_-^\rho \varepsilon$ ; every equation of the random-walk system generalizes through the conversion factor  $\varepsilon_c \equiv \theta_-^{\rho-1} \varepsilon$  and a  $\rho$ -scaling of the marginal-promise recursion.

**State Space, Grids, and Tolerances:** States are discretized, while all within-period choices ( $c, y, n$ , participation, and the continuation multipliers  $\lambda', \Gamma'$ ) are solved exactly at every point by root-finding, and continuation objects are interpolated linearly over  $(\lambda, \Gamma, \ln \theta)$ . The discretization is:

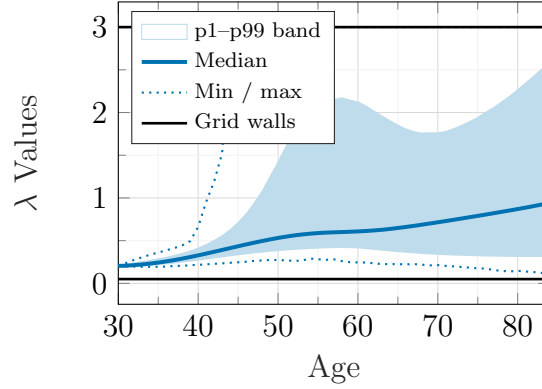
| Variable                           | Type                       | Points | Rule                              |
|------------------------------------|----------------------------|--------|-----------------------------------|
| $\lambda$ (promised-utility mult.) | continuous state           | 44     | [0.05, 3.0]                       |
| $\Gamma$ (marginal-promise mult.)  | continuous state           | 20     | [−0.7, 0]                         |
| $\theta_-$ (ability level)         | continuous state           | 11     | log-spaced, $\pm 4$ sd + buffer   |
| $h_-$ (health)                     | discrete state             | 3      | good / poor / disabled            |
| $P_-$ (lagged participation)       | discrete state             | 2      | worked / not                      |
| $t$ (age)                          | discrete                   | 55     | ages 30–84, annual                |
| $\varepsilon$ (ability shock)      | continuous (never gridded) | —      | truncated at the 1st/99th pctiles |

Each period therefore solves  $44 \times 20 \times 11 = 9,680$  grid points  $\times 3 \times 2 = 6$  health-participation branches = 58,080 within-period optimal-control problems, each an ODE integration over  $\varepsilon$  inside a shooting root with inner allocation roots at  $10^{-7}$ – $10^{-6}$ . Following [Farhi and Werning \(2013\)](#), the shock dimension is never discretized for solution purposes: the integrator selects its own  $\varepsilon$  evaluation points adaptively, and the solved policy paths are subsequently stored at 10 fixed  $\varepsilon$  nodes used only for interpolation in the simulation; retirement-region multiplier bounds widen to [0.01, 100] at the terminal boundary. Simulations use one million agents on the interpolated policies.

**Grid-Bounds Audit:** At each date, the pair of promises ranges over an endogenous feasible set that is not known in closed form; we discretize a rectangle for the corresponding

multipliers and audit whether its boundaries bind. In the headline simulation, the  $p1$ – $p99$  range of  $\lambda$  is strictly interior at every age (Figure 16).

Figure 16: Grid-bounds Audit



*Notes:* The simulated distribution of the promised-utility multiplier  $\lambda$  by age— median,  $p1$ – $p99$  band, and min/max— against the grid’s lower and upper bounds.

**Numerical Accuracy:** Table 15 re-solves the calibrated economy with a  $10\times$  tighter root tolerance and with a 40% denser grid in each continuous state, one state at a time: optimal-wedge moments move by at most 0.002 and participation moments by at most 0.014; the tighter-tolerance re-solve matches the baseline to three decimals in every reported moment.

Table 15: Grid and Tolerance Robustness

|                          | Baseline | Tol. $\times 10$ | $\lambda$ -grid +40% | $\theta$ -grid +40% | $\Gamma$ -grid +40% |
|--------------------------|----------|------------------|----------------------|---------------------|---------------------|
| LFPR at 60 (good)        | 0.549    | 0.549            | 0.542                | 0.563               | 0.558               |
| Labor wedge at 60 (good) | 0.082    | 0.082            | 0.082                | 0.080               | 0.084               |

*Notes:* Every column re-solves the *same* calibrated headline economy, changing one numerical setting at a time. *Tol.*  $\times 10$  tightens the root-finding tolerance  $10^{-7} \rightarrow 10^{-8}$ ;  $\lambda$ -grid raises 44→62 points;  $\theta$ -grid 11→15;  $\Gamma$ -grid 20→28. *LFPR at 60 (good)* = share of good-health agents working at age 60 in the simulation; *labor wedge at 60 (good)* = mean labor wedge across good-health workers at age 60.

## C.6 Incentive Compatibility and the First-Order Approach

This appendix reports the ex post incentive-compatibility checks described in Section 3, following Farhi and Werning (2013).

We simulate misreporting strategies in which agents misreport the transitory shock by a factor  $\delta$  (over- and under-reports) from  $\pm 1 - 2\%$  to  $\pm 5 - 10\%$ , over one-period, and permanently at five start ages (30, 34, 39, 49, 59) and comparing welfare gains against placebo of deviations ( $\pm 0.1\%$ ) whose true welfare gain is zero in theory and whose measured welfare gain captures the numerical accuracy of our algorithm. The statistic we will measure

is welfare gain from these misreports at the top percentile  $p99$  the scaled by the size of the deviation  $p99/|\delta|$ . We scale down the top percentile  $p99$  by  $|\delta|$  to make a 0.1% control comparable to a 5–10% misreport.

Table 16: Incentive Compatibility Check

| Deviation                                  | Mean welfare gain | 99th pct. welfare gain | $p99/ \delta $ |
|--------------------------------------------|-------------------|------------------------|----------------|
| <i>Controls</i>                            |                   |                        |                |
| Placebo $\delta = \pm 0.1\%$ , one-period  | −0.001 to +0.001  | $\leq 0.006$           | 2–6            |
| Placebo $\delta = \pm 0.1\%$ , permanent   | −0.006 to +0.002  | $\leq 0.41$            | 19–406         |
| <i>Treatments</i>                          |                   |                        |                |
| $\delta = \pm 1\text{--}2\%$ , one-period  | −0.009 to +0.007  | $\leq 0.23$            | 2–12           |
| $\delta = \pm 1\text{--}2\%$ , permanent   | −3.6 to +0.001    | $\leq 0.90$            | 1–71           |
| $\delta = \pm 5\text{--}10\%$ , one-period | −0.05 to +0.04    | $\leq 0.43$            | 2–7            |
| $\delta = \pm 5\text{--}10\%$ , permanent  | −2,670 to −0.010  | $\leq 3.1$             | $\leq 31$      |

*Notes:* BASELINE is 10,000 simulated agents who always report truthfully. Each row re-simulates the *same* 10,000 agents with the *same* shock draws (common random numbers, so any difference is the lie, not luck), except the agent misreports his ability innovation by the stated factor  $\delta$ . What each row changes relative to truth-telling: *Placebo*: reports  $0.999\times$  or  $1.001\times$  the truth — a  $\pm 0.1\%$  misreport too small to truly profit, so its measured “gain” shows the level the test’s own numerical noise reaches; *One-period*  $\delta$  misreport for a single period; *Permanent*: the same  $\delta$  applied in every period from the start age on. Every deviation family is run at every duration and at each start age (30, 34, 39, 49, 59). After a finite misreport window ends, the agent’s report returns to the truth from the following period onward. The table is read within durations: each treatment row is judged against the placebo row of the same duration — both scaled columns are divided by their own misreport size, which is what makes a 0.1% control comparable to a 5–10% treatment. Across durations, the treatments’ scaled values lie within or below their control’s range. gain per agent = lifetime discounted utility under the misreport − under truth-telling; *mean gain* = average across agents (negative = lying hurts); *99th pct.* = the luckiest one percent of liars;  $p99/|\delta|$  = that gain divided by the misreport size.

Table 16 reports the mean welfare gain, 99% percentile welfare gain and  $p99/|\delta|$  for our placebo and and large misreporting strategies. Mean welfare gains of placebo one-period misreports are essentially zero, while permanent placebo misreports have welfare gains between −0.006 and +0.002 in levels.

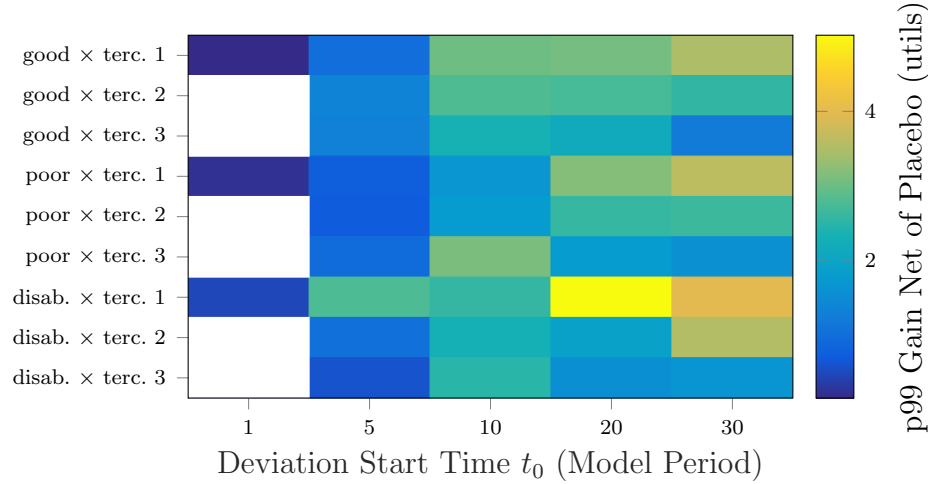
The hypothesis of global incentive compatibility implies that the statistic  $p99/|\delta|$  should be constant or decreasing across  $\delta$ , while a genuine violation would make it grow with  $\delta$ . The  $p99/|\delta|$  column shows that the statistic does indeed not grow when we consider our large deviation treatments. For the large  $\delta = \pm 5 - 10\%$  and permanent misreports, some trajectories hit out-of-state bounds, so we extrapolate the welfare gains and can bound the 99th percentile welfare gains.

In order to understand where there might be tension for the FoA, Figure 17 reports the welfare gains in utils relative to placebo of the top percentile within each health x ability tercile x starting age of the deviation for the large  $\delta = \pm 5 - 10\%$  and permanent misreports. Potential 99th-percentile welfare gains from large deviations net of numerical uncertainty (placebo) concentrate in the lowest-productivity tercile in poor health or with a disability, for deviations starting at model ages 20–30 (calendar ages 49–59). These are the states, if



any, in which large deviations might be welfare enhancing, and the first-order approach may find its limits.

Figure 17: Incentive Compatibility

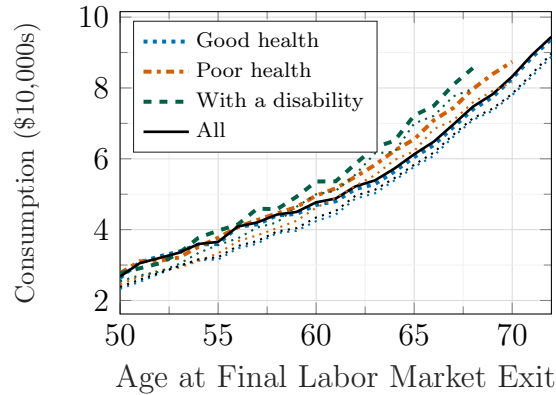


*Notes:* Incentive-compatibility stress map cell by cell (health  $\times$  ability tercile  $\times$  deviation start age), each cell scored against its own placebo. Net-of-placebo 99th-percentile gains concentrate in the lowest-productivity tercile in poor health or with disabilities, for deviations starting at model ages 20–30 (calendar ages 49–59); the worst cell (disabled  $\times$  tercile 1,  $t_0 = 20$ ) has a net p99 gain of +5.0 utils against a placebo of 0.5.

## C.7 Additional Exhibits for the Headline Economy

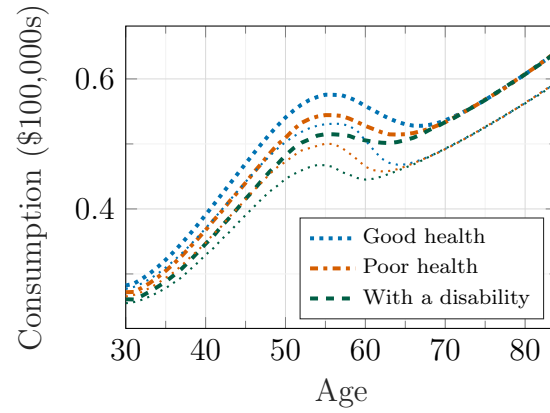
This appendix collects additional exhibits for the headline economy of Section 6.

Figure 18: Retirement Consumption Schedule and the Delayed Retirement Credits



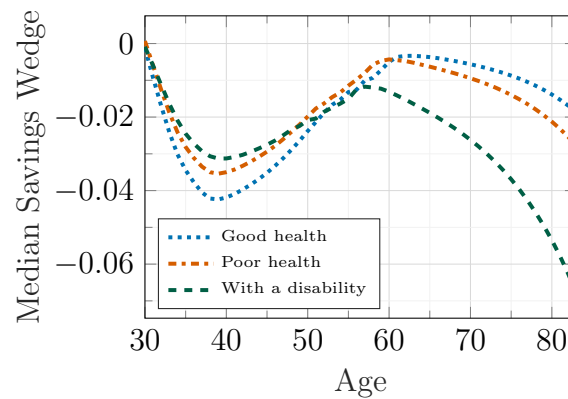
*Notes:* Consumption in the first retirement year, by stopping age and health at exit (thick lines: mean; thin lines: median; units: \$10,000s per year, 2016 dollars). The schedule is upward sloping: stopping at 65 yields 1.7 $\times$  the retirement consumption of stopping at 55. The health ordering at a given stopping age (disabled above good) reflects selection; see the discussion of Figure 4 in Section 6.2.

Figure 19: Consumption by Age and Health



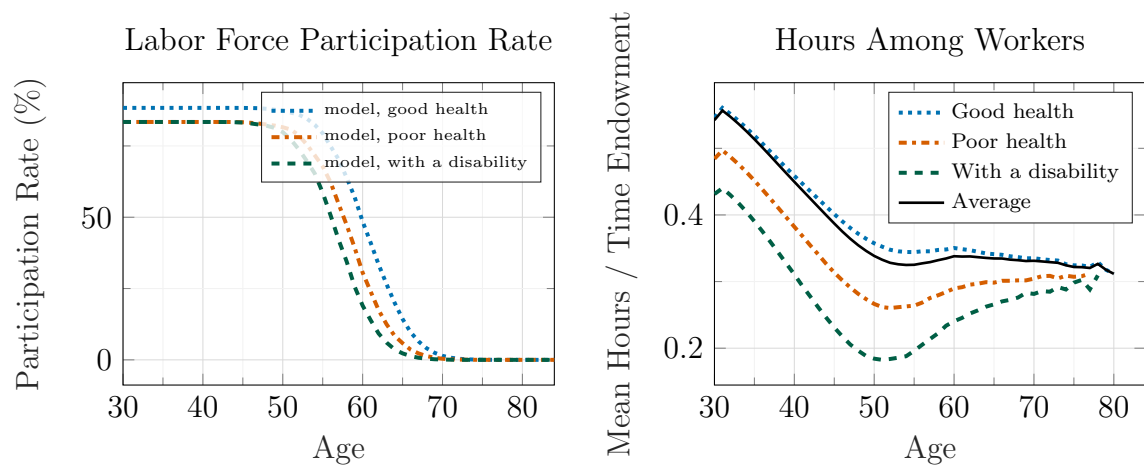
*Notes:* Means (thick dashed) and medians (thin dashed). Working-age gaps across health reflect the composition of ability and promise histories. After exit, the retirement rule is the same for every health state – consumption depends only on the promised-utility state – so remaining by-health gaps reflect only the composition of promises at exit and narrow as health status mixes.

Figure 20: Median Savings Wedge by Health



*Notes:* The median value complements the means of the main text. The two measures nearly coincide from the early 30s through the mid-40s (both fall to  $-0.03$  to  $-0.04$  at 40) and again late in life, where both decrease with mortality, and are steepest for the disabled.

Figure 21: Additional Exhibits of Optimal Policies



*Notes:* Labor force participation and mean hours worked as a fraction of time endowment. Labor force participation is higher in good health and lower in disability. Hours are similarly ordered in health status and bounded away from zero.

## C.8 How Implementable Are the Optimal Wedges?

The optimal wedges of Section 6 are allocations, not tax instruments. Following Appendix A.7, this appendix measures how much of their variation simple tax schedules capture, using a variance decomposition on the simulated optimum (Figure 22).

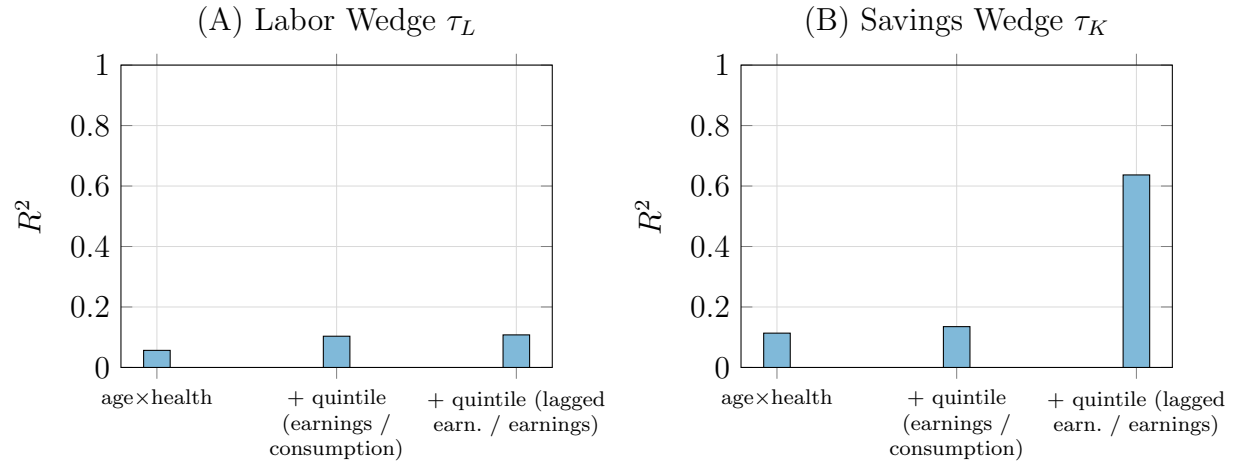
A statutory schedule conditioning on age and health alone captures only 5.7% of the optimal labor wedge variation and 11.3% of the savings wedge variation. For the labor wedge, adding a current-earnings test raises this to 10.3%, and one period of earnings history adds little more (10.8%). For the savings wedge, a consumption test adds almost nothing (13.5%), while an earnings test on top of it reaches 63.7%. The savings side, unlike the labor side, is substantially implementable with earnings-conditioned instruments.<sup>39</sup>

Two lessons follow. First, the optimum is history dependent: most wedge variation occurs in the promised-utility state, so implementation requires history-contingent instruments—the retirement-consumption promises of Figure 4 in our decentralization—rather than a static schedule. Second, an earnings-conditioned savings tax or subsidies captures nearly two-thirds of the savings wedge variation, more than any other simple instrument. Together with the negative median wedge at older ages, these could rationalize earnings- and age-conditioned preferential savings treatment—401(k) and IRA accounts with means-tested clawbacks—as the natural implementation of the savings wedge. This is consistent with Golosov and Tsyvinski’s (2006) asset-tested disability insurance.

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<sup>39</sup>The unit of observation is an agent-period of the simulated optimum (one million agents, ages 30–84). A schedule is a partition of agent-periods into cells. The reported statistic is the  $R^2$  of these schedules, which is the share of the wedge’s cross-sectional variance that the best tax function measurable in those variables could reproduce. No functional form is imposed on the tax functions except their dependence on the cell information. The labor wedge variations are computed among worker-periods for schedules that depend on age  $\times$  health (first bar); then  $\times$  current-earnings quintile (second bar); then  $\times$  lagged-earnings quintile (third bar) which corresponds to one step of history dependence. The savings wedge variations are computed among all alive agent-periods for schedules that depend on age  $\times$  health; then  $\times$  consumption quintile, the proxy for an asset or means test; then  $\times$  current-earnings quintile. Quintiles are computed within each age-health cell, so an earnings test may differ by age and health. Cells with fewer than 50 observations are not split further for numerical precision.

Figure 22: Variation in Optimal Wedges



*Notes:* Share of optimal wedge variation captured by progressively richer tax schedules (cell-mean  $R^2$  on the simulated optimum). The labor ladder (panel A) is computed on worker-periods; the savings ladder (panel B) on all surviving agent-periods.

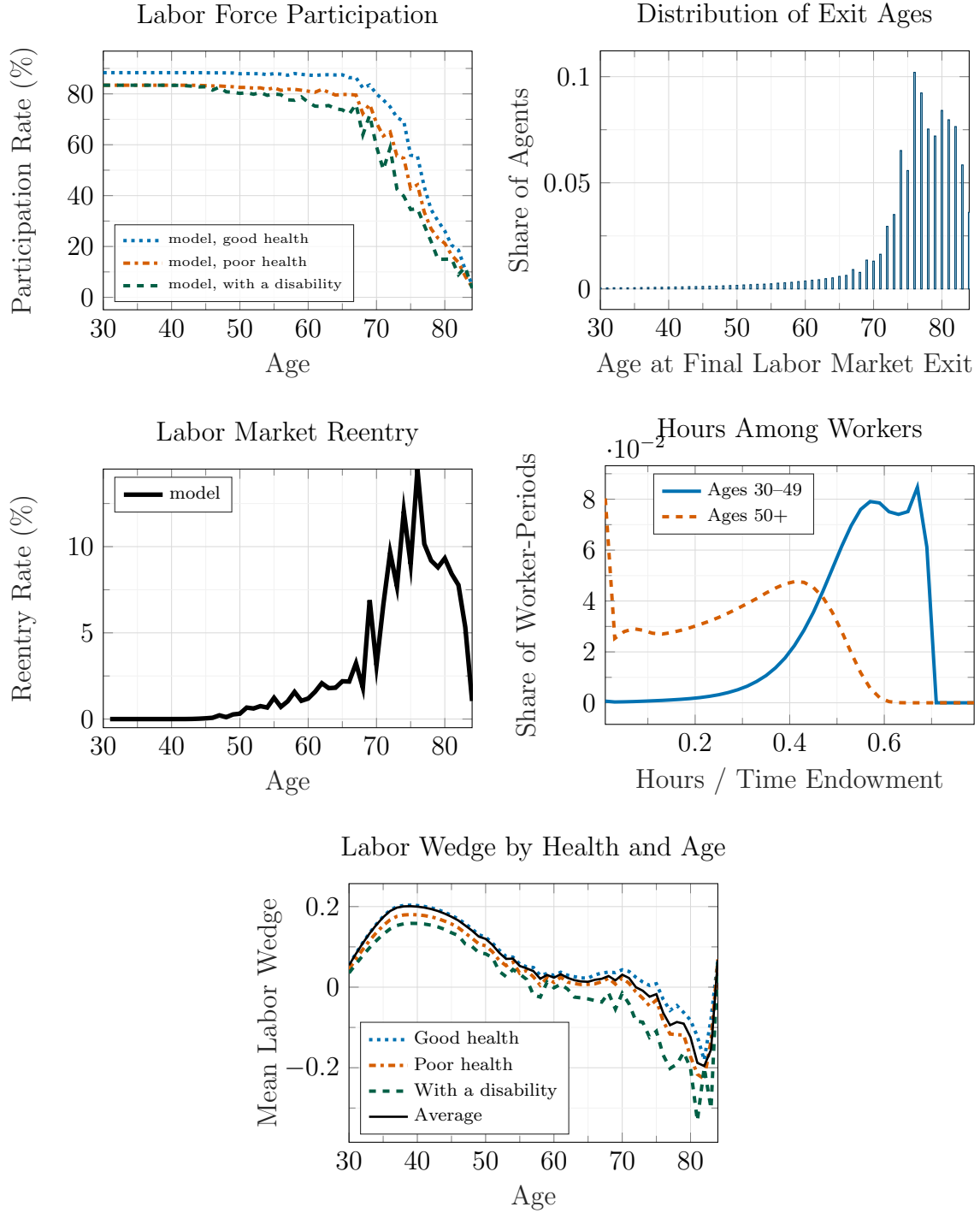
## C.9 Counterfactual: No Fixed Costs and No Reentry Cost

This appendix reports the counterfactual referenced in Section 6, which removes both the fixed participation costs and the reentry cost ( $\phi_{RE} = \phi_n(h) = 0$ ) while holding everything else at the headline calibration. Labor force participation at age 60 rises from 0.549 to 0.989 for the healthy and from 0.338 to 0.961 for workers in poor health or with a disability. With the fixed costs removed, workers work a few hours, and with the reentry cost also removed, exits stop being absorbing, with nearly half of age-70 nonworkers returning to work. Retirement as a discrete, absorbing final exit therefore dissolves once the nonconvexity and the reentry cost are absent.

## C.10 The Random-Walk Economy

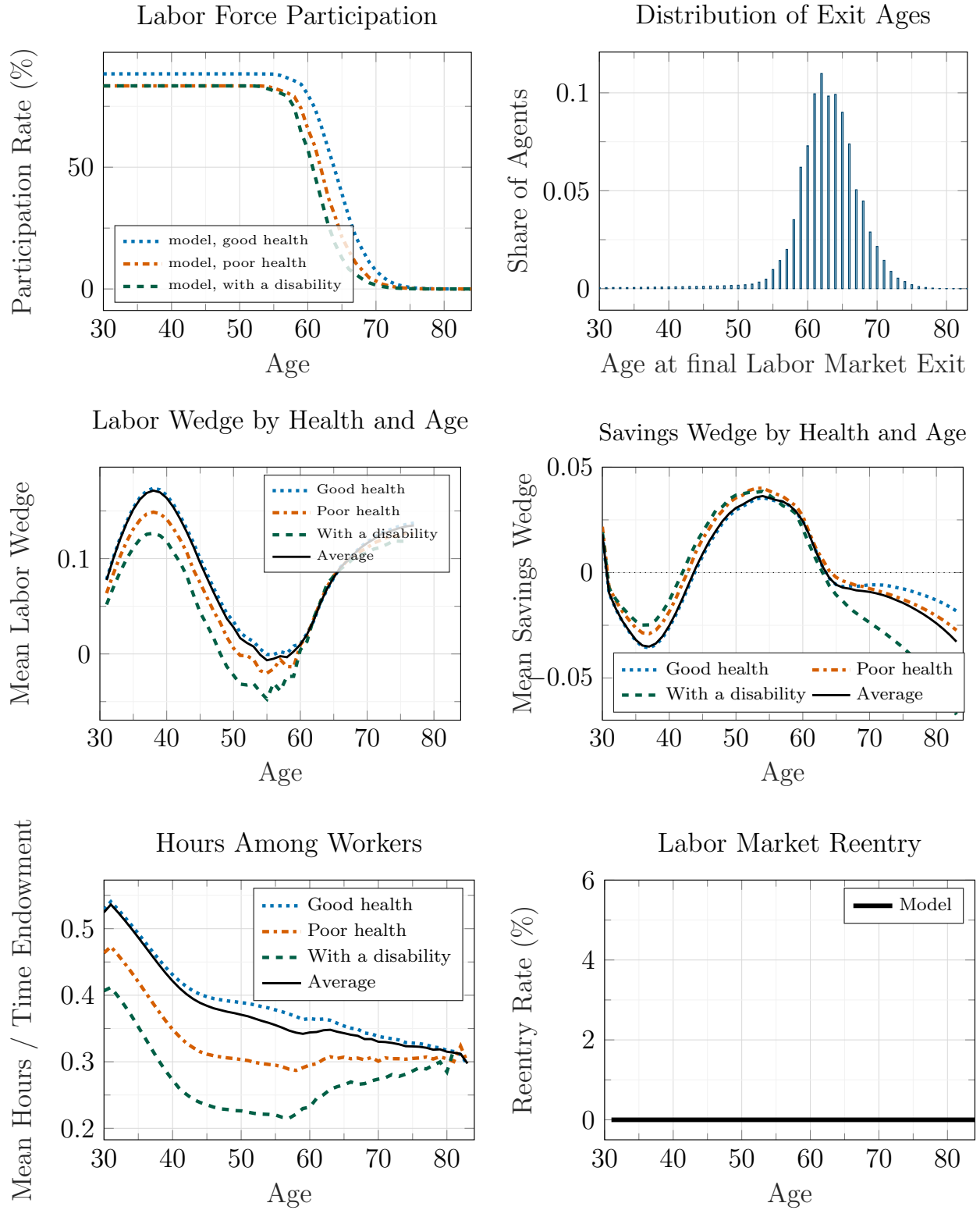
This appendix collects the exhibits for the random-walk economy ( $\rho = 1$ ) of Section 6.5; Table 4 compares it with the headline economy. Relative to the estimated-persistence headline, the random-walk economy raises the mean labor market exit age from 59.0 to 62.8: permanent shocks keep the upper tail of late-life productivity fat, and that tail sustains long careers.

Figure 23: The  $\phi = 0$  Counterfactual



Notes: The  $\phi = 0$  counterfactual (no fixed costs, no reentry cost): participation, the exit-age distribution, reentry, hours, and the labor wedge.

Figure 24: The Random Walk Economy ( $\rho = 1$ )



Notes: participation, the exit-age distribution, the labor wedge by health, saving wedge by health, hours, and reentry.